



ANGLIA RUSKIN UNIVERSITY
FACULTY OF BUSINESS AND LAW

MANUFACTURING PRACTICES AND OPERATIONAL PERFORMANCE: THE ROLES OF SUPPLIER AND INTERNAL INTEGRATION

MOHAMMAD MEZIAD AL-JUNIDI

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Dedication

This work is dedicated to:

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ABSTRACT

FACULTY OF BUSINESS AND LAW
DOCTOR OF PHILOSOPHY

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Operational performance is considered vital and at top priority in all organizations for profit maximization, which is achieved through the cost and flexibility. Following that, important pillars such as TPM, TQM, supplier integration, and internal integration are crucial for organizations. Therefore, this study focuses on the linkage of these organizational factors to improve cost and flexibility, which enhances performance for identifying and understanding the sustainability and expansion goals of the organizations. More specifically, the current research aims at exploring and understanding the relationships of manufacturing practices (i.e., TPM, TQM) and SCM (i.e., supplier integration, internal integration) along with operational performance (i.e., cost, flexibility) of the industrial firms. This study was conducted through fundamental rules and the basis of the positivist paradigm, which favours using the quantitative method in order to investigate the research problem. The research philosophy follows the objectivism approach through deductive reasoning. This research incorporated quantitative data from 224 Jordanian industrial firms through traditional and online questionnaires. This data was analysed by using SPSS and AMOS software in order to obtain valid, reliable, and generalisable findings and conclusion. In this way, the current study contributed to highlight and develop a better understanding of the potential determinants of operational performance through generally proving its significant correlation with the efficient SCM and the proper implementation of manufacturing practices. Furthermore, findings from this research revealed that supplier integration is significantly as well as positively impacting the cost and flexibility performance of the organisation, while internal integration is found to have significant positive effect on flexibility performance of the organisation. Moreover, the study concluded that understanding the different operational performance's determinants (cost and flexibility) allows stakeholders in the industrial sector to make thoughtful decisions that can help them producing more unique products with a competitive quality, which enables the industrial organisations to manage their profitability concerns.

Keywords: Operational Performance; Organisational Performance; Manufacturing Practices; Supplier Integration; Internal Integration

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Abbreviations

TQM	Total Quality Management
TPM	Total Productive Maintenance
SCM	Supply Chain Management
JIT	Just in Time
SI	Supplier Integration
II	Internal Integration
OP	Operational Performance
OEE	Overall Equipment Effectiveness
RBV	Resource-based View
GDP	Gross Domestic Product
SPSS	Statistical Package for the Social Sciences
AMOS	Analysis of a Moment Structures
EFA	Exploratory Factor Analysis
CFA	Confirmatory Factor Analysis
STD	Standard Deviation
AVE	Average variance extracted
MENA	Middle East and North Africa
JD	Jordanian Dinar
FTA	Free Trade agreements
ISO	International Organization for Standardization
JIPM	Japan Institute of Plant Maintenance
SMEs	Small and Medium Enterprise

Chapter 1 : INTRODUCTION

1.1. Overview

Operational performance is seen as a key element of organisational success; however, this success could only be achieved when the key factors, which are contributing to operational performance, are focused on by the organisations. Operational performance implies the effective conversion of inputs into desired outputs in a specified time (Polder et al., 2013). Hence, the inputs, outputs, and their interlinked ratios are the main determinants of organisational performance, which could be fulfilled through the careful management of the operational performance of organisation and its contributing factors. It is essential for organisations to identify as well as understanding the internal and external factors; those are significantly contributing to the production operations. Internal factors such as preventive maintenance, quality management, supplier and internal integration, operational flexibility, and cost decisions may affect the operational performance levels of an organisation (Biener et al., 2016).

To achieve a sustained competitive advantage, supply chain management (SCM from here onwards) practices and total quality management (TQM from here onwards) practices have raised to the top of the priority list. TQM, total productive maintenance (TPM from here onwards), and SCM have been studied as the most significant predictors of company performance by researchers from around the world using various models (i.e Bouranta, et al., 2017; Danese, 2013; Huo, 2012; Kaur et al., 2012). A set of procedures known as TQM emphasizes continuous improvement, ongoing results measurement, meeting customer requests, a team-based approach to problem-solving, competitive benchmarking, lowering workload, long-term planning, and strong relationships with suppliers (Soares et al., 2017). Meanwhile, SCM has become one of the most important and effective competitive strategies for manufacturing companies in recent years. The fundamental tenet of SCM can be summarised as taking input from suppliers, adding value, and giving it to the customer.

Based on the above concepts, organizations need to keep a constant focus on such factors, if they want to attain the desired levels of organizational performance through the enhancement of operational performance (Polder et al., 2013). Moreover, through relying on the

aforementioned discussion, the current study focuses on identifying the key determinants of operational performance in the industrial sector, by exploring the impact of manufacturing practices like TQM and TPM in the operational performance (Cost, Flexibility), while studying the role of supplier and internal integration in the Jordanian industrial sector. The focus of this research is on the significance of internal and external integration in companies as a strategy for improving operational performance and other outcomes. These include the assessment of the identified benefits; those can be derived from the adoption of TPM and TQM in organisations to enhance operational performance and excellence. It focuses on these innovations in the industrial sector, which are designed to improve the efficiency and the operational performance of manufacturing processes. Moreover, it is essential to conduct empirical research to identify how these innovations influence suppliers and internal integration as well as operational performance in the industrial sector.

1.2. Research Context

The core focus of this study is to evaluate the role of manufacturing practices and supply chain integration in enhancing the operational performance of Jordanian industrial sector. Operational performance is considered as a critical success factor for organisations across the globe. It is best indicated as the level at which business units of an organisation achieves organisational goals through their combined efforts (Nabass et al., 2019). Many organisations create specific job functions or departments that mainly focus on the transformation of organisational assets into enhanced performance (Saleh et al., 2018). The enhancement of operational performance requires organisations to modify their manufacturing practices. Such modification leads to agility, precision, and efficiency to consistently outperform the competition (Saleh et al., 2018). Enhancement of operational performance has been widely focused by businesses for organisational transformational initiatives. Innovative approaches, technologies and tools are enabling the firms to enhance their performance with flexibility, agility, and precision (Nawanir, 2016). In recent competitive environment, such transformations have become vital for businesses. Enhanced performance could no longer be achieved through money, time, and resources only. Therefore, most of the businesses are moving beyond traditional options for reducing costs and improved performance. However, the focus is solid; ensuring business success through enhanced performance (Ali et al., 2020).

In this regard, manufacturing organisations primarily concentrate on their manufacturing practices to accomplish their operational performance excellence. Manufacturing practices are

considered as a set of activities that leads towards achieving the targeted organisational goals. Organisations adopting latest manufacturing practices or transforming their traditional practice to the latest required ones, are more viable to achieve success (Nawanir, 2016). In recent globalised markets, the businesses are not limited to the domestic markets only, rather the organisations focus on attaining international standards (Nayal, et al., 2022). To compete in the international markets, operational performance excellence is widely reported as a key factor that is significantly influenced by the manufacturing practices adopted by the organisations. However, in manufacturing organisations across globe TQM and TPM are widely implemented manufacturing practices (Nabasset al., 2019). Therefore, these practices are a key concern for researchers in supply chain field as well. Literature reports evidence of studying the relationships between manufacturing practices and operational performance. In this context, other aspects of supply chains are also reported to have significant relationship with operational performance including supply chain integration (Matar and Eneizan, 2018). The literature lacks the interrelationships of all these main factors including manufacturing practices, operational performance, supply chain integration etc (refer to table 2-1). Therefore, further research needs to investigate and highlight the nature of relationships among these key factors.

Furthermore, the manufacturing organisations and industries are the backbone of any economy. These contribute significantly to economic development of the country. Jordan is not a different case. Jordan is a developing, non-oil producing country that is situated in the heart of the middle east region, where many businesses are interested in venturing, due to the economic growth that this area has witnessed. Choosing Jordan as a venturing destination provides companies and businesses with several benefits; a wide consumer base and higher revenue numbers (Ali et al., 2020). Being the venturing destination, the international business in Jordan is moving rapidly. But for achieving profitability, there is a need for the industrial sector to implement strategies for TQM, TPM, strategic integration, and internal integration through operational performance (cost and flexibility) for profit maximization. Therefore, this research focuses on elaborating how manufacturing practices and supply chain integration influence the operational performance of the Jordanian industrial sector. Jordanian industrial sector significantly contributes to the economic development of the country (Saleh et al., 2018). Such sectors need the critical focus of researchers and policy makers to conduct their critical role in highlighting the factors that can enhance the operational performance of manufacturing industries. However, different parameters and measurements must be thought about in advance for the

effective decisions regarding the selection of the right processes to develop the Jordanian industrial sector (Dahi, 2014).

The economic stability and strategic relations allow Jordan to be among the viable countries to run businesses as well as attracting further relations with other countries across the world. Jordan is also regarded as an emerging destination in which a solid base for the renewable energy exists that aim to achieve clean technology. These advantages constitute Jordan's unique strategic position that helps to facilitate the socio-economic growth and sustainable development (Alrgaibat, 2017; Mugableh, 2019). According to a report from the economic policy council, it seems clear that Jordan's economic growth is one of the reasons that attract businesses and investors. There is an economic growth plan ranging from 2018 to 2022, which puts Jordan in a position to achieve sustainable growth and trajectory. Through the plan, the economic growth in Jordan is expected to be doubled in the coming years. The report also expects an increase in the number of businesses targeting the Jordanian market due to the flexible economic environment for conducting business as well as the political stability and security and the vital and modern infrastructure that give investors surety of getting returns on their investments (Alrgaibat, 2017).

The Jordanian economic status is deemed to be attractive to foreign investments, due to the high skilled workforce and the high levels of employment opportunities that can be achieved from the process. Jordan is a major touristic destination comparing to other countries in the middle east. Among the tourism types that Jordan is known with is the medical care and tours, entertainment adventures and historical site investigations. According to the world bank, there are major reforms that support the strong economic pathway not only for investors, but also for the locals in Jordan (Al-Hawary and Al-Hamwan, 2017). These reforms furnish the ground for long-term prosperity in which employees and working individuals are able to cover the income tax, handle different business regulations as well as understanding investment promotions and laws.

The support of different global free trade, as well as bilateral agreements, is also worth noting. Jordan 2025 vision has been among the guiding principles for the investors and the country at large towards steering solid economic growth. Through this vision, there is an establishment of an economic growth plan, which shows a gradual economic increase from 6.5 recorded in the year of 2021 to 7.5 that should be achieved by 2025. The vision is generally aiming at ensuring a jumpstarting of Jordan's economic development to be beyond the expectation and

support the Jordanian government to grow (Sahoo and Yadav, 2020). These are the main reasons why the industrial sector in Jordan has managed to grow immensely and businesses are coming from foreign directions to be welcomed in Jordan (Lubbad and Adam-Bagley, 2021). Another significant factor that becomes the attraction of Jordanian economy for internal organisations is its educated youth. Educated young people contribute significantly to the success of organisations and understand the latest strategic and structural changes required in industries (Sahoo and Yadav, 2020; Mahafzah, 2015; Lubbad and Adam-Bagley, 2021).

Moreover, Jordan as a country also has the opportunity of offering different location advantages that are vital for the attraction of investments to the Jordanian industrial sector. These include the provision of a high-quality air transit for access; strong railroads infrastructure with better highway infrastructure developed. The Jordanian industrial sector contains a different array of essential sub-sectors. These sub-sectors help the industries to exhibit the different sophistication, and bring about proper dynamisms, especially the economic boost process. The industrial sector is characterised by reciprocity and dealing with the issues complementarily. With these issues, sub-sectors in connection with the industries require the production to be appropriate for a better development (Matar and Eneizan, 2018).

Despite having all the above-mentioned advantages of the Jordanian economy, for attracting the international investors, there is a need to highlight the facts about manufacturing sectors including the ways of enhancing operational performance. Moreover, highlighting the key factors can contribute significantly in enhancing the level of production of these manufacturing industries, which ultimately attracts international investors (Mahafzah, 2015). Therefore, this research is concentrating on highlighting the key factors of Jordanian industrial sector that can contribute positively to enhancing the operational performance of manufacturing firms in Jordan. The main focus of this research is to highlight the contribution of TQM and TPM among other manufacturing practices because these two practices are widely considered as significant contributors of operational performance (Konecny and Thun, 2011; Kaynak and Hartley, 2009; Tiwari et al., 2015; Huo et al., 2018, Phan et al., 2019, Danese, 2013; Sahoo and Yadav, 2020; Ahmad et al., 2019; Sahoo, 2019; Wicaksono and Bambang, 2019; Phan et al., 2019). On the other hand, among different SCM practices, internal integration and supplier integration are found to be comparatively better contributors to the operational performance than the customer integration (Gimenez et al., 2012; Mofokeng and Chinomona, 2019; Swink and Schoenherr, 2015). In this way, this research is a significant contribution as well, as it

reports all these significant factors combined in one study and their accumulated contribution to the operational performance of organizations.

1.3. Problem Statement

In the recent era, no organisation could survive in the marketplace, unless having a keen focus on the enhancement of their operational performance, which also results in attaining the core purpose of stakeholders' wealth maximization. Therefore, almost all organisations shed the light on improving their efficiency in every possible way, but successful are the only ones who have a keen focus on the contributing factors that could have a significant impact on inputs, outputs, and the overall performance of the organisation (Aspinwall and Elgharib, 2013; Phan et al., 2019).

Therefore, there are several studies investigated the contributing factors towards organisational performance and efficiency. In this regards, the current literature reports the role of manufacturing practices on organisational performance in different ways, but among all other manufacturing practices; TQM and TPM are considered as the most significant contributing manufacturing practices in the recent era (Ahmad et al., 2019; Sahoo 2019; Phan et al., 2019; Sahoo and Yadav, 2020), because these are found to have a significant impact in the efficiency of organisational performance, as compared to other manufacturing practices of the firm. Additionally, the literature also reports that the most significant determinants of organisational performance efficiency are operational performance efficiencies like the cost efficiency and the flexibility efficiency (Ahmad et al., 2019; Wicaksono and Bambang, 2019; Sahoo and Yadav, 2020).

Recent studies have reported these manufacturing practices (TPM, TQM) as significantly contributing to the operational performance; cost performance, flexibility performance (Sahoo and Yadav, 2020; Ahmad et al., 2019; Sahoo, 2019; Wicaksono and Bambang, 2019; Phan et al., 2019). In addition, some studies have also reported the interrelationship of TPM and TQM to evaluate their impact in organisations' performance (Ahmad et al., 2015; Konecny and Thun, 2011). However, findings of these studies lack reporting whether manufacturing practices directly affect the operational performance of the Jordanian industrial companies.

In a similar pattern, some other studies including (Konecny and Thun, 2011; Kaynak and Hartley, 2009; Tiwari et al., 2015; Huo et al., 2018, Phan et al., 2019, Danese, 2013) reported that organisational performance and the manufacturing practices of organisation, like TPM and

TQM are influenced by supplier integration, and internal integration (will be discussed in detail in chapter 3), but most of these studies reported the impact of these supplier, and internal integration in manufacturing practices (TPM, TQM) and organisational performance separately. Like the direct impact of internal and external integration on firm performance is reported by (Droge, et al., 2004), and the direct influence of supplier, internal and external integration on business and operational performance of firm is reported by Flynn et al. (2010). However, the impact of supply chain integration in the performance of firms is also reported in different ways (Gimenez et al., 2012; Mofokeng and Chinomona, 2019; Swink and Schoenherr, 2015), while the current literature lacks evidence regarding the impact of supplier and internal integration in manufacturing practices (TPM, TQM) and the operational performance of the firms in a single setting.

Moreover, most of these studies are focusing on the resource-based view of the firm, which indicates that a firm achieves its objectives, due to the internal resources at its disposal and how it organises them. In this respect, the supplier integration is seen as a way of organising the link between the organisation and its suppliers. In the same way, the internal integration involves strategies, such as TPM and TQM, which are adopted as a tool for organising and enhancing the performance of the organisational operations as a whole. Hence, this study is based on the need to undertake research on the relationship between TPM/ TQM and cost/ flexibility by assessing the available resources of the organisation and how they are organised and integrated. Because based on previous studies, it is found that there is a significant gap in the current literature regarding the impact of manufacturing practices in organisational performance, while investigating the role of supplier and internal integration in developing such relationships. Thus, in order to fill this gap, the current study concentrates on investigating the impact of manufacturing practices (TPM, TQM) on the operational performance (cost, flexibility) of industrial firms with the role of supplier and internal integration.

Moreover, this research is mainly focusing on two manufacturing practices (TPM, TQM) because as reported earlier, these manufacturing practices are recently found to be the most significant contributors towards organisational performance, as compared to other manufacturing practices across globe (Sahoo and Yadav, 2020; Ahmad et al., 2019; Sahoo, 2019; Wicaksono and Bambang, 2019; Phan et al., 2019). Similarly, TQM and TPM are also found to be the significant contributors towards Jordanian organisations' performance. Al-tarawnah (2021) reported that TQM not also significantly contributes to enhancing

organisational performance; but also creates a competitive advantage for Jordanian organisations, while (Al-Refaie, et al., 2022) reported TQM and TPM as significant contributors towards performance of firms indicating that TQM and TPM have comparatively larger impact on performance than other manufacturing practices in the Jordanian industrial organizations.

Secondly, operational performance (Cost, flexibility) is the main focus of this research, due to its contribution towards defining organisations' success (Ahmad et al., 2019; Wicaksono and Bambang, 2019; Sahoo and Yadav, 2020). In Jordanian industrial firms cost and operational flexibility are found to be the key determinants of operational performance (Yousaf et al., 2019; Alrwashdeh et al., 2022) that's why the current study is primarily focusing on cost and flexibility in terms of operational performance of Jordanian industrial firms. Operational flexibility is referred to the capability of organisational operations to respond according to the changes in needs and requirements of the market (Yu et al., 2021). In recent era, operational flexibility has become one of the critical requirements for attaining the competitive edge. Operations of the organisations are required to be flexible enough to be modified according to the needs and demands of customers. Operational flexibility is considered to be among the most significant aspects that lead to enhanced operational performance (Yu et al., 2015). The other important aspect being focused by the current research is the cost of operations. Organisations always emphasize on reducing the costs of their operations that is termed as the operational efficiency of the organisation. Cost is also considered as a significant aspect of operational performance as operational performance could never be enhanced while having a control on the costs, otherwise organisations are unable to gain the competitive advantage (Yu et al., 2021). Therefore, this research concentrates on the operational flexibility and cost for operational performance of Jordanian industrial organizations.

Thirdly, while studying the supply chain integration, many previous studies have explored the impact of customer integration in different ways (Chavez et al., 2015; Martinelli and Tunisini, 2018; Alzaydi et al., 2018; Schumacher et al., 2018), but rare evidence is found regarding the impact of supplier integration, and internal integration in the relationship of manufacturing practices (TPM, TQM) on operational performance; therefore, this gap needs to be filled and the current research aims at filling this gap by focusing on these indirect relationships.

Moreover, the current research is mainly focusing on supplier and internal integration while customer integration is not included because of the targeted population of the study i.e., Jordan.

Because in Jordan, supplier and internal integration are found to have greater impact in firms' performance as compared to customer integration. Abdallah et al. (2021) and Sharabati et al. (2020) reported internal integration as more significantly contributing towards Jordanian firms' performance, while (Pakurar et al., 2019) reported supplier, and internal integration as more significantly contributing to Jordanian firms' performance, as compared to customer integration. Therefore, the current study concentrates on the supplier and internal integration of Jordanian industrial firms. In fact, Jordanian business society is not currently in a state to involve customers in their business processes i.e., business culture in Jordan is a bit different than the developed nation's business cultures.

However, the impact of supplier and internal integration has been investigated on organisational performance in different ways, but the inter-relationships of supplier integration, and internal integration in relation to TPM and TQM and their combined effect on performance is lacking in the literature. Especially, in the context of Jordanian industrial firms as on one side supplier and internal integration are found to be the significant contributors towards organizational performance (Abdallah et al., 2021; Pakurar et al., 2019; Sharabati et al., 2020), while on the other side, TPM and TQM are found to have significant contribution to organisational performance of Jordanian firms (Al-tarawnah, 2021; Al-Refaie et al., 2022); therefore, there is a dire need to investigate their interrelationship and inter linked contribution towards organisational performance of Jordanian industrial firms. The integration of internal and external stakeholders in the development of TPM and TQM strategies can be highly beneficial in developing a sustainable organisation in which the material supplied is suitable to the needs and requirements of the entire value chain (Ahmad et al., 2019). This allows reducing the cost and quality challenges that firms may encounter, during their production processes. TQM focuses on the overall quality of the operations of organisations, while TPM focuses on the enhancement of employees' efficiencies to enhance the production from operations, while the combined effect of these results in enhanced performance of operations (Al-tarawnah, 2021). The critical issue in this research is investigating how supplier and internal integration can improve the organisational outcomes in terms of cost reduction and flexibility. There is a dearth of literature on the impact that TPM/ TQM can have on organisational outcomes through internal and supplier integration in line with the resource-based view of the firm.

1.4. Research Questions

The review of previous literature indicates that the manufacturing practices (TPM, TQM), and supplier and internal integration are significantly contributing to the organizational performance particularly the operational performance of organizations. Especially, in the context of Jordanian industrial organizations, TPM, TQM, supplier and internal integration are found to be the most significant contributors towards organizational performance, but it is also found that manufacturing practices and SC integration are separately studied in this particular context. Hence, a huge gap exists in literature for studying their interrelationships and inter-linked contribution towards operational performance of organisations, especially in Jordan. Therefore, the current research aims at fulfilling this gap by addressing the following research questions. From the resource-based view perspective, this research addresses three major research questions:

- 1- How do TPM and TQM jointly impact cost and flexibility performance in the Jordanian industrial sector?
- 2- How do supplier and internal integration jointly impact TPM and TQM in the Jordanian industrial sector?
- 3- How do supplier and internal integration jointly impact cost and flexibility performance in the Jordanian industrial sector?

1.5. Research Aim and Objectives

The current research aims at investigating the impact of manufacturing practices (TPM, TQM) and supply chain integration (specially focusing on supplier integration and internal integration) on the operational performance (i.e., cost, flexibility) of the Jordanian industrial sector. The core purpose is to investigate the interrelationships and inter-linked contribution of manufacturing practices and supply chain integration towards operational performance of Jordanian industrial firms. The research focuses on developing a better understanding of the internal and external factors that influence operational performance. Thus, in order to achieve the main aim of this research, the following objectives must be addressed:

- 1- Investigating the joint impact of TPM and TQM in the cost and flexibility performance in the Jordanian industrial sector?
- 2- Investigating the joint impact of supplier and internal integration in the TPM and TQM in the Jordanian industrial sector?

- 3- Investigating the joint impact of supplier and internal integration in the operational performance in the Jordanian industrial sector?

1.6. Research Motivation

In recent dynamic market environment, when the choices and demands of customers are frequently changing, achievement of competitive advantage has become challenging for organisations. Therefore, organisations concern of improving their operational efficiencies to compete the market. At one end, they need to focus on sustaining their competitive position, and on the other side they also need to attain the core objective of profit maximisation for their stakeholders. Such scenarios have diverted the attention of industrialists at one side, and the researchers on other side to highlight the key factors that can enhance the operational efficiency of organisations. Operational efficiency of the organisations enables them to convert their inputs into desired outputs while having their wealth maximisation as well. In this regard, researchers have started contributing their efforts through investigating and highlighting the key factors that must be the key focus of organisations. Despite having continuous efforts exerted by researchers, still there are key areas that need to be focused by researchers as well as industrialists. This fact is the key motivation of current research. Literature reports different factors that contribute in enhancing the operational performance of organisations, but the interrelationships of many factors are missing in literature. Therefore, the major motivation of this research is to highlight the fact whether the key factors reported by literature, have direct impact on the operational performance of organisations, or the interrelationships of these factors effects the operational performance of organizations. The current study is, hence, exploring the key factors that contribute towards better operational performance of industrial firms. It is also motivated to highlight the fact that the factors reported separately in different studies can have combined impact on the operational performance of organisations. This research is motivated by the need to implement innovative management philosophies in Jordan to support its industrialization. Jordan needs to improve its industrial sector by determining how to best utilize the available resources and connections with its active stakeholders.

Moreover, it is indicated by the previous literature that at one side manufacturing practices (TPM, TQM) significantly impact the organisational performance while on the other hand supply chain integration plays a significant role in enhancing organisational performance. Especially in the context of Jordanian industrial organisations, TPM, TQM, supplier and internal integration are found to have significant contribution towards Jordanian organisations’

performance. But it is also found that these indicators are studied separately by the previous studies while a huge gap exists in literature for studying the combined or inter linked effect of these indicators. Hence, this gap is another motivation for conducting this research so that the key factors could be highlighted, and Jordanian industrialists could focus on these factors for enhancing their organisations' operational performance. Moreover, it is clear from the available literature that TPM and TQM have significant benefits on organisational outcomes, which can be addressed through effective integration of internal resources as well as the suppliers. Furthermore, this research is also motivated by the dearth of literature on the specific context facing the Jordanian industrial sector. It is meant to highlight how the individual aspects of the sector can be improved through the adoption of TPM and TQM.

1.7. Research Contribution

Operational performance of organisations has become the source of competitive advantage for the organisations. Therefore, organisations focus on the key factors that lead to enhance their operational performance. Researchers have also started contributing by highlighting the contribution of key factors. Literature reports different evidence in this regard but lacks evidence about the combined effects of different industrial factors that contribute to the enhancement of operational performance. This research is a significant contribution to fill this gap in literature. Therefore, this research highlights the role of manufacturing practices including TQM and TPM, combined with the supply chain practices including supplier integration and internal integration towards operational performance. It highlights the fact that how different industrial factors accumulatively enhance the organisational outcomes. This research not only contributes significantly to literature but also have significant contribution to the economic development of Jordan. It highlights the facts about the manufacturing practices and supply chain practices being implemented in Jordan industrial organisations. It also highlights further improvements required by Jordanian industrial sector that can help enhancing the operational performance of firms. The focus of the research is to provide information to managers on how they can incorporate supplier integration and internal integration of their operations to improve their performance. It will highlight the relevant practical considerations for improving cost effectiveness and flexibility in organisations by effectively utilizing their internal resources and organising them effectively in line with the resource-based view of the firm. This research also has significant contribution to attract foreign direct investment for Jordanian economy. By highlighting the key factors and their combined influence on the operational efficiency of the organisations in Jordan it enhances the knowledge of key

stakeholders of Jordan's industries. The improvement in Jordanian industrial sector can be the source of attracting foreign investments that ultimately leads to economic development of the country. Hence, this research not also significantly contributes to the literature, but also to the economic development and growth of Jordan.

1.8. Thesis Structure

Before moving to the next chapter, it is worthwhile to describe the structure that this thesis is following. This thesis includes nine chapters that are structured as presented in the following figure 1:

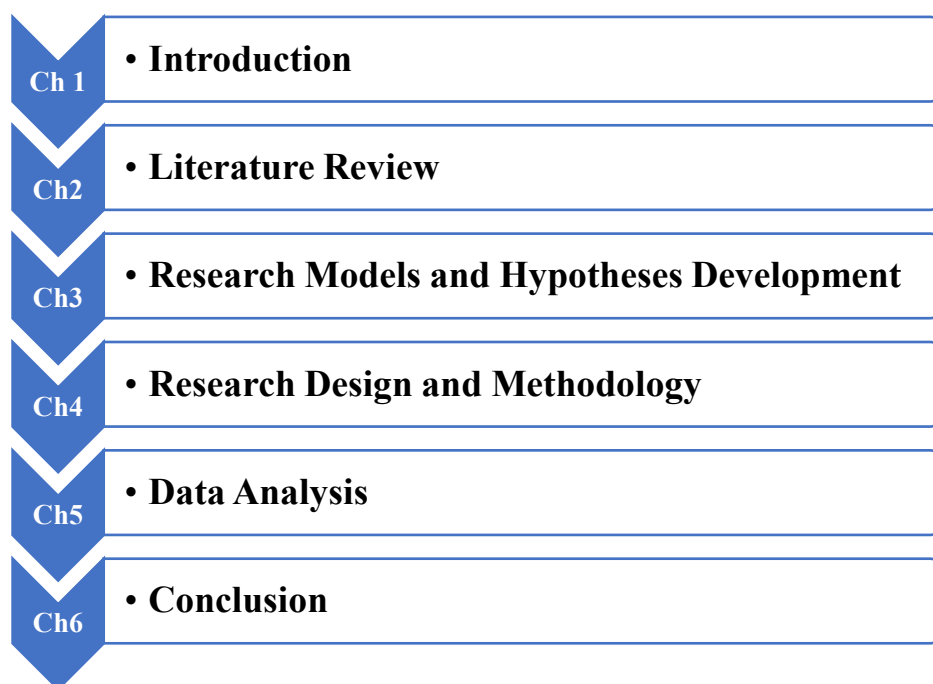


Figure 1-1: Thesis Structure

The above figure exhibits that the thesis includes six chapters, where the first chapter is the introduction, and it provides an overview regarding the research topic and the concerning theoretical boundaries. The chapter also provides a glimpse about the research problem and motivation as well as its aim, objectives, findings, and contributions. The second chapter discusses the available literature, which is focused on exploring the manufacturing practices, and their relationship with operational performance. Thereafter, the thesis moved to discuss the available literature, which is focused on exploring the SCM supplier integration, and internal integration and their relationship along with operational performance. After that in chapter three the thesis focused on discussing the extant models, which are related to this research. It also provides a theoretical background regarding the different research hypotheses. However,

in the fourth chapter, the study provided a detailed description of the methods, which are adopted to collect and analyse the research data as well as the chapter focused on discussing research's design. Additionally, the chapter describes the procedure that was used to obtain the data, which are required to achieve the study aims and objectives. In chapter number five, the study discussed the descriptive analysis of the study's data in order to test hypotheses as well as extracting the related results. The chapter also elaborated the steps through which the data was analysed by using the 'statistical package for social sciences (SPSS) software' to check the missing data and exclude any incomplete questionnaire. It further described and tested the validity, stability and normality of the used variables using the exploratory and confirmatory factor analysis. Finally, the last chapter was concerned with developing a framework in an attempt to describe the impact of the manufacturing practices in the operational performance of firms in the industrial sector. The chapter also concluded with the main takeaways and recommendations as well as suggested future research avenues.

Chapter 2 Literature Review

TQM AND TPM

This chapter critically reviews the extant literature regarding manufacturing practices (TPM, TQM) and their impact on the organisational operational performance. It also highlights the importance of understanding the relationships of manufacturing practices (i.e., TPM, TQM) and SCM (i.e., supplier integration, internal integration) along with operational performance (i.e., cost, flexibility) of the industrial firms. Therefore, the chapter focuses on discussing both TPM and TQM as a base to promote the ability of the industrial organisations to effectively reduce the cost and enhance the flexibility of their operational performance. Moreover, these manufacturing practices are discussed in light of the resource-based view as a theoretical lens that is adopted in research work.

2.1. Total Quality Management (TQM)

The term ‘(TQM)’ is defined as a management approach that is focusing on the long-term success of an organisation by enhancing customer satisfaction through constant improvements. It is a management approach in which all members of an organisation participate in creating the improvement strategies for the products, processes, and overall organisational culture. It is a management system for integrating the quality discipline into the management culture of an organisation (Juran, 1988; Feigenbaum, 1991; Ross and Perry, 2003). TQM is a change management quality approach for improving effectiveness, flexibility, and competitiveness of the business in line with customer requirements. It focuses on the overall quality measures for an organisation with the aim of meeting customer expectations (Karyotakis and Moustakis, 2014). It is a customer centred approach to the management of organisations that focuses on ensuring that their critical issues and needs are effectively addressed. The approach manages an organisation based on quality control, design, development, quality assurance and continuous improvement.

2.2. The evolution of TQM

Some of the first seeds of TQM were laid down as the principles of scientific management in the 1920s. However, the actual principles of TQM evolved from the quality assurance methods that developed during world war I in response to the needs of the war; factories engaged in large scale manufacturing of goods that were poor in quality. In response, quality inspectors

were introduced into the factories to correct the quality issues and ensure that failure levels due to quality would be minimised (Houston and Dockstader, 1998). After the war, quality inspection became a key element of manufacturing and was addressed through Edward Deming's theory of statistical quality control. According to the theory, variations in the production processes contributed to the variability in the end product. This could be eliminated through a more standardised production processes to ensure high-quality products (Ooi et al., 2008).

After the world war II, manufacturers in Japan were producing poor quality goods and needed to design effective ways of controlling quality levels. Their response to the challenge was that the Japanese union of scientists and engineers invited Dr. Edwards Deming to provide training them on how to manage quality processes. This was a key stepping-stone for TQM as the quality control and statistical analysis methods were heavily adopted by Japanese executives and engineers in the end of the 1950s. Joseph M. Juran taught the concepts of quality control and managerial breakthrough with a focus on the reduction of defects. The Japanese managers coined the term quality management systems (Juran, 1989; Ross and Perry, 2003).

By 1970, the Japanese named their approach to quality management, 'companywide quality control'. These approaches were critical to the ascension of Japan as a leader in quality. The aspect of quality control involved all employees in an organisation from the top management to all low-level workers. The results seen in Japanese companies were a critical driver for the interest from other countries in the quality management strategies (Juran, 1989). During this period, the high quality of Japanese products that were made under the conditions of total quality control made Japan a key manufacturer of goods for the international market. The Japanese companies were producing high quality products at low prices, especially electronics and automobiles, which present a stiff competition for the United States and European nations (Houston and Dockstader, 1998). Companies in the United States started exploring the application of the quality control techniques and, thus, initiated the practices of TQM (Arumugam et al., 2008).

The origin of the term TQM was in the 1980s and was inspired by a number of publications such as "*What Is Total Quality Control? The Japanese Way*" by Kaoru Ishikawa and Armand Feigenbaum "*Total Quality Control*". In 1984, the US Navy asked civilian professionals to assess its statistical process controls and make recommendations on how to improve operational efficiency. It was recommended that they adopt the processes and work of Edwards

Deming and, as a result, they branded the approach 'TQM' in 1985 (Houston and Dockstader, 1998). The methods spread from the Navy throughout other arms of the federal government and the private sector. Organisations in the private sector found the TQM approaches to be highly significant in enabling them to compete and recapture their market share and bid for government contracts (Ahluwalia, 2008).

2.3. The value of TQM

TQM is a management framework that focuses on improving the overall capacity of the employees to respond to the needs of customers. They have the mandate to ensure that organisational quality is managed in a systemic manner. It describes the quality policy of an organisation to develop clear strategies for defining how the employees work together to address them. TQM is a management system that focuses on different elements that enhance the value provided to organisations and customers (Bhat, 2010). These elements are highlighted in the eight principles of the practice: customer focused approach; total employee involvement; process thinking; an integrated system; a strategic and systemic approach; continual improvement; fact-based decisions and communication. These elements work together to improve consensus in the organisation, coordinate all aspects of the organisation and engage in continuous improvement efforts (Phan et al., 2019).

TQM is a tool that directly influences operational performance in an organisation. This is achieved by focusing on employees to drive both quality product and service provision and efficient customer handling procedures (Ross and Perry, 2003). Putri, et al. (2017) reported that employees' comfort and satisfaction in their jobs positively influences the firm operational performance. Employee job satisfaction includes actively involving these employees in the firms' activities from the start to the end, which promotes the achievement of positive performance and higher firm operational performance. Firms that involve their employees in every phase of the production process usually report better financial performance and operational performance as opposed to their counterparts (Yu et al., 2021). Involving employees in the different production phases can also reduce wastage of materials and resources on unnecessary renovations. Employees are considered as a critical aspect of the TQM framework that promotes increased operational performance through ensuring that the hands-on work employees are enjoying their jobs and contribute to the achievement of firm objectives (Yu et al., 2015).

Further, TQM management aims at building corporate culture in which customer satisfaction is seen to be the focal point. Customer satisfaction through TQM is maximised through continuous improvements of work processes and the active involvement of employees in production. TQM is a long-term process that requires setting goals and working to achieve them through continuous improvement and revision of the included work processes. Sufa and Ambarsari (2020) observe that customer satisfaction is not only about involving the employees, but also the various stakeholders in the production process such as business partners, society, suppliers and owners. Incorporating and harmonizing the quality dimensions of stakeholders can go beyond increasing customer satisfaction through quality goods to ensure the improved operational performance through the multiple stakeholder investment efforts.

The implementation of TQM in a firm is not enough as it requires establishing key success factors that are needed to support the critical role of TQM in achieving firm goals. TQM among other roles contributes to the quality control in firms, management of time and resource use, customer relationship and employee satisfaction. While these roles are quite clear in most of the firms that implement TQM, sustaining these benefits has proven to be a major hurdle for these firms. Thus, achieving performance or operational performance standards in a firm depends on its ability to establish and develop new attractive products that are competitive to excite customers and/or exceed the expectations of customers, form the ground-breaking climate that promotes business innovation and develop products that incorporate customer needs and concerns (Furterer and Elshennawy, 2005). Such firms can achieve the TQM through the development of business practices that regularly reduce defects in products while reducing waste and cycle time (Qazi and Simsekler, 2021). The establishment of strong relations with key suppliers increasing the number of committed employees in an organisation and equally provides avenues for achieving TQM practices.

Due to such positive characteristics, organisations are currently prioritising the Quality management frameworks such as TQM. But still there is a lack of focus on establishing the contribution of TQM to operational performance due to the absence of the suitable performance indicators. Pertinent to the achievement of the TQM framework is the establishment of performance measurement indicators. This is because the performance indicators establish the guidelines to what to do and by whom it should be done. Juran (1995) established that operational performance can be categorised into five broad categories namely, competitive

performance, quality improvement, the performance of microprocessors, product performance, the performance of macro processes and the cost of poor quality (Truong et al., 2017).

Firms that establish these performance indicators have clear guidelines on what is expected at every stage and therefore can monitor the success rate of the implementation process. The core principles and strategies, on which TQM operates, play a major role in enabling the organisations to achieve their goals of quality and responsiveness to customer needs. One of the key issues or values that TQM offers for organisations is associated with the costs involved. However, studies such as Lam et al. (2011) indicate that the direct and indirect costs of quality challenges present a significant problem that is greater than that of implementing TQM. TQM aids organisations to avoid the price of non-conformance, which consists of costs of prevention, appraisal and failure as presented in the PAF model. The prevention costs are incurred before the processes are undertaken and they consist of the design, implementation, and maintenance of the TQM system. These planned costs include material requirements, quality assurance and quality planning. Training and empowering employees contributes to enhancing the value they can provide for the organisation, including the provision of services (Psomas et al., 2017). Empowering employees to improve quality also contributes to higher levels of innovation in organisations (Matias and Coelho, 2014). This means that TQM contributes to a more motivated workforce that can effectively respond to quality issues in a holistic manner.

Employee empowerment also helps controlling the failure costs of the organisations. The products and materials require appraisals to align with the specifications and this is effectively done by the employees when they are empowered to do this. Failure costs not only arise within the organisations but externally as well. This means that the costs such as waste, scrap and reworking can be eliminated with the development of an effective TQM framework (Mosadeghrad, 2012). Failure analysis is adopted as a key element of TQM and aids in identifying the root causes of quality problems in order to develop positive strategies for addressing them. A key aspect of TQM is that it leads to the development of an optimised supply chain that provides the producer the right material in the right time (Matias and Coelho, 2014). The processes are also optimised to respond to customer requirements and produce commodities that meet their expectations, which lead to higher numbers of sales (Gharakhani, et al., 2013).

Based on the aforementioned facts, it is noted that the major benefit of implementing TQM is the establishment of an employee empowered organisational environment, that leads to the

operational efficiency and ultimately customer satisfaction. In fact the total commitment of employees is critical to the success of TQM and, thus, the organisation as it improves their ability to address customer needs (Matias and Coelho, 2014). A committed workforce understands the vision and goals of the organisation and has the competence to work towards accomplishing them. Employees are given, and accept, the ownership of the different issues in the organisation. An organisation implementing TQM is a continually learning because the employees are encouraged to seek opportunities for improving their knowledge. This organisation should be, therefore, highly coordinated where employees can discuss problems and find suitable ways of how to address them (Gharakhani, et al., 2013).

2.4. Impact of Total Quality Management on Organisational Operational performance

Organisations, across the world, now consider that effective quality management practices have become the need of organisations to stay competitive because globalization has not only brought opportunities for businesses but critical challenges as well. Especially, the manufacturing firms have to face many challenges to compete in the market and they could only cope with such challenges by improving their operational performance especially having a major focus on quality (Lee and Lee, 2014). Moreover, mounting competition in the international markets has compelled manufacturers that operational performance should not be the indicator of organisational efficiency but also the organisational effectiveness. Yet the difference between these two is usually overlooked. For example, if organisation is completely following its process specifications for production it is utilising its resources efficiently but if the organisation is not producing according to the demands and needs of customers it means the organisation fails to utilise its resources effectively. In fact, there is no point in churning out high-quality products that nobody wants (Yunis et al., 2013). Therefore, organisations are now required to majorly focusing on the demands and requirements of the customers which are dynamically changing day by day. Customer satisfaction has become the ultimate competitive advantage for organisations that they need to attain to remain competitive in their market (Wickramasinghe, 2012).

Satisfied customers are the ultimate competitive advantage for the organisations so to retain the existing customers and attracting potential customers the organisations are always focused on delivering value to their customers that should not only be effective but also higher than the competitors. Even the customer satisfaction could be transformed into customer delight when

the provided value exceeds customers' expectations (Wickramasinghe, 2012). Hence, developing and understanding of the customers' wants and demands is crucial for organisations. When the organisations are successfully directing all of their efforts towards attaining such kinds of goals the operational performance would always be according to the expectations and wants of customers. This ultimately results in enhanced sales and operating profits for the organisations because in such cases the customers are ready to buy more and pay more for the quality products meeting their expectations and fulfilling their wants. It also restrains inventory pileup due to rejected and returned products and unsold items as well as saves the warranty claims, recall costs and product liability costs (Nilda et al., 2017).

To enjoy profitability, the organisations must acknowledge value and operational performance from customers' perspective not from their engineering teams' perspective. Therefore, the internal processes of the organisations must be mechanized towards producing the products and services that are useful and valuable according to their customers (Liu and Liu, 2014). The TQM philosophy ensures the effective efficiency and considers operational performance not only in terms of the high-quality end products, but the high-quality products and services valued by customers, and emphasises on enhancing organisational operational performance in this perspective. Hence, the TQM philosophy contributes in enhancing the organisational operational performance in different perspectives (Zeng et al., 2015).

When organisations carefully take customer requirements into consideration, the operational performance of the organisations also aligns with the customer satisfaction. It also provides cost efficiencies by eliminating defective products, reworks cost and eventually saves the throughput time as well. Hence, customer focus significantly contributes to enhanced operational performance by eliminating the internal and external failure costs along with assurance of production of goods that are in demand (Khan, 2003).

TQM practices help organisations attain their key objectives of customer satisfaction and profit maximisation by having a major focus on continuous work improvement. Lee and Lee (2014) indicate that TQM practices focus on continuous work improvement so various organisations implement these practices in various perspectives, but the major purpose remains same that is the satisfaction of organisations' internal and external customers. A major part of the costs of total quality could not be controlled by the management alone unless the workers are involved, so TQM also focuses on employee empowerment and involvement to enhance organisational operational performance. Management alone can only aim for delivering the quality but

employee commitment to deliver quality actually results in achieving the goal (Rachmat, 2019).

When employees are themselves committed to deliver high quality, they take initiatives towards meeting process and product specifications, improvement of process and product designs, detection and elimination of bottle necks, and setting reasonable yet challenging targets. Employee empowerment encourages their involvement at critical stages of production as well as taking required initiatives without any hesitation. Employee involvement results in diversity of opinions and views based on diverse experiences of workers and results in creative ideas for enhancing customer satisfaction and ultimately fosters operational performance (Phan et al., 2011).

In a similar context, Liu and Liu (2014) investigated the relationship between TQM practices and employees' wellbeing and found that TQM practices have significant positive impact on employee job satisfaction and ultimately on operational performance of the organisations. The authors argue that quality management practices bring significant change in workplace characteristics that result in improvement of employees' work-related attitudes and behaviours and their work life outcomes. It indicates that quality management practices not only focus on operational performance but also employee wellbeing that makes the employees more committed towards improving organisational operational performance.

Different researchers have reported the implementation of TQM in different perspectives. The study conducted by Faeq, Garanti and Sadq (2021) offers an empirical basis for measuring and enhancing TQM practises in construction business performance. The study elaborated the TQM adoption in terms of management-leadership commitment, employee interactions, supplier management, project design, training, quality data and reporting, process management, continuous improvement, and customer focus using practical implications. It supports the claim that TQM practises help businesses perform at higher levels in a variety of areas, including the quality of work, external customer satisfaction, safety, market share, the effectiveness of planning, labour efficiency, the percentage of successful tenders or quality contractors selected, competency in managing human resources, risk management, and management (Faeq et al., 2021).

Pambreni et al. (2019) indicate that TQM helps organisations defining their objectives and the ways to attain these objectives through efficiencies attained within the organisations. An

organisation that had implemented TQM had a strategic plan that included broad objectives, a vision and mission, as well as a set of activities that needed to be taken to accomplish the broad objectives. TQM is a method for defining the goals of an organisation, developing its policies, deciding how to get there, and managing the resources needed to carry out those plans. These steps help organisations perform better and more effectively in line with their successfully attained goals. It is used strategically based as one of the independent variables called top management commitment as the most crucial factor for the success of TQM. According to the study, senior management commitment has a crucial influence in the prediction of operational performance (Pambreni et al., 2019).

Wickramasinghe (2012) argued that implementation of TQM practices also bring change in human resource function of the organisation. Human resource management practices become more effective, and employee oriented under the umbrella of TQM because the focus changes from only guiding employees to follow the instructions towards getting them involved and encourage their feedback and initiatives for improvement of processes and products. It ultimately results in enhanced organisational operational performance in effective as well as efficient manner. In a similar context, Lee and Lee (2014) argue that TQM practices progress employee empowerment and customer satisfaction in parallel. The authors argue that TQM practices not only focus on enhancement of customer satisfaction through enhanced quality and operational performance but also have significant focus on employee satisfaction assuming that satisfied employees would be able to satisfy the customers by delivering products that are most desired by customers. Hence, employee satisfaction leads in efficient organisational operational performance.

The enhancement of organisational operational performance is among the key objectives of organisations which are ensured by the TQM practices of the organisations. The systematic management approach focused by TQM ensures that the operational performance of the organisation is quite consistent with the desired targets of operational performance i.e., the organisation is producing what it actually wants to produce. It also ensures the efficient utilisation of the resources as well as the achievement of efficiency through enhanced outputs while declining inputs (Khan, 2003).

Under systematic management approach, organisational operational performance is enhanced through better products, improved responsiveness, shortening of cycle times for novel products and services, effective marketing, production, and engineering strategies. Moreover, by

elimination of errors, wastage and defects, the organisations are also capable of enhancing operational performance by having sufficient control on costs (Nilda et al., 2017). Businesses are motivated to submit applications for employee engagement initiatives in order to improve corporate performance, growth, and competitiveness on the local or international market. The success of TQM deployments depends heavily on the employee focus, which identified as one of the independent variables. According to the study, effective people management significantly correlated with rising customer happiness, employee morale, and improved production performance. One of the four TQM components, people management which is the source of managing flexibility and cost, significantly impacted both financial and operational performance (Pambreni et al., 2019).

Hence, previous literature (Lee and Lee, 2014; Liu and Liu, 2014; Wickramasinghe, 2012) indicates that TQM have significant contribution in enhancement of organisational operational performance by having its focus on different organisational perspectives. Rachmat (2019) argue that TQM practices have significant relationship with operational performance of organisations by bringing changes in organisational strategy, structure, and performance. But at the same time, the effective implementation of these practices is very important. Al-Refaie et al. (2022) argue that implementation of TQM practices changes the employees' level of learning and learning direction, hence leading towards improved knowledge implementation towards their tasks. It also brings the improvements in behavioural factors by influencing the work effort level of employees, hence significantly affecting the organisational operational performance.

Despite having all the key benefits and characteristics reported by the researchers, the researchers also emphasise on focusing on the effective implementation of these practices. Nilda et al., (2017) argue that only adoption of TQM practices is not sufficient for the organisation, rather effective implementation of these practices plays a significant role. Organisations who fail to implement these practices on various aspects of organisation, do not achieve the desired results i.e., if the organisation is solely focusing on improvement of its operations and the end product, it could never be able to get the desired outcomes because TQM implementation focuses on all including operations, products, customers, and most importantly the employees of the organisation. Employees' involvement is significantly important in effective implementation of these practices and achieving the desired outcomes of operational performance.

Some manufacturing companies apply the quality management practises incorrectly, and the primary cause of this situation is internal problems with their knowledge, understanding, and skills related to quality management practises. These issues result in the "use of the wrong tool to solve a problem," "use of the same tool to solve the entire problem," and "use of the same set of tools on each problem." (Sahoo and Yadav, 2018). Unfortunately, a lack of technical skills frequently leads to the ad hoc adoption of certain methods without the corresponding system-wide philosophy and ethos. External consultation may be used to overcome this lack of experience, but this alternative is frequently not available because these manufacturing SMEs would need financial resources both to pay consultants and to support the actual implementation of such ideas (Saleh et al. 2018). Most SMEs lack sound financial management skills and have inadequate funding arrangements. Another issue is that most SMEs offer quality management initiatives shorter time periods than they do elsewhere and apply them more quickly to deal with manufacturing SMEs add more capacity as their product design evolves. Ample financial resources are also needed for teaching individuals how to use TQM tools and processes. Therefore, inadequate training and a lack of resources are two main obstacles to implementing the TQM programme in SMEs (Asad et al., 2020).

TQM techniques for the manufacturing sector require a lot of resources but eventually pay off. Manufacturing SMEs' entrepreneurs in India are unsure of the possible advantages of quality management systems as a result of these divergent points of view. Since the breadth of improvement through the application of TQM within the manufacturing firm is expanding, organisations may want to consider more modern methods of integrating the business operations outside of their physical boundaries (Asad et al., 2020). Businesses find it difficult to persuade and influence business partners to restructure their business operations and embrace a quality management philosophy, especially when dealing with such external ties. Additionally, SMEs might not have the market strength to persuade their networks of business partners, particularly their suppliers, to implement quality management and control procedures (Saleh et al. 2018).

In a similar context, Zeng et al., (2015) argue that although the adoption and implementation of TQM practices is full of advantages for almost all kinds of organisations, but it requires critical analysis of the organisational structure, strategies, policies and working environment. Because any kind of change proposed for the organisation, even it's in the favour of organisation and its staff, has to face reluctance at some initial stages, which most of the time

declines with the passage of time or by positive experiences of the organisations' employees. Hence, adoption of TQM practices also requires effective implementation for getting desired outcomes. Otherwise, organisations may face even more difficulties and negative impact on operational performance.

Emphasising on the effective implementation of the TQM practices, Khan (2003) argues that TQM practices could not be implemented without sufficient and relevant training of the employees being involved in the process. So, the training period may have negative impact on the operational performance of the organisation in a way that employees have to stop their regular work for the time of training, but the impact lasts for a short-term period i.e. only for the training period unless the employees again start accomplishing their routine tasks. Yunis and Chen (2013) also argue that TQM practices may also have negative impact on organisational operational performance in a way that it mostly leads towards incremental modifications on regular basis that may result in unfavourable reaction of individuals who are comfortable with the existing working environment. Their reactions or feelings may have negative influence on their performance and eventually affecting the overall operational performance of the organisations.

Focusing on the effective implementation of the TQM practices, Wickramasinghe (2012) highlighted another key factor, the reluctance of employees. The author argued that employees never welcome change immediately, although some of them adopt it with the passage of time while others still stay reluctant. Similarly, the structural and strategic changes due to the adoption and implementation of TQM practices may result in developing a feeling in employees that they may lose their jobs due to such changes. Such negative feelings also result in negative influence on the operational performance of these employees and having a significant negative impact on organisational operational performance.

Contrary to the above arguments, Rachmat (2019) argue that negative impacts due to the adoption and implementation of TQM approaches are not long lasting. Some initial investments in terms of employee trainings and lost operational performance ultimately results in positive outcomes. Such practices not only improve the production processes by eliminating defects and delays but also help employees in improving their operational performance at individual levels. So, employees initially reluctant for adoption of such change gradually become the part of the process and contribute positively to enhancing organisational operational performance.

Researchers have reported different views about the adoption and implementation of the TQM practices. Some focusing on its key features, while the others focusing on its effective implementation. Some have even mentioned it as a fact that could never be ignored by the organisations for their survival in the market. Zeng et al. (2015) argue that TQM has become the essential element for the survival of organisations in manufacturing and services industries. Organisations who are successfully achieving efficient operational performance along with effective operational performance are able to gain the competitive advantage in the industry and efficient operational performance is achieved through effective implementation of TQM practices. Nilda et al., (2017) indicate that TQM practices significantly contribute to enhancing organisational operational performance through employee and customer satisfaction, so the benefit is enjoyed at the both ends. The aforementioned discussion indicates the significant role of TQM practices in the enhancement of organisational operational performance. Having significant focus on various determinants of organisational operational performance and providing an effective direction to all these resources, TQM not only enables organisations to achieve effective operational performance levels but also efficient operational performance according to the needs and wants of customers.

2.5. Total Productive Maintenance (TPM)

The term ‘(TPM)’ is known as a system of plant maintenance for improving the effectiveness of production processes where maintenance is put into focus as a necessary part of the business and manufacturing process (Chandran, 2015). This element of productive maintenance is achieved through integration of assets, people, and processes to maximise the availability of equipment. TPM is defined as a productive maintenance approach that is accomplished by small groups of employees in an organisation (Madanhire and Mbohwa, 2015). In other words, TPM implies an innovative strategy to maintenance, which optimises systems functionality by including every employee in the maintenance practices. While increasing the effectiveness of systems, implementing TPM eliminates breakdowns by ensuring daily maintenance during routine operations. Ahuja and Khamba, (2008a) observe that TPM was developed from various productive maintenance concepts. The concepts behind the development of TPM include breakdown maintenance, preventive maintenance, corrective maintenance, productive maintenance, and computerised management system (Chandran, 2015).

These concepts were combined under TPM as one holistic maintenance tool that focuses on the proactive approach to maintenance during daily operations in organisations. The theory

behind TPM holds that performance of a system depends on its capability and availability. TPM incorporates all employees into maintenance activities to enhance capacity and operations performance, which contributes to the overall system performance. Reliability, maintainability, and support performance are all functions of TPM (Madanhire and Mbohwa, 2015) that seek to provide a lasting solution to the system breakdown problems in firms. System performance theory provides two different approaches through which TPM can be implemented. First, a deductive approach, which starts by defining the quality standards to be maintained to ensuring that maintenance, is optimised. Secondly, the analytic approach where the areas of defect in a system are diagnosed and the right procedures for corrective measures are deployed to enhance the performance of operations (Singh et al., 2013).

2.6. The Evolution of TPM

Seiichi Nakajima (1950) developed the concept of TPM in the 1950s although it was not known by that term. Nakajima is a Japanese engineer who worked as an interpreter for presentations by George Smith on preventive maintenance. Since Smith was at the forefront of the maintenance improvement approach of management, he provided inspiration to Seiichi Nakajima (1950). He focused on integrating the concepts of American preventive maintenance with other maintenance practices from his experience such as quality management, reliability engineering and operator assisted maintenance (Singh et al., 2013). The new approach that he developed was based on the 5S workplace organisation methodology. The approach consisted of five elements: sort; set in order; shine and organise; standardise and sustain. Considering the success of the approach, it was observed that a leadership mindset that involves frontline teams was essential to enhance small group's performance in the maintenance of operations (Díaz-Reza et al., 2019).

The Japanese Institute of Plant Maintenance was a key player in developing the methodology beyond the work of Nakajima. It accomplished this by integrating the concepts of lean manufacturing into the new approach. The TPM process became widely accepted in 1971 with the first company to apply it as Nippondenso, which created components for Toyota. It was the first company to win the PM award, which was an internationally accepted TPM benchmark developed by JIPM (Díaz-Reza et al., 2019). The JIPM developed the eight pillars of TPM, focusing heavily on preventive and proactive maintenance techniques (Asefeso, 2014).

The eight pillars of the method were autonomous maintenance, planned maintenance, focused improvement, quality management, equipment management, education, Administrative and

supportive TPM and safety (Prabhuswamy et al., 2013). The effectiveness of the method in developing more reliable plants promoted its increased adoption in manufacturing operations across the globe.

2.7. The Value of TPM

TPM is about involving the employees in maintaining their equipment through emphasising proactive and preventive maintenance techniques. The basic goal of TPM is to develop a better production environment in which no defects are detected in the production and no breakdowns are happening to the equipment. It is also meant to improve operational efficiency by reducing downtime and enhancing the overall equipment effectiveness. The approach of TPM introduces the aspect of preventive and proactive maintenance where the operation of machinery is put at the core of production processes to maximise its availability (Asefeso, 2014). Efficient and effective use of the machinery in question is a key aspect of TPM and is addressed by having small multidisciplinary teams addressing core aspects of the production environment. The integration of a proactive and predictive maintenance approach is a key advantage of implementing TPM because it lowers the costs of repairs and reduces the expenses associated with unplanned downtime (Díaz-Reza et al., 2019; Ahuja and Khamba, 2008).

A core component of TPM, 5S is a method of setting up the workplace to ensure consistent performance while including the workers. It is built around the following five fundamental tasks: (Sort, Set in Sequence, Shine, Standardize, and Sustain). Sorting entails tidying up the workspace and removing everything unnecessary. Determine the location and number of the necessary objects before setting the order. Shine entails fixing problems in the workplace and cleaning it up. Using standardised tools and practises, standardise documents and the work technique. Sustain refers to continuing to improve, assisting staff in forming ethical work practises, and finally incorporating 5S into corporate culture (Xiang and Feng, 2021).

To close the skills and knowledge gap and boost employee morale and experience, education and training are essential to TPM (Méndez and Rodríguez, 2017). Adesta, Prabowo, and Agusman (2018). Employees progress through many skill phases across the pillar to become proficient. The level of education and training must be evaluated by the training system in order to select the best training modules and schedule. Equipment operators, for instance, learn how to spot anomalies during routine and infrequent inspections, whereas equipment maintenance staff learns maintenance concepts and practises and develops specialised maintenance abilities.

The goal of the safety, health, and environment pillar is to find and remove occurrences that correspond to these goals in order to create the perfect workplace (Asad et al., 2020).

The overall equipment efficiency arising from the implementation of TPM results in an environment, where production schedules are not altered. This means that the potential of customer complaints declines significantly with the adoption of TPM. The focus of the approach on health and training of the employees is a major aspect that contributes to reduction of workplace accidents and their associated costs. With this in mind, the overall manufacturing costs of a plant decline considerably (Méndez and Rodríguez, 2017). The implementation of TPM has also been found to have a wide range of indirect benefits such as increased product quality and employee confidence levels. Since the employees are trained and responsible for the maintenance of the different equipment, they are more confident about the capacity of the machinery as well as their skills for maintaining the working order of the equipment (Arthur et al., 2015). The adoption of TPM incorporates ownership over all core elements of production processes. This means that employees have positive attitudes towards the organisation and production processes in general. Since TPM involves cross-disciplinary teams, it triggers higher levels of knowledge and experience sharing across departments, which contributes heavily to promoting the capacity of the organisation and its innovative practices.

The establishment of cross disciplinary teams also helps boosting employees' morale. The breakdown of machines on repeated occasions negatively affects staff morale and reduces the production capacity of the firm. Initiating the TPM approach to business modifies the employees' mindset and promotes a work culture those values getting the work done in a more efficient manner (Ahuja and Khamba, 2008). This is achieved through engaging functions from all organisational levels in the production equipment to enhance overall production efficiency. Initially, maintenance is the role of the maintenance department to ensure the prevention of machines collapse and production systems failure. Recently, this narrative is seen to be less viable and, thus, it has been changed through using TPM equipment to engage every individual in an organisation in the maintenance activities (Asefeso, 2014). Furthermore, TPM involves all employees in the overall maintenance approach of an organisation. Agustiadhy and Cudney (2018) added that the key strength of TPM is the ability to coordinate all vendors, operators and maintenance personnel together in various events. Team working in the various level of the organisations, as well as promoting zero breakdowns and product defects, helps to ensure that the operational performance is achieved. The outcome for the organisation is zero defects

and breakdowns and, hence, higher efficiency. The higher efficiency coupled with the reduced wastage and costs that are associated with the absence of defects make the operations more profitable (Chandran, 2015).

Based on these facts, the businesses need to focus on TPM for effective management of their operations and production. TPM is highlighted as the process of using different machines, equipment, employees and supporting processes. When these mechanisms are used, they help in maintaining or ensuring there is an improvement in quality as well as integrity in the production units. In simple terms, TPM is the mechanism or the process of getting the different employees involved in ensuring that they maintain their own equipment. In the process, the employees are trained or emphasised that they should be proactive and have preventive maintenance techniques. The TPM is an aspect that focuses on the achievement of perfect maintenance within the production unit and brings about growth, not only among employees, but also in the institution at large (Al Maazmi and Al Hamadi, 2021). The main point of focus is that the process will ensure the following:

1. There will be no breakdowns.
2. The process will be smooth with no stops or slow procedures in the organisation.
3. There will be strategies to ensure that there are no defects.
4. There will be elimination and, thus, no accidents at all.

In general, improving the Overall Equipment Effectiveness (OEE) through the application of TPM is something that is done after having to form small and effective teams. These teams must be multidisciplinary to help addressing the different core areas as preventive and automotive maintenance processes (Lackie and Murphy, 2020). The institutions also must train the employees who can operate machinery and improve on security in the institution. These combination processes bring about standardization in work processes. The issue of TPM is an aspect that focuses on efficient as well as effective use of the different means of production. It means that all the departments should be combined and involved in the process. In this case, small teams can work together for the purpose of increasing operational performance and, at the same time, can help in decreasing downtime mechanisms through a focus on equipment reliability.

From the above statements, the goal of total productive management is to ensure that there is an improvement of operational performance through having to reduce any aspects of drawdown that might occur in the company. Downtime is reduced by making sure that there is an implementation of strong TPM programs, which has a great impact on the equipment effectiveness. Therefore, for the success of these mechanisms, the idea of preventive mechanisms and maintenance should be prioritised in everybody's mind (Lackie and Murphy, 2020). Working as a collaborative approach enhances better growth and brings about satisfaction in the institution (Shinde and Prasad, 2018). An example to note is the running of machines with having the mindset of 'we'll fix it' when the machine breaks should not be considered as options when handling the TPM. Using the TPM programs helps in eradicating the mind-set and turns the statement by putting the machinery having strong core focus on two main issues that is operation and attainment of available resources.

2.8. Impact of Total Productive Maintenance on Organisational Operational performance

Thakur and Panghal (2021) argue that in contrast to TQM in Lean philosophy which emphasises on improved quality through having a major focus on relevant processes, TPM emphasises on improved operational performance depends on the performance of equipment's. In fact, in recent advancements the major focus is to make the production process more labour free (unmanned) owing to automation and robotics in manufacturing industries. Such automation has also made the equipment's even more complicated hence the resulting production outcomes are becoming dependent on the quality and performance of this complex equipment's.

Attaining perfect levels of targeted organisational operational performance along with zero, defects, break downs and accidents, looks like a dream for most of the organisations, but they can make this dream come true through the effective implementation of TPM. Nallusamy (2016) argue that TPM facilitates organisations in achieving their desired operational performance levels through reducing unplanned maintenance time, improving working environment safety, enhancement of quality outputs and proven impact through equipment effectiveness.

But this is also a key fact that the deployment of TPM requires a complicated system with significant capital outlay. Procedure and TPM implementation should be carefully chosen in order to obtain the best economic benefits. To successfully apply TPM in Indian SMEs, top

management must possess a high level of maturity. At the sub-system level in this study, the major enablers are motivation, total staff involvement, employee empowerment, team spirit, good attitude, varied maintenance systems, education and training, available resources, etc. Therefore, it is crucial to comprehend the nature of different enablers and their effects on the effective implementation of TPM in Indian SMEs (Jain et al., 2018).

Therefore, TPM have a significant focus on the maintenance methodology and improvement of maintenance skills hence leading towards enhanced operational performance and has become the need of the era. TPM significantly influences the organisational operational performance in different ways. Just like TQM, total operational performance maintenance approach also greatly emphasises on employee empowerment and encourages providing the ownership of machinery to the employees so that the production uptime could be enhanced (Chaurey et al., 2021).

Therefore, instead of allocating the jobs separately to the maintenance personnel, these particular jobs are allocated to the plant personnel so that individuals operating machinery or plants become responsible for up keeping their equipment's and machinery. It enhances operational performance in two ways; the operators take care of basic manufacturing maintenance and keep their equipment's and machinery in operating condition which saves maintenance time as well as eliminate delays for maintenance. On the other hand, it frees up the maintenance staff for performing more value-adding services. In this way, organisations are able to reduce unplanned maintenance time by adopting TPM approach (Mendez and Ramon, 2017).

To enhance the efficiencies of the maintenance staff, TPM approach incorporates four stages, including preparation, introduction, implementation, and institutionalisation (Eugen, 2010 as cited by Xiang and Feng, 2021). Since the introduction of the TPM, its implementation process has gone through several developments, including the model developed by Sharma and Sharma (2013) using DMAIC approach. Similarly, the twelve implementation steps of Nakajima's (1988) TPM approach were expanded upon by (Jain et al., 2018) to create a rather comprehensive TPM technique. Despite having bit differences in the focus of its implementation, the process starts in a similar manner in all contexts. The process begins with a suitable announcement of TPM, followed by education, a support system, the establishment of policies and goals, a deployment strategy, the implementation of several pillars, and, lastly, continual development. A year later, they (Jain et al., 2018) presented a considerably simpler

approach that consists of six steps: data gathering, staff training via technical seminars, trial runs, fine-tuning, and final implementation.

Despite having detailed elaboration of the approach, its adaptation and implementation is not an easy task for organisations. The key hurdle for its adoption is the reluctance of employees due to their hesitation at the very initial stages of the process. The key people mainly reluctant about its adoption are the operators due to their perception of additional responsibilities being assigned to them in terms of maintenance of machinery. But the positive outcomes with the passage of time make the employees convinced for adoption of this approach because it enhances their operational performance. So, employees initially hesitant of the adoption of this approach, later on feel proud of taking the possession of their machines as their own asset that results in enhanced operational performance (Thakur and Panghal, 2021).

In this way, the TPM approach contributes positively to organisational operational performance by providing a better and safer working environment. The continuous up keeping of machinery and equipment's by the operators' results in clean and vigorous machines that are less likely to have issues like lubricant leakage, loose ball bearings, or any other kinds of hidden issues. In this way, it reduces the risks of injuries which are more likely to happen in dirty working environments with poorly working and disruptive machineries and equipment's (Nallusamy, 2016). Moreover, the disruptions of equipment's and machineries also negatively influence the resulting operational performance from these equipment's and machines; hence continuous maintenance by the operating staff through TPM approach eliminates such delays and contributes positively in operational performance (Chaurey et al., 2021).

Another significant contribution of the TPM approach to the operational performance of organisations is its significant focus on quality maintenance which leads to quality enhancements of the production processes. Therefore, TPM approach is found to have significant positive contribution in customer satisfaction through improved quality of organisational operational performance. By having a focus on enhanced quality, the approach emphasises on detection of defects within the production processes and then assigning the quality component tasks for individual points of production (Mendez and Ramon, 2017). These checkpoints at different levels of production processes result in quality outputs which help in eliminating customer complaints and attaining the core purpose of customer satisfaction. It also helps enhancing employees' morale through positive experiences of eliminating reworking and headaches of dealing with defects and complaints (Thakur and Panghal, 2021).

By enhancing the efficiencies of the workers in eliminating the defects and complaints, TPM approach helps organisations in attaining their objectives of perfect production through overall equipment effectiveness which facilitates organisations in quick identification of loss areas, benchmarking the progress and enhancement of equipment operational performance. It also helps identifying the performance of organisation in comparison to the industry to find out the effectiveness of the TPM approach adopted by the organisation (Chaurey et al., 2021). Hence, TPM approach enhances equipment steadfastness by taking its ownership and developing the feelings of pride in its up keeping. Implemented effectively, the TPM approach leads to quantifiable sustained outcomes including enhanced quality outputs, enhanced manufacturing maintenance regime, declined change over and a proactive environment of taking pride of the machinery (Thakur and Panghal, 2021).

For motivation of employees to be the proud part of such proactive environment, TPM has a major focus on eight key areas, which are the pillars of TPM approach including Autonomous maintenance, scheduled maintenance, quality maintenance, targeted improvement, equipment management, training and education, environmental health and safety, and administration. The full equipment life cycle is the subject of TPM procedures. Support from top management is a crucial differentiator for TPM. Every level of the organisation needs to be on board. As a result, top management must closely coordinate with front-line managers, middle management, and support roles. Through involvement in important events, TPM brings all employees together, including vendors, operators, engineers, and maintenance staff. TPM encourages and puts into practise team-based activities at all organisational levels with the same objective in mind: zero breakdowns and defects. (Nurprihatin, et al., 2019)

Although maintenance of equipment by the operators doesn't result in avoiding every failure, but the development of checkpoints facilitates the production personnel in determination of the best course of action that should be taken when machines go down, help them identification of the reason of breakdown and assessment of the requirement for external advice (Nallusamy, 2016). TPM is a Lean tool that has been widely used. The overall equipment efficiency (OEE) indicator serves as its foundation. Three distinct factors—availability, performance, and quality ratio—are used to create this index. TPM is suited for economic analysis thanks to the OEE index since it allows for the quantification of benefits in monetary terms that management can comprehend. The cost of quality (COQ) model, commonly incorporating opportunity costs (Zeng and Matsui, 2015).

By enabling the employees to have a significant focus on the elimination of failures and empowering them to take the best course of action, TPM helps organisations in enhancing organisational operational performance in different perspectives like eliminating defects, reducing the maintenance time and cost through continuous up keeping of the equipment's, enhanced working of the equipment's and machinery due to their good working condition maintained by the operators, hence resulting in increasing the resulting outputs generated by these equipment's and machines(Mendez and Ramon, 2017). Moreover, it enhances the efficiency of the operators because when it becomes the responsibility of the operators, they operate the equipment's and machines with enhanced care, hence reducing delays.

Employees' self-maintenance helps eliminating the equipment issues as well. Equipment issues reduce productivity, leading to a lot of workarounds and extended production timelines. When deadlines are barely reached, workers get overworked, scheduling is poor, and processes are of poor quality. In such cases, equipment will start to deteriorate. These dangers can be avoided by decreasing or eliminating downtime and improving your effective working time. Failures of tools, unforeseen maintenance, overheated bearings, and motor failure are a few examples of breakdowns. There are many setups and adjustments, such as setup/changeover, material or operator shortages, significant adjustments, and warm-up periods. Next, minor adjustments, sensor blockages, delivery blockages, component jams, and cleaning/checking are examples of small stops. Incorrect setting, equipment wear, alignment issues, start-up problems, scrap, and rework are a few examples of slow running. Finally, rework and scrap are examples of production problems (Agustiady, and Cudney, 2018). Longer manufacturing activities with many more stoppages and modifications characterise traditional maintenance, whereas TPM involves operator interaction and communication, results in fewer stoppages, and emphasises more predictive and proactive maintenance.

On the other hand, Thakur and Panghal (2021) indicate that although TPM contributes positively to organisational operational performance, but at its initial stages it is not free from risk. As the operators become responsible for regular cleaning and maintenance of the equipment and machines, some of them may be just aware of the particular working of machine while lacking knowledge about its parts. Hence, making them responsible may create some technical problems at initial stage unless the operators become completely aware of the handling of equipment's and its parts. Hence, adoption of such approaches requires appropriate

planning and effective implementation otherwise organisations may also face loss instead of desired benefits.

TPM encourages group activities, such as sharing responsibility for routine inspection, an immediate repair procedure, cleaning, and organisation, to increase the effectiveness of the machines and equipment. TPM helps enhance employee morale within the organisation while also playing a significant role in scheduling the maintenance process and enhancing production. It increases employee accountability for the equipment they utilise, which lengthens the equipment's lifespan (Chaurey et al. 2021).

Focusing on the contribution of TPM to the operational performance of organisations, Chaurey et al. (2021) argue that approaches like TPM are always in favour of the organisations by contributing positively towards organisational operational performance. Despite facing some challenges at initial stages of the adoption of such approaches, when effectively implemented these approaches successfully attract employees of the organisations towards the resulting benefits. The enhanced operational performance level at employee as well as organisational level itself becomes the convincing tool for organisational employees and they proudly become the part of the approach. Hence, initially getting employee satisfaction, the TPM approach leads towards customer satisfaction through effective and efficient operational performance particularly targeting customers wants.

Hence, all above discussion indicates that TPM contributes significantly to enhancing organisational operational performance. In recent competitive environment, organisational operational performance may help organisations attaining the competitive advantage if organisations are able to utilise their resources effectively and efficiently, so TPM helps organisations in transforming their effective organisational operational performance into efficient organisational operational performance.

2.9. Operational Performance

Operations performance can be maximised through the modification of several factors in the firm. On the top of these factors is the SCM through integration efforts. In the production process, supplier integration has been established as a paramount strategy to enhance flexibility and contribute to the achievement of operational standards. Coupling technical and managerial aspects with supply chain integration enhances the streamlining of day-to-day operations and improves operations quality (Juran, 1986). Businesses face several challenges that need supply

chain intervention in the initial stages to reduce the adverse effects associated with these challenges (Abualoush et al., 2018). For example, the breakdown of machinery and its parts has been recognised as a rampant cause of halt in many stoppages of work processes in firms. Top management of a firm plans for the allocation of financial, building, and physical material and human resources. They plan for eradication of resource wastage practices; however, they find these efforts futile when the supply chain is eliminated from the equation (El-Kassar and Singh, 2019). Supply chain integration contributes largely to the reduction of the breakdown of the production processes those results from resource scarcity and wastage.

Supply chain integration incorporates the interaction of key stakeholders at different end of the supply chain, to achieve the enhanced efficiencies at different levels. Supplier integration contributes positively to the reduction in the breakdown of the production process by controlling inventory. For example, some production lags are usually caused by the breakdown of machines or some of their parts. Supply chain integration provides avenues for achieving organisational goals through managing inventory; that is the firm does not have to keep excess inventory for the supplies that are not needed in the production process. Instead, firms can only order the most important parts of the production process whenever the need arises. Again, supplier integration in firms promotes corporation among various employees in an organisation and also reduces the costs of mistakes that are committed at work due to faulty processes. Further, it ensures that the total cost of quality is kept to its minimum levels (Dubey et al., 2018). Supplier integration, thus, promotes operational performance in firms by bringing all essential stakeholders together in the production process.

Operational performance of the organisation is focused through different perspectives by different organisations, including cost, quality, speed, flexibility, and dependability. Out of which when businesses directly compete on rates, cost is a crucial consideration. A corporation can cut its customer-facing prices by keeping production costs as low as possible. Information for internal users is produced by cost management. Cost management identifies, gathers, measures, classifies, and reports data that is helpful to organisational management in figuring out the cost of goods, clients, and suppliers, as well as other pertinent objects, and in planning, managing, making continuous advancements, and making decisions. The focus of cost is substantially broader than that of conventional costing systems. In addition to the price of anything, it is also interested in the elements that influence the price, such as cycle time, product quality, and process productivity. Consequently, cost management needs a thorough grasp in

industrial sector (Zhang et al., 2018). Therefore, effective cost management necessitates a thorough knowledge of the organization's cost structure. The ability of management to calculate both the long-term and short-term costs of actions and processes which helps to examine the root causes of these costs, and these must also be properly investigated.

Following that, flexibility is a consistent indicator of manufacturing business performance and a crucial component of efficiency (Nayal et al., 2022). SCs can maintain their efficiency and competitiveness in a globalised market without sacrificing performance by maintaining flexibility inside their organisations. Due to changing demand, manufacturing organisations that don't invest in innovation will experience operational delays (Nayal, et al., 2022). Therefore, firms consider flexibility as an important process in determining the strategies for TQM and TPM. The other factors like speed, quality depends on the cost and flexibility somehow, and if the cost would be less, the stakeholders will motivate to produce the good quality products and they will buy products frequently which help them to improve their profit maximization.

For this purpose, firms tend to keep their suppliers closer to amplify the realisation and acceptance of new realities in the business field. The supply chain is an important source of information needed to increase firms' capabilities and the chance of meeting and exceeding the current demands. During the design phase, supplier participation enhances customised quality in products due to their rich information on customers. Large firms in the global market space have reported increased success of operations due to the established strong relationships with suppliers, which have not only provided them with quality supplies, but also gained from the large economies of scale. While this integration environment serves to advance the competitiveness of businesses, small and medium-sized enterprises have reported fewer positive contributions of supplier integration due to their inability to establish such stable relationships. In this regard, SMEs adopt contrary perspectives on the benefits associated with supplier integration due to their sizes.

All organisations including SMEs mainly focus on increasing quality and productivity, decreasing inventories, the production line, processing space and time and equipment downtime, lean procedures lower total operating expenses (Bento and Tontini, 2018). They also ensure excellent operational performance. Internal lean methods have good associations with quality, delivery, flexibility, and cost. Quality, delivery, cost, and flexibility are four operational performance factors that have been linked favourably. According to prior research,

operational performance is comprised of cost, new products, flexibility, quality, delivery, overtime, inventory, turnover, lead time and setup (Bento and Tontini, 2018).

However, the effectiveness of operational performance depends on the implementation of different strategies to achieve superior performance. This has been an issue of frequent concern among managers who are seeking ways of ensuring operational and organisational success. An important issue in assessing the performance of operations is the consideration of performance. The impact of different management practices in assessing the performance of operations is mixed according to a wide range of studies focusing on financial performance (Bronzo et al., 2013). Some of the key reasons behind the absence of a unified view about the impact of the financial measures on organisational performance are that a wide range of factors affects financial performance. This makes it difficult to isolate and control the main effects associated with management practices or other factors (Hwang et al., 2020). Identifying the interactions among the financial variables is difficult and raises the complexity of adopting the different performance measures (Bronzo et al., 2013).

Organisations are well aware of the key fact that operational performance is the backbone of excellence for any organisation, which involves the capacity to fulfil the mission and dedication to meet its objectives. According to Van Looy and Shafagatova (2016), operational performance refers to the performance of an organisation in line with its standards such as operational performance, cycle time, waste reduction and environmental responsibility. Operational performance involves achieving a high level of effectiveness and efficiency in achieving the operational standards in order to meet customer needs. Neely (2003) focused on financial performance as a factor that negatively affects the capacity of a manager to identify areas of weakness and opportunities for operational improvements. As a result, an organisation tries to improve areas of operational performance to aid in improving its corporate strategy. In line with the corporate strategy, an organisation should identify the relevant operational performance objectives. The five main performance objectives for a firm include quality, speed, flexibility, costs, and dependability (Dumas et al., 2013). While the five aspects of operational excellence help to achieve better operational performance, this review focuses on the *cost* and *flexibility* as the most significant factors that are associated with TPM and TQM implementation (Hwang et al., 2020).

2.10. Cost

In recent dynamic market environments, organisations are not only focused on sustaining their competitive position in the market, but at the same time the core purpose of profit maximisation is also the key focus. For this purpose, the operational performance of organisations is also focused in terms of the costs associated with them (Thakur and Panghal, 2021). The objective of using the cost includes the assessment of how much the unit cost of a product varies. This is measured based on changes in a variety of factors such as the volumes produced and the diversity of the product portfolio. Products with greater variety are produced in lower volumes and have higher unit costs. The cost of a product includes factors such as the use of raw material, equipment, and correction of errors. The cost of a product affects the sales price and the margins obtained by a firm from the particular product (Van Looy and Shafagatova, 2016).

On the other hand, cost as a measure of operational excellence refers to the assessment of the expenses incurred by an operation in achieving its objectives. The day-to-day operations of an organisation have to be addressed through strategies that enable maintaining costs at relatively low levels. The cost performance as an aspect of operational performance involves the assessment of additional costs such as reworking and warranty costs that are associated with errors and defects. Cost is cited as a key aspect of the quality of output because it refers to the ability of an operation to produce consistent quality at low expenditure. The cost performance of an operation involves the assessment of its alignment with the objective of achieving cost savings by eliminating unnecessary expenses (Hwang et al., 2020).

The cost incurred in producing an output involves a wide range of considerations such as the cost of material, equipment operation and maintenance and conformance with quality standards. Since organisations seek to generate financial returns for their shareholders, they operate with the aim of lowering and optimising their operating costs. Optimising costs is another key element of operational excellence that leads to efficiency. Cost competitiveness of an operation contributes to its efficiency and effectiveness in addressing the needs of customers and achieving the goals of the shareholders. As a result, the aspect of cost performance is given a lot of significance as an objective of operational performance (Van Looy and Shafagatova, 2016).

Following the beliefs of TQM that prevention of defects is better than dealing with failures, hence the cost of defect prevention is cheaper for organisation than the cost of quality failures.

In other words, if organisations successfully attain zero defects level, then the total quality costs incurred by the organisation are minimised. Some important quality costs incurred by organisations are prevention cost, internal failure cost, appraisal cost and external failure cost (Zeng and Matsui, 2015).

In this regard, Rachmat (2019) argue that organisations must incur cost to make efforts for reducing poor quality termed as the prevention cost. It includes improvements in the designs of the products to make them durable, providing training to the employees for enabling them to perform a good job, certifying suppliers for ensuring the supply of high-quality products and services by the suppliers. Prevention costs also include the costs incurred on the preventive maintenance of the machines and equipment's as well as the documentation of the quality procedures and improvements in the production processes. Organisations that do not adopt TQM practices, fail to focus on the prevention cost that usually incorporates a very minor proportion of total quality cost (Khan, 2003).

Researchers have identified different types of costs in this regard. Phan et al. (2011) indicate that another major type of quality cost incurred by organisations is the appraisal cost which includes the costs incurred on the testing and inspection of raw material, finished goods and the work in process. Moreover, appraisal cost also incorporates the costs incurred on statistical control processes, sampling, and quality audits. These are some kinds of costs that organisations must incur for getting desired outcomes like testing and inspection of raw material is important because quality products could not be achieved with substandard raw material. Moreover, if quality defects are identified in the final product before delivering the product to the customer, organisations incur this quality cost known as the internal failure cost. Internal failure costs are incurred in reworking of the product, scrapping the product, or it may be in the form of lost operational performance due to labour errors or any kind of machine break downs (Nilda et al., 2017).

Rachmat (2019) argue that internal failure costs are usually more expensive than other types of quality costs because before the detection of defect a great deal of labour and material has already been invested on the product. Hence, organisations implementing TQM practices could eliminate such costs through continuous improvement at various stages of production (Chaurey et al., 2021). Moreover, empowered employees could also help eliminating these costs by detecting defects and taking appropriate actions at any initial stages of the production process before reaching the product at its final stages.

Focusing on the internal failure costs of operations, Nilda et al. (2017) argue that TPM helps organisations eliminating the internal failure costs especially that are incurred due to machine break downs because when equipment's and machinery are under supervision of the operators, they keep these equipment's well maintained and in efficient working conditions hence eliminating the defects in operational performance. Similarly, if any problem occurs, they are able to resolve it immediately without disturbing the production process and the final products. Hence, as a result the final products are always defecting free. Similarly, researchers have also highlighted the impact of productive maintenance on external failure costs.

Khan (2003) argues that TQM practices and TPM also help organisation eliminating the external failure costs. External failure costs are the quality costs bear by the organisations in case the defect in products is identified by the customers after product delivery. It is considered as one of the most expensive classes of quality costs. It includes product returns, warranty claims, repairs and most importantly the lost reputation and ultimately the loss of business of the organisation. Hence, quality management and TPM help organisations eliminating these costs by ensuring high quality end products being delivered to the customers.

Highlighting the impact of manufacturing practices on the quality of products that ultimately helps organisations achieving their cost objectives (Rachmat, 2019) argue that organisations must adopt TPM and TQM approaches for ensuring the quality of products being delivered to the customers because in other case they must face loss in the shape of quality costs incurred in different forms. More importantly, the costs associated with the satisfaction of customers must be borne by the organisations by fixing the problems at design or some initial phase, rather than dealing with it when the product is in customers' hands.

In a similar context, Phan et al. (2011) argues that if implemented effectively and consistently over time, TQM can diminish costs across the organisation especially in the areas of field work, scrap, re-work, and reduction of warranty costs. Such kinds of costs reduction do not require any additional costs incurred by the organisations, so in this way the TQM approach helps organisations in enhancing profitability. On the other hand, Yunis et al. (2013) argue that despite having several benefits in the form of reduced costs by implementation of the TQM practices, but such practices initially require significant training of the employees being involved in the process. Since the training requires employees being taken away from their regular responsibilities so it may have a short-term negative impact on the costs. Similarly, TQM practices also require incremental changes on a continuous basis so it may also result in

adverse reaction of some employees who are in favour of prevailing system, or by developing a feeling of losing their jobs. Such kinds of negative reactions may result in enhanced costs by increasing employee turnover and increased costs of hiring and training of new individuals.

2.11. Flexibility

Operational flexibility refers to the ability of operations to respond to the changing needs and requirements of manufacturing and production (Yu et al., 2021). The changes in manufacturing and production of organisations occur due to the changes in the market. For compete the market and to sustain their competitive advantage organisations need to be flexible in their operations. Operational flexibility could never be achieved by the organisations unless they have effective and efficient manufacturing practices (Yu et al., 2015). Flexibility in operations refers to the capacity to configure and vary product lines in order to deal with different customer requirements. An operation is effective if it has capacity of an operation to alter its product lines quickly to respond to new customer requirements (Bronzo et al., 2013). Quick adjustment to requirements is associated with the speed objective. A high performing operation should be able to produce different quality products and adapt its operations in line with market conditions and demands. Changing the operational strategies of a firm improves its ability to suit different delivery schedules without suffering stock outages or supply chain issues (Dumas et al. 2013; Hwang et al., 2020). Some of the key performance indicators in this dimension of operational performance include the time it takes for the operation to respond to customer demands (Dumas et al. 2013; Van Looy and Shafagatova 2016).

Other key aspects of flexibility, as a key aspect of operational performance, include volume flexibility and delivery flexibility. These refer to the ability of an operation to alter its volumes or production levels in line with the demands in the market. Volume and delivery flexibility are related since delivery flexibility focuses on the altering of production schedules to ensure that products are delivered to customers at the most appropriate time. Product/service flexibility involves the adaptability or agility of a production system to change the product mix in line with customer expectations (Bronzo, et al., 2013). The element of product mix flexibility focuses on having a production system or process that is agile in terms of quickly changing what it produces at low costs. Product introduction flexibility as an element of the flexibility dimension is a complex aspect because it involves having the speed and capacity to introduce a new product/service to the market. This aspect affects the time to market for a product by having systems that are dynamic and responsive enough to ensure that a new product can be

integrated effortlessly to the product portfolio. The production processes should be agile in terms of equipment and personnel interaction to enable a new product to be quickly researched, designed, developed and introduced to the market (Van Looy and Shafagatova, 2016). All this depends on the operational flexibility of the organisations.

Operational flexibility helps organisations remain competitive in dynamic market environments. According to the competition-based approach, an organization's primary goal is to maximise long-term profit and create a durable competitive edge over rivals in the external marketplace. This shows that using a marketing-focused strategy helps the business keep on top of its rivals and find ways to serve clients more effectively than they do in order to increase performance. The majority of studies on flexibility have focused on the market rather than the larger strategic environment, and little has been done to investigate market-related flexibility, including marketing (Alamro et al., 2018). They suggested three strategies for organisations seeking flexibility: resource reduction, investment in underutilised resources, and diversification employed for certain objectives (Alamro et al., 2018).

Phan et al. (2019) argue that operational flexibility is considered as the competitive advantage and have become the dire need for enhancement of organisational performance. In recent era, the dynamic market demands compel manufacturing organisations to be operated under flexible and dynamic systems that could be highly responsive towards uncertainties and changes. Moreover, operational flexibility provokes a global trend of offering extensive product variety. In addition, as the organisations are trying to attain high quality at lowest possible production cost, so flexibility is considered at the ending stages towards productive excellence, but it is also a significant fact that operational flexibility has become vital for organisations competing in global markets to cope with the changing demands (Maria and Verdu-Jover, 2005).

Other key facts for which operational flexibility is required by the organisations include customer satisfaction and responsiveness towards customers' demand which have become the source of competitive advantage for organisations especially in manufacturing and service industries. At the same time, customer demands are also changing frequently so organisations having operational flexibility may win the race (Chaudhuri et al., 2018). Enhanced competition has also changed the success criteria for businesses in almost all industries because customers are always surrounded by lots of options in the market hence the ones who are able to attain their satisfaction may get the competitive advantage. But such satisfaction is not a onetime

phenomenon, but organisations must make lots of efforts to sustain their satisfaction level by responding quickly and innovatively towards their changing needs and demands (Mishra et al., 2018).

Such kind of quick response and innovative behaviour is only possible by the organisations having operational flexibility so that they can modify their operations according to the changing demands of the markets, their existing customers as well as to attract the potential customers. Operational flexibility not only provides reactive capability to the organisations but also it enables the organisation to satisfy mounting sophisticated customers' needs and wants without incurring higher costs, quality defects, process disruptions and elongated delivery times (Abdallah and Nabass, 2018).

Operational flexibility is not limited to the manufacturing of products but also incorporates the product mix and volume flexibility so that firms are able to respond according to the fluctuating demands of the market through elimination of excess inventory and time delays. Hence, organisations having higher flexibility may achieve higher customer satisfaction. By and large, improving flexibility is essential for building up the innovative capabilities and ensuring the organisations' competitiveness (Alolayyan et al., 2013).

Operational flexibility is found to be closely associated with the TQM practices implementation since the two concepts have some common orientations. Organisations adopt TQM practices for ensuring improved products' quality, reduced costs and enhancement of customer satisfaction. Likewise, organisations try to attain flexibility performance that may eliminate unnecessary expenses and excessive production costs (Escrig-Tena et al., 2012). In this regard, Alolayyan et al. (2011) argue that TQM practices provide organisations' higher adjustment ability in their strategies and structural administration, according to the changing market requirements. Moreover, it is found that TQM emphasises on moving beyond the firm-centric approach towards expansion of quality management practices to up stream and down stream supply chain partners so that organisations could be able to cope with high velocity business environments.

2.12. TQM/ TPM and the Resource Based View

2.12.1. Theoretical foundation

The resource-based view (RBV) is a managerial framework that determines the strategic resources a firm can exploit for sustainable competitive advantage. The development of the

resource-based view is associated with Barney's (1991) seminal article '*Firm resources and sustained competitive advantage*'. The focus of the theory is on how firms develop sustainable competitive advantages based on the resources at their disposal. It proposes that firms differ because they possess heterogeneous resources (Armstrong and Shimizu, 2007). Firms adopt different strategies because of their different resource mix (Barney, 1991). In this theory, the managerial attention is given to the firm's internal resources of firms in an attempt to identify the competencies, assets and capabilities that help to deliver competitive advantage.

The RBV is an interdisciplinary approach to strategic planning that presents a change in thinking by focusing on internal, as opposed to external, resources. The internal resources and capabilities of a firm are analogous to the entry barriers cited in other theories. The resources become sustainable drivers of competitiveness if there are factors isolating them and preventing their diffusion to the rest of the industry. According to the theory, internal resources are a means of organising the processes of an organisation to achieve sustainable competitive advantages (Tozan and Ertürk, 2015). The theory posits that firms need to develop unique, firm-specific resources and competencies that enable them to compete and outperform their competitors. The resources must be valuable, rare, and imperfectly imitable and supported by the organisation (Hunt and Derozier, 2004).

2.12.2. The evolution of RBV theory

The resource-based view presents a beneficial approach in assessing how firms compete and their focus on the available assets and resources. This theory provides a significant way of looking at how companies develop sustainable competencies in their operations. These competencies are based on how firms organise their resources and utilise them to generate sustainable competitiveness. The focus on sustainable competitive advantages is highlighted in the theory as a key element of the strategies that should be adopted in response to the needs and changes of the market (Armstrong and Shimizu, 2007).

At the heart of the RBV is the understanding that firms can have or create a resource advantage. Resources present an advantage that is sustainable and effective in generating competitiveness if they enable the firm to adopt unique measures. The development of the theory is based on the work of a number of scholars such as Day (1994), Prahalad and Hamel (1996). Achieving competitive advantage lies at the heart of strategic planning in the areas of marketing and management. These must be addressed through resources and strategies that can be effective

in developing the necessary products and deploying the right strategies for better positioning in the market (Hunt and Derozier, 2004; Richard et al., 2009).

Organizations are being forced to reengineer their cultures, processes, and systems to support customer-focused and quality-driven competitive business strategies as a result of rising customer sophistication and the globalisation of company activities. In order to support their strategic decisions, firms implement quality management (QM) practises, such as TQM (Ahmed, and Ferdousi, 2020). TQM is a comprehensive strategy aimed at enhancing the processes, goods, and service quality continuously to meet or exceed customer expectations. It is "a method of conducting business that seeks to increase an organization's competitiveness via continuous improvement of the quality of its people, processes, and environments," (Ahmed, and Ferdousi, 2020).

The pursuit of improvement is an ongoing effort in TQM. All organisational stakeholders, including managers, employees, suppliers, and customers, must fully participate in the continual quest for improvement; otherwise, the TQM effort would be ineffective. According to the literature, TQM adoption helps organisations achieve a number of advantages, such as efficient internal communication, better problem-solving, increased employee commitment and motivation, stronger relationships with suppliers, a greater understanding of customers' needs, increased customer satisfaction, fewer mistakes and fewer defects, productivity gains, cost savings, a reduction in defects, and enhancements in quality (Ferdousi et al., 2019). After the discussion of the RBV, the following section focuses on explaining the reason of why this theory has been chosen.

2.12.3. Why RBV?

The resource-based view of the firm posits that an organisation competes and is able to develop effective competencies on the basis of the resources at its disposal. The competitiveness of a firm arises from the presence of organisational resources and assets including managerial skills and employees' knowledge and experience. These resources are applied in response to the challenges in the market where the organisation can generate relevant products and processes to meet the needs of the customers. In this theory, the assets and resources held by an organisation help to develop sustainable competences as long as they are unique. Limitability of the valuable and rare resources of a firm means that competitiveness is not sustainable and that others have the ability to come up with similar products/services (Armstrong and Shimizu, 2007; Richard et al., 2009).

The resource-based view focuses on providing better understanding of how firms utilise the available resources to manage their movements in the market. With this in mind, the internal resources of a firm play a major role in promoting the development of business. Firms need valuable and rare resources to take advantage of the opportunities in their market. In this review, the focus of the theory is on justifying how the presence of resources and capabilities of the firm enhance its competitiveness. For a firm to have sustainable competitive advantages, it needs capabilities that are critical to meeting the needs of the customers. These resources must be acquired or developed internally and organised in a way that makes them difficult to be inimitable (Sanni, 2018).

TQM has the ability to produce a long-lasting competitive advantage. An open culture, employee empowerment, executive commitment, top management commitment/leadership, teams, culture, training/education, and process efficiency were identified as potential sources of sustainable competitive advantage when considering TQM aspects (Ferdousi et al., 2019). The relationship between TQM and competitive advantage has been researched from a resource-based perspective. Researchers have known for a while that competitive advantage depends on both the external environment and unique internal organisational resources and talents. Resources, both internal and external, are essential for competitive success. The ability to establish a competitive position belongs to the company with valuable resources that are distinct from and superior to those of rivals. Resources are said to need to be both valuable and uncommon (Ahmed and Ferdousi, 2020).

On the other side, the competitiveness could be achieved through the development of managerial strategies. The adoption of strategies such as TQM and TPM can noticeably improve the internal competitiveness and the competencies of a firm. Adopting TPM and TQM supports organisational competitiveness by giving a firm the internal competencies and operational strategies to achieve excellence (Neely, 2003). Firms achieve operational excellence by adopting these strategies because they increase the capacity to achieve high quality outputs and reduce the incurred costs. The assumption is that adopting the principles of TQM involves having coordination among employees and managers to achieve a high level of quality in all aspects. These methodologies also focus on the customers as key decision-makers who influence how products are made. The TQM approach involves continuous improvement and communication across all ranks. These issues contribute to the competitiveness of firms by

considering employees as key influencers in the decision-making framework (Bronzo et al., 2013).

Employees are the key people involved in the operations of organisations; therefore, they are the key focus of TPM approach. The concepts of TPM are designed to make a firm highly efficient in its operations. The TPM strategies and principles involve knowledge sharing to raise the commitment of workforce (Halawi et al., 2006). These factors, in addition to the fact that TPM makes an organisation proactive in its approach to maintenance and responsiveness to customer needs, make TPM a key internal resource. The management approaches and strategies of TPM and TQM are central to the development of internal capabilities. By adopting these measures and their strategies in an effective way, managers can make their organisations more responsive to the needs of their customers (Chandran, 2015). Issues such as organisational learning, knowledge sharing, coordination across departments and commitment of the employees can be considered as critical aspects of the RBV theory. This is because they influence the internal capabilities and processes for responding to the needs of customers. The operational excellence associated with these interventions and management approaches increases the efficiency of firm performance. The outcome, thus, is that firms can develop both tangible and intangible assets, enabling them to better compete in the market (Hwang et al., 2020).

Due to the relevance of the theory and how the resource-based view can impact securing quality internal resources from the management approaches, it is appropriate for the assessment of organisational performance. The formulation of the RBV as a way of assessing internal resources makes it beneficial as a conceptual framework for evaluating how the adoption of TPM and TQM can enhance organisational competitiveness. The approaches contribute to the adoption of management strategies that improve the internal coordination and other aspects of organisation's performance such as the commitment of employees, the efficiency of equipment and the nature of approach to quality management. Quality management and commitment to the strategies of the two approaches is critical to achieving organisational competitiveness (Hwang et al., 2020). This theory will provide a beneficial framework for assessing how organisations develop their competitiveness and respond to market pressures. The adoption of TPM and TQM can be considered as a way of organising the internal resources to capture value. If the organisation is effectively using both management approaches of TPM and TQM, it can generate sustainable competitive advantages (Armstrong and Shimizu, 2007).

Based on the above-mentioned key facts of these approaches, and the understanding of manufacturing fundamentals and business needs, a number of manufacturing organisations in both developed and emerging economies utilise either TPM or TQM solely, while others adopt both TPM and TQM at the same time. In order to survive and thrive in the current, highly competitive global environment, international organisations must implement ISO standards and quality management procedures (Sahoo, 2019). TQM is a management philosophy and a set of management techniques that strive to achieve world-class status and provide consumers with superior value by identifying their expressed and hidden demands. TQM is a powerful example of success. However, it is countered that TPM efficiently handles TQM's modest impact on machinery and equipment performance (Sahoo, 2019).

However, critical evaluation and decision making is required by the organisations to adopt and implement such approaches to lead to the achievement of core business objectives. An organisation can develop world-class manufacturing competencies by integrating TPM concepts with an ISO-certified quality standard framework. Alternate functional approaches, like as TPM and TQM, are available for enhancing the effectiveness and efficiency of an organization's operations. Compared to the solitary application of TPM and TQM principles, TPM and TQM have consistently shown to work synergistically, leading to enhanced performance. These two manufacturing techniques are complementary, which means that when one is used, the marginal returns of the others increase as well. The adoption of manufacturing strategies across various sectors may vary due to changes in industry structure, environment, and operational success criteria. Businesses must be supported by strong manufacturing practises. The researcher therefore assumes that TPM and TQM efficacy may differ from industry to industry based on their policies, rules and regulations (Sahoo, 2019).

In this regard, RBV serves as the theoretical underpinning for a sizable and expanding corpus of supply chain research, emphasising the growing significance of resources and capabilities. First, it is stated that strategic resources used in supply chain organising processes can be found within businesses or beyond boundaries in inter-firm routines and resources. Intangible resources, such as information and knowledge, are more challenging to acquire than physical resources. Second, resources should be integrated to support capability development because they are thought to be unable to deliver value on their own. The procedures by which resources are employed to gain a competitive advantage are the second theme. Researchers investigated how a combination of resources and capabilities affected performance (Yu et al., 2018).

Supply Chain Management

2.13. Supply Chain Management (SCM)

SCM refers to how organisations manage the flow of services and goods across the value chain. The process of SCM can also be defined as the design, planning, execution, and control of supply chain activities to create a net value and build a competitive infrastructure. It involves taking advantage of global logistic networks to synchronise supply and demand (Wieland et al., 2016). SCM practices draw heavily on aspects of systems engineering, industrial engineering, procurement, and logistics. It involves a wide range of activities from the storage and movement of raw materials, work in progress and finished goods. It involves the interrelated networks or channels that combine in the provision of goods and services to final consumers. SCM is a major element of risk management that focuses on ensuring that

commodities are available, and that the value chain is not disrupted by issues affecting any part of the value delivered to customers (Tozan and Ertürk, 2015; Zhang et al., 2018).

2.14. The Themes of SCM

Recent developments in SCM have presented a wide range of themes that are considered highly relevant. The main themes of SCM that contribute to enhancing the value of the practice to organisations include issues such as collaboration and communication. Effective internal collaboration and communication among the various players within an organisation are critical to influence the supply chain performance. Collaboration plays an important role in enhancing the performance of a supply chain and its operational performance in response to the needs of internal and external stakeholders (Zhang et al., 2018).

Since it has become popular in recent years, SCM has been the focus of an increasing number of academic papers. However, Polish scholars have not sufficiently examined the intersection of these two topics, i.e., the role of small and medium enterprises in supply chains and chain management from the standpoint of such organisations. Although the writers of earlier research used the terms "supply chain" and "small and medium firms" in their titles, the real goals of those studies had little to do with those concepts. As a result, these studies make little mention of how small and medium-sized businesses participate in supply chains (Kot et al., 2018)

Small and Medium Enterprises can successfully link the chain and acknowledges their role as links in the supply chain. He thinks that because of characteristics like flexibility, adaptation to consumer needs, adaptability to environmental changes, and reduced plant costs, SMEs can reach strong positions in chains, albeit typically it won't be the responsibility of the chain's leader (Kot et al., 2018). For businesses to thrive in this dynamic environment where operations like product development, manufacturing, acquiring materials, and distribution must be adjusted both reactively and/or proactively, they must increase their capabilities. Either internal or external information integration can be improved to accomplish this. The external method, however, may be more successful than the internal one when it comes to these two types of information integration. Although internal information integration methods like ERP can also increase flexibility, their long-term impact on a company's performance still depends on how well the various divisions can work together as opposed to just purchasing expensive gear. Contrarily, businesses must implement a traceability system to integrate with their channel

partners in order to establish information integration, which improves operational performance (Yu et al., 2018).

Supply chain quality management incorporates key components including Quality Management (supplier support, customer focus and orientation, strategic planning and leadership, continuous improvement and learning, empowerment and teamwork, human resource focus, management structure, quality tools), SCM, and quality tools. (Logistics and transportation, marketing, best practises, supply base integration, alliances and partnerships, and strategic management). Organizations work to integrate quality into the supply chain and identifying the critical success factors (CSFs) is one efficient strategy to raise the standard of SCM. By using enterprise resource planning systems, CSFs are thought to enhance data exchange, optimise material flow, and enhance quality processes. However, they frequently differ in various situations (Chau et al., 2021).

Cost saving is another main theme of SCM that influences its effectiveness as well as how its performance is measured. Managers in SCM are interested in understanding how to measure and improve the expenditure of costs. Tracking the SCM network is a critical aspect of organisational evaluation that influences its performance including the impact it has on overall success of the organisation. Managers are interested in assessing how they can save on costs and use the SCM network to improve their overall operations (Tiwari, Tiwari, and Samuel, 2015).

An advancement in SCM is the 2019 publication "Supply Chains for the Twenty-first Century" (SC21) aims to increase competition throughout the whole supply chain, from the prime contractor to the SME suppliers (Manville et al., 2021). The SC21 programme seeks to use the TQM principles to significantly advance organisational improvement and effectiveness throughout the aerospace supply chain (TQM). It aims to correct the flaws in earlier TQM implementations where tools and processes were valued more than softer factors like management commitment, leadership, and culture. Additionally, it aims to reduce the obstacles SMEs have in implementing comprehensive performance improvement frameworks through stakeholder relationships throughout the supply chain (SC21, 2019).

SCM practices compel managers to focus on different aspects of the supply chain. In this regard, an important element of SCM is the focus of managers on having a continuous tracking of inventory levels and supply chain. Having an up-to-date numbers of inventory levels and

system operational performance positively affects the supply chain and the efficiency of its management. A clear view of the entire organisation by the supply chain managers helps them to make confident decisions about the supply chain. These decisions should not be made in silos and, thus, their positive impact on the organisation is guaranteed (Zhang et al., 2018). A centralised end-to-end view of the supply chain is known as a control tower and acts as a point of coordination for all key stakeholders. This may involve multiple capabilities such as logistics, data analytics, forecasting and contract management. This element of control means that organisations can have full visibility over the different functions, which helps determining how to address the operational performance of the supply chain and the value it delivers to stakeholders. The control tower concept is highly valuable in assessing the delivery of services and products across organisations and, therefore, can be used to improve the level of collaboration (Kaynak, and Hartley, 2009).

To enhance the collaboration at different levels of supply chain, the main focus is to establish the interrelationships among different stakeholders. Therefore, enhancing supplier relationships is another key theme of SCM. This requires moving from the perception that suppliers and their customers only need a transactional relationship with a simple focus on compliance with procurement rules. Supply chains with limited relationships among their stakeholders are ineffective and focus only on short-term gains. Instead, organisations nowadays are interested in developing long term strategic relationships with suppliers to deliver value beyond low prices and mere availability of the products. The supplier relationship can be enhanced through sharing demand signals with them, having a relationship management framework and pursuing outcome-based approaches. The value of strategic relationships with the supply chain players is a critical part of the themes being considered by modern supply chain managers (Huo et al., 2018).

Another major element of the SCM is the issue of having strategic control over the resources. In the field of SCM, organisations seek new ways of strategically controlling their inventories to enhance spending. The stockholding levels are not determined arbitrarily in this case. Instead, they are determined based on the demand levels and forecasting by the different departments. This can eliminate the holding of excess inventory and improve the performance of the organisation. This also pools the inventory risks across the entire organisation and reduces the strategic burden associated with the value chain (Huo, Gu, and Wang, 2018). The consideration of SCM as a strategic resource with the capacity to enhance organisational

performance is a key theme of organisations contributing to how supply chains are currently perceived (Chen et al., 2018).

Organisations emphasise and focus on these approaches to sustain their competitive position. In order to achieve a sustained competitive advantage, SCM practises and TQM practises have risen to the top of the priority list. TQM and SCM have been studied by researchers from around the world using various models as the key factors influencing corporate performance. A set of procedures known as TQM emphasises continuous improvement, ongoing results measurement, meeting customer requests, a team-based approach to problem solving, competitive benchmarking, lowering workload, long-term planning, and strong relationships with suppliers (Soares et al., 2017). Meanwhile, SCM has become one of the most significant and effective competitive strategies for manufacturing companies in recent years. The fundamental tenet of SCM may be summed up as taking input from suppliers, adding value, and giving it to the customer (Basheer et al., 2019).

2.15. The Value of SCM

Research on the relevance and value of SCM as a part of organisational success has focused on a wide range of benefits. Some of the key elements of the value associated with the presence of effective supply chains for organisations include the possibility of making data driven decisions. For example, effective adoption and management of the supply chain for an organisation creates a room for its optimisation. Optimising the supply chain provides a room for improving the bottom line and overall financial performance. Some of the most important aspects or value components of SCM include better collaboration among the players (Chen et al., 2018). Oracle argues that most companies lack automated information flow across their value chains. This leads to lost sales and doubts the overall value of the organisations. The critical concern for firms is to have integrated software solutions for SCM that allow seamless information sharing and an end-to-end view. With this in place, the effective SCM approach provides better access to information and the opportunity for making more informed decisions (Zhang et al, 2018).

While focusing on SCM practices, organisations face some critical challenges as well. Quality issues present a significant challenge for companies when they arise across the different parts of the supply chain. The cost of replacing or repairing an item in the supply chain poses a significant control issue for firms with no effective SCM networks. Firms with greater control over their suppliers and other players in their supply chains benefit from better quality control.

A firm can implement the standard minimum quality criteria for its suppliers, motivating them to partner with secondary suppliers who meet their requirements. This heavily contributes to accomplishing the agreed quality standards. Additionally, companies can implement process guidelines for their suppliers and, hence, enabling them to comply with quality requirements (Flynn and Flynn, 2005). In order to enhance the delivery of value for the organisations, the SCM can be used to ensure compliance with the quality guidelines, while at the same time, assessing compliance steps through frequent audits. The SCM strategies provide the opportunity for a managerial operating system to monitor the key performance indicators of the supply chain and its players. Some of the KPIs used for assessing the SCM players include on time delivery, product quality, scrap rates and the result of supplier quality assessment. Through these assessments, the best vendors and suppliers can be considered for ensuring quality control (Koech and Ronoh, 2015).

The core purpose of SCM is to develop an integrated network among different members of the supply chain. Hence, SCM is a set of techniques used to effectively integrate distributors, manufacturers, storage facilities, and stores so that the services or products can be produced and distributed at the proper time, quantity, and location to reduce costs and meet service level requirements. Contrarily, the supply chain is dynamic and involves a continuous flow of data, merchandise, and money between various stages for customers. SCM is greatly impacted by SMEs because they serve as suppliers, distributors, producers, and clients (Setyaningsih, and Kelle, 2021).

SCM is particularly helpful for SMEs to increase their competitive advantage through in-the-moment collaboration with partners, such as customer service enhancement, forecast improvement, logistic cost reduction, enhanced planning and scheduling. However, SMEs frequently face disadvantages because they lack a skilled staff for the supply chain or a sophisticated IT infrastructure to support the SCM system because of their small size and scale (Wu et al., 2006). Limited number of studies has examined various SCM systems using cross-country comparison views from various continents. Second, few research have examined the use of SCM in nations where micro-businesses make up the majority of SME (Setyaningsih, and Kelle, 2021). Therefore, there is a need to further elaborate the role of SCM in SMEs, so that it can help enhancing the operational efficiencies of SMEs as well.

SMEs, MNCs or any other type of organisations, SCM contributes to the efficiency of operations by ensuring that real time data is available on all critical issues such as availability

of materials or demand. It also ensures that the managers are aware of the possibility for manufacturing delays that may arise from lack of material. With real time SCM data, companies can initiate backup plans and prevent adverse outcomes of stock outages, late shipment, and customer dissatisfaction. SCM implementation and optimisation can provide opportunities for smart automation where operational performance can be enhanced through strategies for leveraging data and effective communication (Abdallah et al., 2014). With SCM, efforts to keep up with demand changes are quickly and effectively communicated and, hence, reduce the bullwhip effect phenomenon. This phenomenon involves the views that an increase in customer orders can result in inflated orders at a factory due to communication delays. These delays are addressed through real time data and communication that can better predict demand and the necessary orders to meet them (Koech and Ronoh, 2015).

SCM helps enhancing operational efficiencies in different ways. It plays a major role in optimising logistics because the effective implementation of SCM can contribute to the identification of efficient logistic methods and strategies. Having an effective SCM approach provides a framework for determining the most efficient logistic methods for different types of cargo to minimise costs and time of delivering commodities. These considerations present savings that can be passed on to customers through lower prices, which boost the bottom-line of the firms (Ronoh, 2016). The SCM network promotes the overall efficiency of determining the needed measures for handling different types of inventories (Kin et al., 2017). This can reduce the overheads for a business and develop an environment where high service and quality levels can be achieved at relatively lower costs (Abdallah et al., 2014).

Another significant value for the SCM framework is providing opportunities to mitigate high risk level. Risk mitigation for firms is an essential factor to respond to potential issues that may affect the delivery of products/services. With an effective SCM framework, a firm can put in place the necessary back-up systems or strategies to respond to unexpected circumstances (Flynn and Flynn, 2005). The SCM provides avenues for proactive action where disruptions and quality control issues are addressed before they arise to avoid adverse outcomes. For example, according to Kin (2020), 87% of firms believe that they can reduce their inventories by 22% if they better understand the risks in their supply chains.

Focusing on particularly SMEs, SCM has a somewhat different connotation for SMEs than it does for LEs. To increase the value of the finished product and, consequently, to ensure a long-term partnership, SCM in this instance could refer to a set of business activities that include

purchasing from the open market, manufacturing or processing of subcomponents within the plant, and delivery to using hired transportation. By combining the processes of diverse partners on three levels—strategic, tactical, and operational—SCM is defined in SMEs as a method that helps the organisation run more quickly and efficiently (Asad et al., 2020). SCM may improve the performance of SMEs and raise their profitability by enhancing their capacity to receive supplies of the appropriate quality and at the right time. Globalization forces every firm to serve items at cheaper prices. Even still, the majority of SMEs do not fully utilise SCM, despite the fact that they are aware of its advantages. Strategic planning and supply chain integration are two major areas where SMEs' level of SCM adoption can be broken down (Asad et al., 2020).

In general, supply chain integration entails information exchange, operational, tactical, and strategic planning, coordination, and control of inputs, components, and final items. The advantages of SCM integration include better customer service and lower expenses for managing inventory. The complexity of a SME's operations in terms of size and scope increases as its firm expands. SMEs might incur higher costs and run a greater risk (Setyaningsih, and Kelle, 2021; Yvon et al., 2019). Therefore, it requires basic financial plans and budgets so that SMEs can start planning for the future rather than reacting to changes in the industry. Strategic planning is essential to the survival and expansion of SMEs. It can be dealt with by working well with partners and sharing information more effectively. The following are the four techniques for strategic planning that SMEs might employ. (1) network optimization, which involves creating the least expensive network while concentrating on customer demand; (2) network simulation, which involves comparing various models to forecast supply chain behaviour; (3) policy optimization, which entails creating the best operating rules; and (4) robustness designing, which entails foreseeing unforeseen events and possibilities (Setyaningsih, and Kelle, 2021; Yvon et al., 2019).

2.16. Integrated Supply Chain

Integration of supply chains implies the collaboration of different stakeholders of the organisations involved at different points of the supply chain, with the core purpose of enhancing their performance through collaborative work towards creation and distribution of the organisational products while implementing management techniques. In large corporation, stakeholders are considered as divisions of business who are all bound by shared objectives and trust (Kwasi et al., 2019).

Although all types of supply chains are somewhat integrated, but this integration is further strengthened for coordination of the relevant resources offered by all participants fulfilling the needs of the supply chain for enhancement of the overall performance of that chain. The integration process incorporates processes, techniques, and skills to be implemented in structured manner for collaborating the capabilities and characteristics for better availing the prevailing business opportunities (Lotfi et al., 2013). The shared objectives of all involved participants of the supply chain are elimination of associated risks and enhancements of their interests (Yang et al., 2009). Therefore, all stakeholders focus on effective management of supply chains.

Supply chains if not managed appropriately incorporate less consideration towards sharing risks and benefits and are focused on short term achievements while having little or no consideration towards long term achievements. Moreover, cost and delivery of products are prioritised over the value of the products and there are limited interactions among firms and their suppliers those results in restricted communications (Yang et al., 2009).

In contrast to the traditional supply chain, the integrated supply chain focuses on the shared benefits of the concerned participants of supply chains and focus on enjoying such benefits at a longer period of time. Such long-term relationships are managed and sustained through development of open communication among all the key participants which helps them responding efficiently to the changes in surrounding environment and facilitate them in attaining the competitive advantage for them (Barbara et al., 2010).

The integration of supply chains is an ongoing process where stakeholders jointly put their efforts for improvement of relationships, and they are always well informed about the prevailing activities of the supply chain at any level. In such integrated supply chain, suppliers, retailers, manufacturers work in a progressive manner through effective information sharing and feedback to both ends of the supply chain (Amoako-Gyampah, 2019).

2.16.1. Why Integration is Necessary?

Nontrivial contradictions have also been found when reviewing the supplier integration literature findings. Supplier integration has been linked in certain academic studies to new product development processes, product performance, supplier development, and visibility-related metrics. Others, however, have discovered either no significant association or even an adverse relationship between supplier integration and operational performance. The question of whether SCI should be simulated with operational performance in general without limitation

is another topic of debate in the literature (Lu et al., 2018). Therefore, the following global trends and forces are driving supply chains towards greater integration (Marian and Soare, 2020):

Improved cost competitiveness. Organisations are looking to significantly improve the efficiency of their internal operations and further reduce costs through enhancement of efficiency and synergies within the supply chain.

Product life cycle is shortened. Mounting competitions due to frequently changing market demands have shortened the life cycle of products. Products being competitive for many years may have been at the forefront for less than a year and continue to tend to shorten product lifecycles.

Faster product development cycle; For achieving the competitive edge, organisations are required to shorten their product development cycles because organisations who successfully introduce new products at early stages could better capture the market share and are able to gain the cost efficiency as well.

Globalization leading to customised products; Globalisation has made the accessibility of products so convenient that customers have a huge variety of products for meeting their needs so mass customization has become a new marketing mantra.

Enhanced quality; As customers become more affluent and the competition to meet their needs intensifies, there is a need to improve overall quality.

Hence, these increasing demands compel the organisations to focus on the having better controls over cost, design, distribution, and support of their products.

However, integration of supply chains is a very challenging task especially in terms of associated complexity, risks, and costs for managing a highly integrated chain which should be focused by the organisation in a similar manner as focusing on the operations of their organisations. Therefore, very limited numbers of organisations go for integrating their supply chains especially due to the associated costs that create a challenge for the organisation. Some associated costs include time spent for the training and support of management and other stakeholders, efforts to have loyal customers, making investments in compatible information

systems for the entire supply chain, opportunity costs that may come from sacrificing other business opportunities and the risks involved in production downtimes (Ragatza et al., 2002).

In contrast to the benefits associated with the integration of supply chains, there are certain challenges as well. The enhanced levels of integration in the supply chain may also result in enhanced confusion due to the degree of interdependence and interoperability, so organisations could not ignore the risk of production downtime. Such integrated supply chains also cause interdependence especially on the suppliers who, most of the time, are having minimal levels of inventory due to just-in-time practices so the organisations become dependent on their offerings. Moreover, the delivery also becomes dependent on suppliers as if there are delays in delivery, so organisation have to face delays in their production processes as well (Talib et al., 2011).

Some other challenges associated with the supply chains are the external factors. As, supply chains may also have to face potential threats like weather conditions, technical problems like computer hacks, power outages, environmental threats, terrorism, equipment failures, and communication failures etc. which could not be predicted, and they may have to incur heavy costs for these. So, such kinds of risks are required to be assessed at individual levels of the supply chains but could be better managed through tighter integration, enhanced procurement, controlling for inventory and development of strong relationships among the stakeholders (Gassmann, 2010).

The most significant and attractive benefit of integrating the supply chain is having a better control on inventory levels, because such integration facilitates the speedy flow of materials and stocks among different levels of supply chains. Therefore, it helps organisations in controlling carrying costs of inventory by reducing the levels of inventory. Moreover, transaction costs are also reduced through having appropriate flow of information, eliminating the number of transactions, reducing the costs of transactions etc. that creates substantial benefits for all participants in the form of cost savings (Marian and Soare, 2020).

Product costs could also be reduced by eliminating the supplier dismissal while enhancing the levels of production by other suppliers which also helps eliminating the costs incurred on management of supply chain. Although in this way the suppliers could be overburdened by the enhanced management and investment responsibilities, but it also results in improved decision making by appropriate maintenance of the flow of information among remaining parties (Amoako-Gyampah, 2019).

Therefore, in order to focus on the operational efficiencies of the organisations, they must also ensure that sustainability practises are integrated across their supply chain (SC). As "a strategic and transparent integration designed to achieve social, environmental, and economic goals through the systemic coordination of important inter-organizational business processes for improving the long-term economic performance of the individual company through supply chain integration. Supply Chain (SC) partners can improve their sustainable procurement, Sustainable design, and distribution systems through collaborative processes, practises, structures, methodologies, and technology Both internal and external integration assist businesses in overcoming inefficiencies brought on by resource scarcity and capacity limitations in order to create a sustainable supply chain integration (Shashi et al., 2018).

Supply chain integration also provides some other significant benefits like less barriers, elimination of the waste of resources, less friction, elimination of procedures that are unable to add value, improved functional and procedural synergy, enhanced responsiveness for changing in market trends, reduced costs of operations, shorter product development cycles, reduced investment in manufacturing capacity, reduced costs of product development, enhanced competitiveness and financial benefits (Barbara, 2010; Lotfi et al., 2013).

Consequently, today, organisations of all sizes inside the planned operations industry approach a full set-up of inventive and cost-efficient programming arrangements that empower full visibility and constant sharing of data from one end to another in their supply chain networks (Yang et al, 2009). The shift to paperless work processes doesn't need to involve disturbance of the existing frameworks or work rehearses. Arrangements as of now exist that offer the ability to tap in and quickly associate with a broad organisation, permitting you to hold your past speculations. We really want to recall, notwithstanding, that chain integration requires a change in mentality that starts with an agreement that traditional cycles and arrangements are unprepared to satisfy the needs of the present hyper connected clients (Lotfi et al., 2013). Supply chain integration has become the need of the era for almost all kinds of businesses. It incorporates integration among all stakeholders including supplier integration, customer integration and internal integration.

Customer integration implies the acquisition of marketing, technological, production and inventory related information from the customers. This information is used by the manufacturers for making products that are able to fulfil customers' expectations and can gain customer satisfaction. Customer integration is also termed as downstream integration as it

incorporates flourishing the flow of products and services towards customers by getting better information from customers about the changing trends in the market (Gassmann, 2010).

Customer integration refers to developing healthy relationships between organisations and its customers for developing a clearer and better understanding of customers' wants, preferences and changing demands. It also implies the incorporation of customers' opinions and feedback in the decision-making process about goods production, so it emphasises adopting the ways and methods for enhancing coordination among customers and manufacturers (Gassmann, 2010).

SMEs enterprises are able to develop into new markets and streamline corporate processes thanks to integration with upstream and downstream relationships. Uncertain or divergent sustainability objectives, for instance, can undermine the SCs' members' cooperative environmental efforts (Shashi et al., 2018). Synergy with suppliers promotes the development and use of cutting-edge environmental technologies. As a result, the majority of the top manufacturing companies choose to work with partners who can provide sustainable goods and services. Customers, suppliers, and the media all have an impact on a company's sustainable practises and, in turn, operational effectiveness. The government, rivals, and the logistics network are further motivating variables for better operational performance (Shashi et al., 2018).

Supplier integration incorporates the flow of technical, operational, and financial information between suppliers and the manufacturers. Moreover, manufacturers also share information related to demand forecasts, production plans and levels of inventory with the suppliers of the organisation. Sharing such kinds of information becomes beneficial for the organisations because it facilitates the estimation of product and production requirements as well as enables the organisation to better utilise the production plants and suppliers' capabilities and the cost structure of the organisation (Talib et al., 2011).

Supplier integration implies the collaboration of suppliers and manufacturers for making decisions related to collaborative planning, inventory management, forecasting, flows of physical resources and replenishment. It is also termed as upstream integration because it incorporates the relationship between upstream suppliers and the organisation for getting information from suppliers and making them the part of decision-making process about products. Such relationships and the communication flow between the participants contribute

effectively to the production process because suppliers have better knowledge about the components being supplied for the production process (Marian and Soare, 2020).

Supplier and customer integration are the parts of external integration. Contrary to these two, internal integration implies the integration of internal departments of the organisation from material intake to the distribution of final product. It involves the integration across different functions and departments under the control of manufacturer for meeting the customer requirements according to their expectations (Yang et al, 2009).

Internal integration suggests that management should give more consideration to interplay among different functional departments including the production department, procurement, inventory, logistics, marketing, distribution, and sales. It demonstrates the degree, to which an organisation can build all of its practices and functions in more organised and collaborative manner for meeting customers' expectations. Therefore, the departments and the functions within a manufacturer facility as a single integrated and coordinated system that works together for improvement of performance and meeting customers' requirements (Amoako-Gyampah, 2019).

In internal integration, some elements contribute significantly in enhancing performance including shared information among different departments and functions, functional coordination, joint planning, coordination teams, and collaborative decision making for ensuring the elimination of delays and defects as well as enhancing customers' satisfaction (Marian and Soare, 2020). Lotfi et al. (2013) argue that information sharing has become a vital tool and proves to be effective in supply chains for attaining effective and efficient organisational operational performance. One of the most significant dimensions of operational performance of the organisation is the quality of product that seeks more attention to compete in the market. Therefore, collaborative activities between all members of the supply chain contribute significantly to enhancing the quality of product and facilitate the organisations to improve its operational performance. Lotfi et al. (2013) argue that organisations that adopt collaborative and cooperative environment among all stakeholders are the ones who can successfully attain the competitive advantage than those who do not prioritise this. Additionally, Amoako-Gyampah (2019) argue that organisations having non-integrated supply chain has disconnected departments and functions who are all busy in accomplishing their own tasks without knowing about the functioning of other departments. In this way, each function or department functions in a way that it generates benefits for any single aspect of the business

i.e. cost, or timing or the accuracy, but not in all these three aspects. Such functioning may result in saving and efficiency in one area, but it may cost high to some other part of the business. It also restricts the flow of information among different units that results in difficulties in identification of issues and adjustments of operations (Kwasi et al., 2019).

In contrast, Lotfi et al. (2013) indicate that organisations having integrated supply chains have enhanced flexibility in their operations for adjusting according to the customers' requirements, competitors' actions or even during any uncertain event in the market or industry. Amoako-Gyampah (2019) argue that integrated supply chains help organisations eliminating waste and reducing costs, hence gaining the competitive advantage and benefits for their businesses.

2.17. Supply Chain Integration and Manufacturing Practices

The core purpose of implementing manufacturing practices like TQM and TPM is to enhance the quality of products and attaining efficient operational performance according to the demands and expectations of customers. On the other hand, SCM is practiced by the organisations for controlling manufacturing processes for enhancing product quality and eliminating the risks of recalls as well as building a strong customer brand. Hence, both approaches have a similar purpose of quality enhancement and customer satisfaction. The only difference between both elements is the way to approach the quality i.e., within the organisation in case of manufacturing practices (TQM) and across the supply chain in SCM (Marian et al., 2020).

SCM and the TQM have been considered as two most significant approaches for firms like services, manufacturing, and small and medium enterprises (SME). These approaches have become the vital ingredient for organisational success and attaining competitive advantage in the industry. Initially these two approaches were not considered to be adopted at a same time in an organisation due to similarity of purpose but with the passage of time, the integration of these two approaches in an organisation proved to strengthen the competitive advantage of the organisation as well as enhanced customer satisfaction (Talib et al., 2011). Hence, in recent era, applications of TQM principles in supply chains have become common practices especially in the field of SCM (Marian et al., 2020). Anyway, Marian et al. (2020) debated that although all types of supply chain integration like supplier integration, internal integration, and customer integration result in adding benefits for the organisations but when it comes to the application of quality management practices, customer integration is proved to add less benefits than

supplier integration and internal integration. In quality management practices, customer integration impairs the effectiveness of innovative activities either at early innovative phases of the production processes or at few sub-phases (Gassmann et al., 2010).

In recent dynamic environment, innovations have become vital for the survival of organisations in the market, while such innovative behaviour entails high research and development costs. Therefore, organisations have to rely on integrated external innovation sources like customer integration that can guide the organisation that what they need and want and ensure that the new developed product would satisfy their needs. Early innovation phase is the initial phase of the whole innovative process leading to two other phases of technical product development and then in last the commercialisation of the product. Hence, customer integration may have significant impact on the early innovation phase having significant impact on research and development activities. But customer integration has nothing to do with further stages of product development and commercialisation (Amoako-Gyampah, 2019). Contrary to the above discussion, Bassmann et al. (2010) argue that customer integration even at the initial stages of innovation or product development results in negative side effects especially when the focus is on quality enhancement with efficient operational performance. Therefore, the inclusion of customers is not advisable because it hampers the achievement of other targets of the organisations.

Nowadays, the majority of firms view adopting sustainable practises as both a means of competitive advantage and a requirement for long-term survival. As a result, it is crucial that sustainability to be considered when evaluating a company's supply chain performance. The effect of sustainability on supply chain performance has, however, only been covered in a very small number of research (Katiyar, et al., 2018). Kwasi et al. (2019) argue that although customer integration is also one of the most significant and effective aspect of supply chain integration, but when it comes to the implementation of manufacturing practices like TQM, customer integration does not prove to have significant positive outcomes. Because TQM not only emphasises on the enhancement of quality of products, but it also has a major focus on production efficiency through controlling the production costs while integration of customers mostly results in increased production costs. The dilemma is the short life of innovation cycles that entails the risk that early customer integration may result in such innovations that may not be able to meet customers' wants later on, hence increasing the cost. On the other hand, organisation may face delays in market introduction of the product according to customers'

expectations that may be fulfilled by some other substitutes which imposes significant pressure on the organisation and its production efficiency (Amoako-Gyampah, 2019).

While on the other hand, supplier integration and internal integration are found to have significant positive contribution in new product development when combined with quality management practices (Ragatz et al., 2002). Supplier integration allows the flow of information from suppliers to the manufacturers hence using the knowledge of suppliers and incorporation of their expertise for complementing internal capabilities helps reducing the concept to customer cycle time, quality issues, costs and enhance the overall product design (Amoako-Gyampah, 2019).

TQM practices are not only focused on enhancing the quality of products but also have significant emphasis on reducing the costs to improve profitability. Hence, supplier integration contributes significantly towards quality management practices because being in mutual beneficial relationship suppliers also strive for saving costs at different stages of product development (Lee and Lee, 2014). Supplier integration not only helps reducing the delivery costs but also helps controlling the quality issues, availability problems and delays leading to a better service for customers. Moreover, development of a good communication flow between suppliers and manufacturers results in developing a thorough understanding of the end products which enhances suppliers' ability to serve the business accordingly (Amoako-Gyampah, 2019).

Supplier integration helps improving the flow of operations by eliminating the delays from suppliers' sides which ultimately results in elimination of delays in final product delivery. It also helps controlling the fluctuations in market prices of the products. As price fluctuations negatively affect the customer expectations, but the core purpose of TQM practices is to meet the customers' expectations, so supplier integration also helps eliminating the negative impact of such fluctuations (Marian and Soare, 2020). Through developing healthy relationships with the suppliers through supplier integration, manufacturers may take advantage of fixed costs of materials for lengthier contract terms. Similarly, having a clear cost structure of the supplies helps organisations structuring their prices without any ambiguity that may hurt the customers' expectations in future (Liu and Liu, 2014).

A long-term relationship between both parties through supplier integration allows for free flow of ideas and feedback over time. Such kind of streamlined effective chain results in positive impact on both; customer services and costs at both ends. Implementing TQM practices, the

aspects of new product development, initiating new ordering processes and control of inventory becomes a joint venture that results in significant operational and financial benefits at both ends of the association. Working closely with the suppliers, organisations can tailor their supply chain according to their individual needs which also helps them consolidating the processes, reduction of costs and ultimately improvements in the end product for the customers (Yunis, 2013).

Nilda et al. (2017) indicate that internal integration enhances the effectiveness of TQM practices implemented in the organisation. The main focus and emphasis of TQM practices is to improve the quality of products while reducing waste, costs and delays. Therefore, when management systems or different functions within the organisation are integrated, then each function lines up towards a single goal enhancing the overall operational performance of the organisation. Internal integration incorporates coordinated efforts which are much higher than the sum of individual parts which not only improves efficiency but also effectiveness of the organisational operations. A study by Amoako-Gyampah (2019) indicated that internal integration has significant positive impact on the implementation of TQM practices in the form of enhanced quality, safety and resulting operational performance. It also helps eliminating risks associated with the production efficiency as well as helps eliminating defects at different stages of product development. Another significant positive impact of internal integration is the uniform understanding of organisational objectives, policies, resources, and processes that enhances efficient utilisation of organisational resources.

Moreover, the implementation of quality management practices requires training and assessment of employees for effective implementation, so internal integration enhances the effectiveness of training through appropriate implementation of practices by the flow of information among employees from different functions of the organisation (Rachmat, 2019). Integration of multiple functions establishes cohesive processes, resources, objectives that align with the system and improve accountability that ultimately helps improving productive efficiency. However, Rachmat (2019) argue that internal integration improves the consistency of management systems in the organisation which eliminates the complexity of management system and helps employees developing a better understanding of organisational objectives. Employees having good understanding of organisational objectives work together to achieve these objectives; like optimising resources and processes of the organisation that result in

reduced costs. It also helps reducing the time taken for performing certain activities by employees as well as eliminating disruptions and delays that also contribute to cost reduction.

Marian et al. (2020) argued that quality management systems like TQM and TPM are all focused on enhancing the product quality while meeting customer expectations, but when these practices are combined with the information and knowledge gained through supplier integration and internal integration the benefits are multiplied in the form of reduced costs, enhanced quality, and optimization of the processes and resources.

In a similar context, Chaudhuri et al. (2018) argued that companies having good supplier relationships along with integrated internal systems have enhanced flexibility to respond according to the customers' changing requirements as well as fluctuations in the market and industry. Organisations' ability to be adaptive and flexible to varying market situations is significant in attaining the competitive advantage in industry. Integration at supplier and internal levels facilitate organisations to respond so quickly and effectively that could never be possible without such integration. Beyond that, Supplier integration is debated to help organisations in attaining the inventory efficiency like only carrying the needed inventory while ensuring the further availability as soon as it is required by the production facility. It helps eliminating carrying costs as well as saving the space at manufacturing facility. Organisations could not enjoy such benefits in the absence of supplier integration because in such case they may have to face certain delays as well (Kwasi et al., 2019).

These approaches not only add to the efficiencies of operational performance of the organisations in terms of enhanced production, but also focus on eliminating waste to reduce costs and improve productive efficiency; hence supplier and internal integration help organisations eliminating waste at different stages of the production process. Better route management through such integrations not only helps saving the warehouse space but also saves the emissions leading to meet low waste environmental goals. Moreover, reducing the level of inventory also helps eliminating logistics costs and save money hence, being flexible while eliminating waste helps organisations in saving huge amounts that could be better utilised in other operations (Yang et al., 2009).

Focusing on the impacts of internal integration, Amoako-Gyampah (2019) argue that internal integration improves the effectiveness of TQM implementation because it not only helps the free flow of information among different departments but also helps employees in sharing their

expertise when required by other especially at equipment handling stage. So, it helps developing a team-based environment that is more dedicated and knowledgeable which enhances the operational performance by improving the workflow across organisation. Moreover, instead of working independently the cost sharing attitude helps cutting the costs of operations without sacrificing the quality or deadlines (Kwasi et al., 2019), hence, internal and supplier integration better help organisations attaining the competitive advantage than the customer integration.

2.18. Business Impact of Supplier Integration

Flynn et al. (2010) define supply integration as a process where firms partner with their suppliers to align their operations and share pertinent information for growth. Integration of suppliers into a firm aim at restructuring inter-organisational strategies to develop asynchrony in the production processes through efficient sharing of information between suppliers and their firms. While Zhang et al. (2018) observe that supplier integration is a critical tool for enhancing a firm's operational performance. An integrated environment is also important to the supply chain. The supplier provides firms with important information regarding the requirements in the market based on the changing trends. Businesses on the other hand directly connect with the consumers where they get important information that informs the suppliers on the current needs of the consumers.

The concept of supplier integration is supported by the Information processing theory supports the idea that an integrated business environment increases the competitive edge of businesses. The theory suggests that the capability to process the quality information that is provided for firms is an important determinant of success. The business environment today is a complex system that requires strategic decisions and accurate moves from firms to survive the harsh competitive pressures. Munir et al. (2020) argue that supplier and customer integration is a pivotal point in the provision of quality information needed to increase a firm's operational performance. Suppliers rely on the information provided to plan the production schedules and manage their inventory. Suppliers access to quality information from the market and businesses enhance their ability to align their internal operations and capability to provide relevant products that will suit the market. Again, supplier ability to interpret the information provided by the market players enhances their ability to direct businesses to the right decision. Suppliers are also able to determine the proper supplies to deliver to businesses. Contrary to suppliers, businesses benefit from the information gained from the integrated environment to align the

internal capabilities and promote information sharing within the firm to enhance planning and operational alignment. Supplier integration is an important part for an organisation to provide information intelligence for efficient operational performance.

An integrated environment aids businesses in attaining their goals and providing quality products to customers through custom-designed products. Supplier integration is also essential to inform organisations on the anticipated changes for them to use total maintenance practices to align the technologies with the current market demands. Lack of supplier integration has two implications on business. First, it leads to reduced operational performance and increased production costs. Second, it leads to a longer channel of distribution between the supplier and business, which reduces the costs of production. On the contrary, an integrated environment reduces the length of the distribution channel and, consequently, reduces the cost of production for the business. Suppliers benefit from the integrated environment by establishing stable relationships with their buyers to increase their sales. An integrated business environment, thus, creates a stable business environment for suppliers, which improves their business performance (Zhang, Lettice, Chan and Nguyen, 2018).

Strategic partners must make a commitment to achieving their shared supply chain goals in order for a business transaction involving supply chain partners to last. Business partnerships become weak and insecure in the lack of commitment. Supplier integration emphasises the importance of having a relationship with a significant supplier that includes information sharing, teamwork during planning, and cooperative production development to overcome inter-organizational boundaries (Afshan and Motwani, 2021). It enables businesses to collaborate more successfully with a select group of significant suppliers who are ready to share accountability for the goods' success. In order to collaboratively improve service capabilities at lower supply chain costs, supplier integration synchronises the key competences and capabilities of buyers and suppliers (Afshan and Motwani, 2021).

Researchers have highlighted the key features of supplier integration. In this regard, Flynn et al. (2010) define supplier integration as a partnership between an organisation and its suppliers that focuses on aligning operations and sharing information. This aims to structuring inter-organisational strategies and to develop synchronised processes in which information is shared and knowledge is gained for achieving efficient operations. Zhang et al. (2018) argue that supplier integration is a critical tool to optimise a firm's competitive advantage through improving inter-enterprise performance. Firms can integrate with suppliers through enterprise

integration or interoperability. Enterprise integration occurs when enterprises get into long-term collaboration in which they homogenise their processes and harmonise their operations. In interoperability integration, on the other hand, each firm remains independent, but develops its own capability to collaborate and synchronise strategies, skills, and processes with each other (Kin, 2020).

The importance of supply chain integration for firm performance is supported by the 'information processing theory'. Information processing theory bases its arguments on the information processing capability of firms. The theory suggests that firms need quality information and enhanced information processing capability to provide a ground for reacting to the daily changes of the complex business environment. The theory adds that the greater the risk that a firm is predicting, the greater the amount of quality information processing needed to manage risk impacts and aspire performance outcomes. Supplier and customer integration allows for the fast gathering of quality information to guide task execution (Munir et al., 2020). The information acquired from supplier integration enhances firm's capability to detect traditional risks, prevent the extent of harm caused by the risk and to react to the risk posed by the complex business environment. This is aided by the internal capabilities and alignments that promote information to be shared within the firm to enhance planning and operational alignment. Supplier integration is critical for organisations to achieve information intelligence for efficient operations performance (Chen et al., 2018).

Despite the widespread use of supplier integration, research in the field has shown inconsistent results about the relationship between supplier integration and performance (Duhaylongsod and De Giovanni, 2019). More research on supplier integration and performance relationship is needed, in particular, to better understand the causes of supplier integration and the results of integration efforts that are related to suppliers, according to several researchers who expressed concerns about the inconsistent findings (Lu et al., 2018). It is emphasised the need to look into the causes of supplier integration as well as supplier-related outcomes from integration efforts. With the growing significance of supply chain integration, it is crucial to record the performance results of various supply chain integration dimensions. This research by creating the scale for the immediate performance outcome of supplier integration, specifically contributed to the literature on supply chain integration (SRPO). Given the favourable correlation between SRPO and supplier integration, supply chain managers should have direction regarding the immediate performance benefits they can anticipate from supplier

integration, which will ultimately improve the operational performance of the company (Afshan and Motwani, 2021).

Supplier integration also complements a firm's effort in meeting consumer needs and helps it to respond to changes in the business environment. Customer needs vary with the changing technology, which makes the organisational operations obsolete and unable to provide the right requirements to fit the current market. Supplier integration is important in this respect to inform organisations on such anticipated changes for them to utilise the total maintenance practices to align the technologies with the market demands (Chen et al., 2018). A lack of supplier integration can not only lead to reduced operational performance, but also increase the costs of production. This is because creating products that do not meet the target market forces firms to modify the products at a later date, which increases the costs of production. Relying on suppliers to provide this timely intelligent is, thus, important to inform the management on the needed technological maintenance (Koech and Ronoh, 2015).

Supplier integration creates several benefits for the organisations. The business environment fluctuates on a day-to-day basis providing both opportunities and threats for a firm. Supplier integration enlightens firms about the existing risks and, thus, improves the operational performance. For instance, an anticipated shortage in the supply of certain products is crucial for firms to modify their operations and to increase their operational performance. In instances where firms lack such information, they may not be able to align their operations to gain competitive advantages (Wong et al., 2011). Just-in-time practices and TQM requires quality information to provide the right adaptive measures to mitigate the risks in the external environment. Integration with suppliers enables firms to gain insights into supplier capabilities, processes, and constraints, which as a result inform internal planning and forecasting (Danese, 2013).

Supplier integration can be done in different forms, including process integration. Process integration with suppliers is also important in enabling firms to meet their goals such as financial goals. Process integration enables firms to create synchronised work processes across the supply chain. This provides an advantage to a firm by bringing suppliers into the internal circle of the business. Suppliers are considered as a force in the business external environment, which also known as the macro-environment. Suppliers as a force in the macro-environment factors have an adverse effect on the firm's performance. Maintaining a good relationship with suppliers is a move that promotes firms success ambitions (Wieland et al., 2016). Porter

explains that high bargaining power of suppliers pose threats to a firm. Higher bargaining power of suppliers implies reduced competitive advantage in a firm; therefore, firms need to increase their capabilities to gain competitive advantages. These strategies include process integration with suppliers to increase transactional efficiencies to solve problems. Integrating the supplier in the internal operations of all stages of product development and planning enables both the firm and suppliers to coordinate better decision-making (Tozan and Ertürk, 2015). This includes the desired maintenance in the firm to arrive at increased gains in operational efficiencies. Process integration with suppliers, hence, positively influences the performance of firms.

2.19. Business Impact of Internal Integration

Internal integration for a firm involves the consideration of how key stakeholders work together to determine and achieve the value that is expected from the final output. Put differently, internal integration for a firm refers to the adoption of strategies that work across organisational boundaries or departments. The focus of internal integration is the elimination of departmental silos that adversely affects the achievement of the overall organisational goals. Organisational success nowadays relies on healthy relationships between stakeholders to enable achieving the internal goals. Success and excellence require the accomplishment of internal collaboration among all stakeholders to ensure their full participation in responding to the changing needs of customers (Chen et al., 2018; Ahmad et al., 2019). Alongside the goal of internal integration for organisational success, firms engage in TQM as a tool for ensuring that they address their quality needs and issues across the value chain. The focus on ensuring the total quality is associated with the goal of securing the continuous availability of the most used resources. The focus of the research has been on the significance of responding to organisational needs by integrating employees and getting them to work together. The strength of employee's integration can have a serious impact on organisational goals and excellence in several aspects such as quality, cost, and responsiveness (Wieland et al., 2016).

Internal systems are critical to the achievement of organisational goals because of the need to work together to avoid inefficiencies. Since internal systems enhance the performance of firms by enabling the value chain to operate effectively, their role is considered significant to respond to the weaknesses of inefficient coordination that adversely affects organisational outcomes. Inadequate coordination and communication among internal stakeholders (e.g. employees, departments) can influence the operational performance of the organisation. The functional

departments of an organisation need to coordinate their activities to ensure that they effectively respond to all issues and that they can communicate effectively regarding challenges in the operations. Open communication systems and strategies can influence the costs incurred by firms because they eliminate the likelihood of error occurrence (Flynn and Flynn, 2005).

Although internal integration is proved to be beneficial in all types of organisations but it's even more beneficial for the manufacturing firms (Ahmad et al., 2019). Manufacturing organisations' internal integration can improve the supply chain's (SC) collaboration to balance supply and demand. A system that is installed inside in a business to integrate all internal functions is known as internal integration. Customer and logistics collaboration integration, which is an example of external integration, has an impact on SC partnerships (Tarigan, et al., 2021). SC resilience is impacted by internal integration in Malaysian manufacturing companies. In order to promote organisational agility, internal integration is performed in businesses through coordinating business information systems at the operational level, specifically, structural and social collaboration (Tarigan, et al., 2021).

When internal integration is targeted to improve the operational performance of the organisations, it results in enhancing the profitability of the operations as well. Some of the most important factors that are considered when determining the profitability of an operation are the capacity to reduce costs while improving the operation efficiency. Cost effectiveness of an operation arises from having integrated and coordinated internal functions. The internal functions of a firm need to work together in coordinating responsiveness to market changes as well as communicating how to compete effectively. The cost performance of a firm presents a significant point of competitiveness as it affects the capacity of using the available resources to capture higher value of money (Phan et al., 2019). Internal integration involves ensuring that all employees are onboard and that they can work together to achieve quality and excellence. With close coordination and cooperation among the internal stakeholders, firms can benefit from better responsiveness to customer needs and lower costs associated with stock errors and outages. Employees can coordinate activities as with the case of maintenance and order processing to avoid delays that may cause loss of customers (Danese, 2013).

Organisations now do not adopt any one approach to achieve their core objectives. Rather different approaches are adopted and implemented in parallel to have even more efficient outcomes (Ahmad et al., 2019). Linking internal integration of stakeholders and the application of TQM helps in determining and accomplishing organisational excellence. As indicated in

prior studies, such as Ahmad et al (2019), TQM is strongly linked to business performance. This indicates to the success of TQM as a tool that promotes responsivity to organisational needs by eliminating unnecessary costs and wastage across all levels. With internal integration, stakeholders can work together without departmental boundaries towards the same excellence goals. Functional boundaries do not present a barrier to knowledge sharing and the implementation of strategies for improving quality along the value chain. This means that internal integration can support the development of effective TQM strategies and their application in achieving the expected excellence levels such as high profitability, low costs, and improved quality (Huo, GU, and Wang, 2018).

2.20. The Role of Supplier Integration

Strategic integration of a firm with its suppliers is important to reduce internal conflict through aligning targets and objectives. Involving suppliers in the strategic planning in an organisation creates analogy in attaining goals between partners in the supply chain (Lu et al., 2018). This creates a platform for building long lasting relationships and alignment of TPM practices that enhance firm operations. Existing studies indicate that there is an improvement in the operational performance for firms that implement supply chain risk management. Performance benefits that occur to firms that implement supply chain risk management include the prevention of disruptions and low operational deficiencies. Supply chain risk management provides firms with the opportunity to react to external environment and to improve operational performance. This is made possible by integrating suppliers and customers in the TPM, which increases operational performance through mitigating uncertainty in the business environment (Gurahoo and Salisbury, 2018).

Strategic interdependence and formulation of common goals between firms and their suppliers reduces opportunistic behaviour and minimises the risks within the supply chain. In addition, strategic integration reduces the costs of production and transaction, while still promoting relationship investments (Flynn et al., 2010). This is also enhanced by the leveraged resources and knowledge. Making the supplier part of the strategy when planning allows firms to be functionally fit and achieve their goals. This creates trust between suppliers and firms to ensure that attaining a competitive advantage that enable firms to respond to the changes in customer needs and preferences (Ahmad et al., 2019).

Firms that do not focus on the supplier part of their strategic plans ignore many competitive advantages because they fail to detect the changes that occur in the supply chain and risk losing

their market share to the competitors. Strategic performance, therefore, directly associates with better firm outcomes. Strategic contingency theory holds that responding to changes in the business environment depends on the firm's strategic decision to adopt market fit strategy (Flynn et al., 2010); that is, the ability to analyse the changes in the external environment and equally responding to them by adopting the right strategies. TPM advocates for proactive approaches to addressing the anticipated changes in the supplier and customer tastes and habits. Moreover, supplier is vital sources of market intelligence to advice firms on the most important aspects of products to enhance its absorption in the market. In this case, collaborating with suppliers promotes market intelligence in firms and, hence, supports the adoption of relevant strategies to enable improved operational performance in a firm (Flynn and Flynn, 2005; Ahmad et al., 2019).

The RBV supports the supplier integration to manage cross-firm business processes through entailing strategic information sharing and collaboration between a focal firm and its suppliers. When referring to services, capacity can be thought of similarly to inventory in the manufacturing industry in that it enables a supply chain to raise production in response to demand. As a result, information flows in the service supply chain, including feedback and information sharing, are crucial because they enable efficient management of uncertainties surrounding client demand. For the coordination of all actions between service providers and their supplier partners, it has been determined that information flow is very important. For the efficient coordination of all major operations in the service supply chains, information flow and technology management are regarded to be crucial (Teng, and Tsinopoulos, 2021).

Supplier integration helps associations in accomplishing the stock proficiency like just conveying the required stock while guaranteeing the further accessibility when it is needed by the manufacturing facility. It helps reducing costs just as saving the space at assembling facility. Associations couldn't appreciate such advantages without any supplier incorporation in light of the fact that in such case they might need to confront specific postponements also (Kwasi et al., 2019). A study of Amoako-Gyampah (2019) contend that organisations having great supplier connections alongside coordinated inside frameworks have upgraded adaptability to react as per the clients' changing prerequisites just as vacillations on the lookout and industry. Associations' capacity to be versatile and adaptable to fluctuating business sector circumstances is critical in achieving the upper hand in industry. Coordination at supplier and

interior levels work with associations to react so rapidly and successfully that would never be conceivable without such joining.

Integrated networks with suppliers help achieving organisational objectives in different perspectives. Supplier combination helps decreasing the conveyance costs as well as helps controlling the quality issues, accessibility issues and postpones prompting a superior assistance for clients. Additionally, improvement of a decent correspondence stream among providers and makers brings about fostering an intensive comprehension of the finished results which upgrades suppliers' capacity to serve the business appropriately (Amoako-Gyampah, 2019). TQM practices are not just cantered on upgrading the nature of items yet additionally have critical accentuation on decreasing the expenses to further develop benefit. Henceforth, supplier combination contributes fundamentally towards quality administration rehearses in light of the fact that being in shared helpful relationship suppliers likewise take a stab at saving expenses at various phases of item advancement (Barbara et al., 2010).

Supplier integration also helps achieving the cost objectives of the organisations. Cost can give a service provider a competitive advantage, just like it can in manufacturing. A company is more likely to receive reliable supply information with a high level of information exchange and collaborative planning with suppliers, which will improve service delivery plans and save inventory and capacity costs (Wang et al., 2018). Consequently, supplier integration allows service providers to take advantage of service delivery economies and reduce service prices. Additionally, supplier integration encourages cost effectiveness. Given that one of the cornerstones of SCM is information sharing, this outcome was anticipated. In contrast to typical industrial supply chains, the relationship between supplier integration and cost performance in services is built on the flow of information rather than tangible items (Teng, and Tsinopoulos, 2021).

SCM practices especially focused on supplier integration, not only creates benefits for the manufacturing firms, but the service sector can also achieve their targets through implementation of these approaches. Results clearly show that the service sector can benefit from lessons learnt about the function of supplier integration in SCM research. Managers in service businesses will benefit from the findings' insights into the operational effects of developing levels of supplier integration. Without effective service supply management, a company is probably spending too much, losing leverage, becoming more complex, and running a higher risk of going out of business (Liu et al., 2019). Furthermore, in this context,

services shouldn't be seen as a single homogeneous category. For instance, there are disparities between shops who maintain some physical inventory and consulting businesses whose people costs account for the majority of costs. When seeking to benchmark integration procedures across service industries, managers must exercise prudence. When constructing a level, it is crucial for service organisations to take sector-specific variables into account in relation to supplier integration (Prajogo et al., 2018).

Supplier integration helps working on the progression of tasks by taking out the postponements from providers' sides which eventually bring about end of deferrals in end result conveyance. It additionally helps controlling the variances in market costs of the items. As value vacillations adversely influence the client assumptions (Yang et al., 2009), however the centre reason for TQM practices is to meet the clients' assumptions so supplier incorporation likewise helps dispensing with the adverse consequence of such changes. Through creating sound associations with the suppliers through their incorporation, producers might take advantage of fixed expenses of materials for lengthier agreement terms. Also, having an unmistakable expense construction of the provisions helps associations organising their costs with next to no equivocalness that might hurt the clients' assumptions in future (Marian and Soare, 2020).

A drawn-out connection between the two players through supplier incorporation considers free progression of thoughts and criticism over the long run. Such sort of smoothed out compelling chain brings about sure effect on both; client administrations and expenses at the two ends. Carrying out TQM focuses on the parts of new item improvement, starting new requesting cycles and control of stock turns into a joint endeavour that outcome in critical functional and monetary advantages at the two ends of the affiliation. Working intimately with the suppliers, associations can fit their store network as indicated by their singular requirements which additionally helps them solidifying the cycles, decrease of expenses and eventually upgrades in the finished result for the clients (Amoako-Gyampah, 2019).

2.21. The Role of Internal Integration

The issue of internal integration is associated with the performance of businesses because it promotes knowledge management and sharing. Effective integration across the organisation reduces the potential of organisational silos, where departments and functions are working in isolation from each other. Internal integration has been cited as a beneficial tool for promoting organisational integration and improving the collaboration within an organisation (Tozan and

Ertürk, 2015). In relation to TPM, internal integration is a key mediator that can promote organisations' capacity to adopt a holistic approach (Swink and Schoenher, 2015).

The conceptual link between TPM and internal integration comes from the assumption that they both require the collection of internal information for stakeholders to address critical organisational issues. It is expected that effective integration of internal functions and strategies can improve the performance of an organisation to better respond to its core issues including maintenance practices, planning operations and responsiveness to market needs (Huo, 2012). It is expected that internal integration contributes to the practice of TPM; having all internal functions and stakeholders working together allows implementing the key elements of TPM in the operation such as knowledge sharing, effective maintenance planning and the involvement of all employees in autonomous maintenance. The focus of TPM is, therefore, to empower all employees within an organisation to participate in proactive maintenance and develop a fault free operating environment. This requires the involvement of all organisational stakeholders in determining the occurrence of faults and maintenance challenges and mitigating them, which promotes effective production planning and low maintenance and capital expenses (Iqbal et al., 2018).

Internal integration is majorly focused on information sharing and integrating data associated with different levels of operations of the organisations. It incorporates looking for supply, operation, and demand information in real-time, and integrating operations across several functional areas. Cross-functional collaboration, or teaming up across several functions for process improvement, is another aspect of internal integration (Feng et al., 2013). The value of internal integration in raising performance has been stressed more and more. Lack of internal integration may lead to conflicting efforts, resource waste, and wasted effort, all of which can be damaging to the effectiveness of a business. Internal integration enhances the connections between various functions by removing spatial and hierarchical boundaries. People from many departments within the company can collaborate and share information thanks to improved horizontal linkages (Feng et al., 2013).

The approaches of quality management and productive maintenance are considered to be incomplete without internal integration (Amoako-Gyampah, 2019). Wakchaure et al. (2014) point to the urgent need to treat TPM as an integrated effort across the organisation. This integration raises the ability of stakeholders to succeed through effectively intervening in all production processes to attain better benefits for the organisation. Effective internal integration

is, therefore, necessary as a tool for promoting the implementation of TPM. Since the modalities of TPM require actively involving all internal stakeholders, having a high level of internal integration can improve the effectiveness and relevance of TPM in promoting organisational success (Huo, Gu, and Wang, 2018). High levels of internal integration allow firms to respond to internal maintenance, which improves the outputs of the production lines. The degree to which departments and employees collaborate to address organisational challenges decides their ability to effectively implement TPM and derive benefits from it (Iqbal, Huq, and Bhutta, 2018).

Internal integration does not only establish the benefits at operational level, but for all integrated departments as well. Through improved information exchange and cross-functional collaboration, internal integration is said to create better intra-firm goal alignment among various functions (such as purchasing, planning, manufacturing, marketing, and logistics). Internal integration also makes it possible to use the skills and strengths of each function, which can enhance organisational performance (Huo et al., 2018). Internal integration also broadens the information that manufacturers can access, which lessens their level of uncertainty when responding to consumer demand. These benefits of internal integration offer a foundation for wise decision-making. A time-efficient preparation of the resources required to meet client demand, for instance, can be made possible by early information from the marketing and buying departments. Additionally, regular exchanges and debate with opposing viewpoints may lessened errors and waste, do away with unreasonable procedures, avoid delays, present opportunities for simplification, and prompt market response (Huo et al., 2018).

Internal integration further ensures the effectiveness of quality management approaches. Amoako-Gyampah (2019) believes that internal integration improves the success of TQM implementation since it not only allows for the open flow of information across departments, but it also allows employees to share their expertise when it is needed, particularly during the equipment handling stage. As a result, it aids in the development of a more focused and competent team environment, which improves operational performance by streamlining workflow across the organisation. Furthermore, rather of operating alone, the cost-sharing mentality aids in lowering operational expenses without sacrificing quality or meeting deadlines (Kwasi et al., 2019).

Internal integration also increases the consistency of management systems in the business, reducing the complexity of management systems and assisting employees in developing a

better knowledge of corporate goals. Employees that have a clear knowledge of the organisation's goals collaborate to achieve them. Employees that have a clear knowledge of the organisation's goals work collaboratively to achieve them, such as optimising the organisation's resources and processes to save expenses. It also aids in minimising the time it takes for staff to complete particular tasks, as well as eliminating disruptions and delays, all of which lead to cost savings (Talib et al., 2011). Furthermore, Internal integration entails coordinated efforts that are far more than the sum of their parts, resulting in increased efficiency and effectiveness of organisational operations. (Amoako, 2019) show that internal integration has a considerable beneficial influence on TQM practice implementation in terms of improved quality, safety, and operational performance. It also aids in the elimination of risks related with production efficiency as well as errors at various phases of product development. Internal integration also has a substantial favourable impact on the consistent understanding of corporate objectives, policies, resources, and processes, which improves the efficient use of organisational resources.

Internal integration improves the efficiency of overall quality management strategies used in the organisation. TQM methods are primarily concerned with improving product quality while minimising waste, expenses, and delays. When management systems or diverse operations within an organisation are integrated, each function aligns itself toward a common goal, hence increasing the organisation's total operational performance (Thakur and Panghal, 2021).

Furthermore, because effective implementation of quality management principles necessitates employee training and assessment, internal integration improves the efficacy of training by facilitating the flow of information among employees from various areas within the business. Integration of numerous roles creates a unified set of processes, resources, and goals that are aligned with the system and improve accountability, all of which contribute to increased operational performance (Mendez and Ramon, 2017).

2.22. SCM and the Resource Based View

The RBV enables the organisations for determination of their core competencies which are really critical for attaining the targeted outcomes of enhancement of organisational performance (Huo et al., 2016; Yang et al., 2018). So, the RBV is also applicable to the SCM of the organisation because the resource-based view emphasises maintenance of key activities through collaboration of different stakeholders i.e. supplier collaboration and internal key functions that can create competitive advantage for the organisation (Shibin et al., 2020). So,

the resource-based view considers all stakeholders, involved at any stage of the supply chain, as a key resource for organisation that could be incorporated efficiently for enhancing the efficiency of supply chain. Although active collaboration of these stakeholders i.e., suppliers and customers are not necessary at all stages of decision making process, but still their collaboration can enhance organisational performance and could create a competitive advantage for the organisation (Huo et al., 2016; Nandi et al., 2020).

As seen by the increasing emphasis on environmental sustainability in SCM research, sustainable SCM has elevated to the top of the research agenda. Sustainability, and in particular environmental sustainability, has come to be seen as a crucial component of contemporary supply chains and SCM paradigms. As a result of recent events, such as the COVID-19 pandemic and the ensuing supply chain crisis, which highlighted the risks of pushing the planet's limits and continuing with the traditional approaches to SCM, there has been a growing recognition in the field that the world cannot afford unsustainable supply chains (Karamchandani et al., 2021).

2.22.1. Resource Based View towards Supply Chain Integration

The resource based view has major focus on integration of internal resources of the organisation for gaining the competitive edge but it also emphasises that the actual value of internal resources could only be determined through the environment surrounding the organisation including external resources i.e. suppliers, hence integrating these external resources with the internal sources of the organisation could provide efficient results in terms of enhanced organisational performance (Shibin et al., 2020). Thus, in this way the resource-based view could be implemented for the management of supply chain for achieving the targeted outcomes in effective and efficient manner (Nandi et al., 2020).

The resource-based view is now not limited to focusing on the internal resources of organisation and their efficient management rather it also emphasises on considering the external sources of information as a significant resource for organisation. That's why successful organisations emphasise on information sharing among different stakeholders involved at different points of the supply chain (Huo et al., 2016). Organisations even rely on information sharing in supply chains for developing coordination among processes at different points of their supply chains including internal production processes, customer and sales services and procurement processes as well (Dong et al., 2009). In fact, in dynamic business

environments, the acquisition of appropriate and sufficient information has become vital for the survival of organisations and the flow of information in supply chains helps smoothing the flow of other business operations like inventory management, finances etc. Moreover, the flow of information helps creating integration among internal and external resources of the organisations e.g. internal integration requires strong coordination among internal resources while supplier integration requires strong coordination among suppliers and the organisation, and all such coordination depends on the quality of information being shared at both ends which serves as a significant resource that contributes significantly in enhancing the performance of business operations (Huo et al., 2016). Based on the above literature review, the previous researchers have been summarized in the form of table which is as follows:

Related studies	Theoretical perspective used	Antecedence	TPM measures	TQM measures	Performance measures	Empirical results / key findings
Hendrick and Singhal (2001)		Time of implementation of TQM practices		Awards from independent award	Firm size, the degree of capital intensity, the degree of diversification	Less capital-intensive firms and more focused firms outperform more capital-intensive firms and more diversified firms respectively. TQM effectiveness is similar for earlier and late adopters.
Cua et al. (2001)		Socially- and technically oriented practices	TPM practices	TQM practices	Manufacturing performance	Manufacturing performance is depends on level of implementation.

McKone et al. (2001)		Just-In-Time (JIT) practices	TPM practices		Manufacturing performance; Low cost, high levels of quality, strong delivery performance	TPM is significantly associated with enhanced quality levels, delivery performance and reduced costs. Significant indirect association between TPM and MP by Just-In-Time (JIT) practices
Huarnget al. (2002)				TQM philosophy, and TQM tools	Cost reduction, business performance	TQM indicates positive association with cost reduction and business performance
Eti et al. (2004)	Maintenance management	Just-in-time (JIT), Total quality-management (TQM)	Maintenance management of physical assets and productive systems		Manufacturing industry performance	TQM and JIT positively influence TPM. TPM can improve industry performance.
Dahlgard and Dahlgard, (2006)	Quality Management			Defect reduction, reduction of waste	Customer satisfaction, stakeholder satisfaction	TQM has significant influence on performance Defect reduction which leads to customer satisfaction. Reduction of waste leads to stakeholder satisfaction
Ahuja and Khambha, (2008)	Supply chain management	Competitive strategy	Leadership and involvement of top management, traditional maintenance practices		Manufacturing performance	TPM improves manufacturing performance.

			and holistic TPM implementa tion			
Motias and Coelho, (2011)	Zero defects theory	Innovation MS		Integrated TQM	Quality enhancement	Innovation MS results in enhanced efficiency and attaining competitive edge.
Konecn y and Thun, (2011)		HR practices	TPM implementa tion	TQM adoption	Improved plant performance	TQM and TPM significantly influence plant performance.
Lam et al., (2011)				TQM Adoption	Market performance	Market performance and learning orientation of service organisations is enhanced by TQM.
Kaur et al., (2012)			TPM initiatives	TQM initiatives	Continuous improvement	TQM and TPM significantly improve business performance, growth an sustainability
Kober et al., (2012)				TQM practices adopted by SMEs	Financial performance	Financial performance is not improved by TQM after controlling for size.
Fening, (2012)	Quality Management			Quality managem ent practices	Organisational growth	TQM significantly influences the performance and growth of SMEs
Lazim et al., (2013)	Resource- emphasized approach	TPM strategy; communicati on and leadership of TPM team	TPM practices		Manufacturing performance	TPM reduces equipment related losses or defects.

Belekoukias et al., (2013)	Lean manufacturing principles	JIT, automation	TPM practices		Operational performance	JIT have the strongest significance on operational performance while kaizen, TPM and VSM seem to have a lesser, or even negative, effect on it.
Fening et al., (2016)		Top management commitment		Committed leadership, closer customer relationship, closer supplier relationship, increased training, employee empowerment, process improvement	Organisational performance	TQM significantly influence the organisational performance.
Alolayyan et al., (2013)	Theory of Quality Management			TQM practices	Operational flexibility	TQM has a significant impact on the OF.
Kumar et al., (2014)			TPM implementation		Manufacturing performance; overall equipment effectiveness (OEE)	TPM implementation significantly influences the manufacturing performance.
Ngambi and Nkemkifu, (2015)	Quality management		Management commitment	Leadership commitment, Quality control, inspection, employee training, customer focus, benchmarking as the basis	Organisational performance is measured by Customer Satisfaction, Corporate Social responsibility, Cost Reduction, and Employee Satisfaction	Leadership commitment, quality control and inspection have a significant impact on cost reduction.

				for enhancing product quality		
Madanhira and Mbohwa, (2015)	Total Productive Maintenance (TPM) philosophy	Work force	Total Productive Maintenance (TPM), Skill acquisition and training, Decision making, problem solving, Management support system.		Operational performance, quality level.	TPM affects the operational performance but work force should also be focused by the firms.
Modgil and Sharma, (2016)	Theory of Quality management	Quality data and reporting	TPM practices	TQM practices	Operational performance; R&D, product innovation and technology management	TPM has significant impact on TQM which in turn helps to achieve operational performance.
Anil and Satish, (2016)	Change management quality approach	Knowledge management		Leadership and top management commitment, Education and training, Quality culture, Continuous improvement, Strategic management,	Operating performance, financial performance	TQM indicates significant relationship with organisation performance.
Bourant et al., (2017)		TQM elements		Top management, strategic quality planning, employee quality management, customer fo	Financial performance, customer focused performance and service quality performance	Hotel business performance is influenced by TQM implementation.

				cus and e mployee knowledg e and edu cation		
Bu et al., (2017)		Companies' size, capital intensity, degree of diversificatio n, time of awards announceme nt		Quality awards of firms	Market share, Return on total assets, ratio of cost to sales, the asset turnover rate	Firms' quality awards have significant impact on financial performance of the firms.
Putri et al., (2017)		Development and management of tool		TQM implemen tation	Employees' operational performance	TQM positively influences the operational performance
Hooi and Leong, (2017)		Top management roles and commitment	TPM initiatives		Manufacturing performance improvement	Manufacturin g performance is enhanced by TPM and not by top management.
Dhaafri et al., (2018)				Leadershi p, strategy, people, resources	Export performance; financial performance, strategic, satisfaction	TQM have significant relationship with EO.
Shannak et al., (2018)	The resource based view	Knowledge Management		Leadershi p, strategic planning, training & learning, managem ent by fact, process focus	Organisational performance; job satisfaction	The paths leading from TQM to organisational performance is interlinked with KM.
Albuhisi et al., (2018)	Balanced score card perspective	Non- financial balanced scorecard perspectives		Customer perspectiv e, internal process perspectiv e, innovatio n and learning perspectiv e	Financial performance	All BSC perspectives are positively influenced by soft TQM.

Habidin et al., (2018)		Kaizen	TPM practices		Innovation performance	Relationship of TPM and IP is not influenced by KE.
Ahmad et al., (2019)	Supply Chain management		TPM practices	TQM practices,	Financial performance, customer satisfaction, operational performance, employee satisfaction	TPM acts as mediator between business performance and TQM.
Phan et al., (2019)		JIT	TQM practices		Flexibility performance	TQM and JIT are interlinked with flexibility performance.
Sahoo et al., (2019)			TPM approach	TQM approach	Manufacturing business performance.	Integration of TPM and TQM creates benefits for organisations.
Pham, (2020)	Supply Chain management		ISO adopters, non – ISO adopters	Leadership, quality information, employee training, supplier management.	Financial performance and non – financial performance	The relationship of TQM with financial performance is affected by non-financial performance.
Maazmi and Hamdi, (2021)	Holistic approach	Process management	Organisational factor, human factors, materials, equipments, finances	Leadership, customer satisfaction, performance measurement,	Manufacturing industry performance	TPM and TQM affect both internal and external environment of organisation.
Xiang and Feng, (2021)		Top management commitment	TPM deployment		Production efficiency	TPM deployment helped accelerating the implementation of critical equipment.

Table 2-1: Summary of Empirical Studies on TPM/TQM-Performance Relationships

Chapter 3 Research Models and Hypotheses Development

After discussing the theoretical foundation of the different research constructs, this chapter sheds the light on discussing the various research models, which are related to the current study. Additionally, the chapter incorporates an investigation regarding the previous work efforts, those are reviewed for the aim of developing the study's hypotheses. The current chapter is also focusing on providing the basic theoretical underpinning for data analysis, research findings and research contributions.

3.1. Hypotheses Development

3.1.1. The impact of TPM on operational performance

3.1.1.1. Impact of TPM on Cost

TPM is seen as a comprehensive program for the improvement that is originated from TQM's idea of zero production defects applied to equipment. In this regard, Sahoo (2018) argued that TPM is a program that has a major concern of reducing the number of losses in an organisation. Therefore, TPM declines the availability losses, product quality losses and performance losses by optimising the overall equipment performance. Reducing the aforementioned losses in a firm cut down on reworking the defects in products and the production process. Sahoo and Yadav (2020) found that TPM significantly contributes to a firm's operational performance. More specifically, positive impacts were observed on the firm's quality, operational performance, and costs. While TPM impact on costs reduction among other benefits that are celebrated in many industries across the world, the Indian manufacturing firms still hold divergent perspectives; the implementation of TPM is capital intensive and that results are long-term (Sahoo and Yadav, 2020). TPM is a proactive problem-solving program that reduces equipment defects to ensure consistent operational performance and supply of quality products. The timeliness approach to problem-solving in a firm through continuous or autonomous maintenance ensures that operational performance costs are minimised (Huo, Gu, and Wang, 2018).

The practice of TPM involves a proactive problem-solving strategy, through which the focus will be on reducing production defects that may adversely influence the performance of an operation. The approach addresses maintenance in a highly dynamic way through the pillars of autonomous maintenance, planned maintenance and focused improvement and training of the employees. TPM concentrates on improving the operating environment and operational processes in an organisation. The approach is geared towards addressing the operational challenges facing an organisation such as equipment failure losses, idling and minor stoppage, adjustment losses, defects and rework losses and start-up losses (Ahuja and Khamba, 2008). During the TPM implementation, the maintenance department allocates more resources and time to preventive maintenance with the aim of addressing the identified losses.

The focus of these interventions is to address the losses an organisation may suffer by providing cost-saving benefits. The implementation of TPM strategies in operations provides cost-saving benefits that are associated with having zero waste and reduced maintenance costs that are associated with higher maintainability of the equipment. According to Wakjira and Ajit (2012), planned maintenance allows lower maintenance costs by 20% and improves the reliability of equipment by 50%.

The relationship between TPM and business performance has mostly been associated with the development of a working organisational environment in which a competitive advantage is achieved through cost and quality improvement. These issues provide savings in terms of the higher performance and reduced equipment failure and stoppages. Management focuses on integrating total quality and maintenance into their operational strategies. The critical aspect from the empirical evidence on the practice is that TPM has a synergistic effect on business performance as it acts as a moderator of the impact of TQM implementation on the operational performance of businesses by managing their overall costs. TPM was found to help achieving these goals by improving the operational performance, safety, hygiene, and delivery time of organisations (Ahmad et al., 2019). In this regard, researchers including Ahmad et al. (2019) argued that TPM helps organisations eliminating the internal failure costs especially that are incurred due to machine break downs because when equipment's and machinery are under supervision of the operators, they keep these equipment's well maintained and in efficient working conditions hence eliminating the defects in operational performance. Similarly, if any problem occurs, they are able to resolve it immediately without disturbing the production process and the final products. Hence, as a result the final products are always defect-free.

TPM also helps organisation eliminating the external failure costs. External failure costs are the quality costs bear by the organisations in case the defect in products is identified by the customers after product delivery. It is considered as one of the most expensive classes of quality costs. It includes product returns, warranty claims, repairs and most importantly the lost reputation and ultimately the loss of business of the organisation. Hence, quality management and TPM help organisations eliminating these costs by ensuing high quality end products being delivered to the customers (Marian and Soare, 2020). In addition, research on the use of TPM in varied contexts has focused on the effects of the interventions on cost aspects.

Aspinwall and Elgharib (2013) evaluated the impact of TPM on different aspects of operational performance. Their results indicated that cost reduction was scored as the most important factor or variable. It was among the most significant aspects of measuring the operational performance of organisations followed by service delivery and quality. The lower costs incurred by organisations were derived from having lower water bills and cost of energy, lower maintenance costs and lower inventory levels. The implementation of TPM was found to be effective in producing the expected benefits as it focused on cost optimisation activities that are significant in aligning the goals with the mission and vision of organisations (Ahmad et al., 2019). From the explored literature, it is apparent that the implementation of TPM produces organisational benefits by effectively improving the management of their costs. These improvements occur from eliminating unnecessary complications and concentrating on cost optimisation. Thus, best on the above discussion proposed research hypothesis is as follows:

H1a: Total productive maintenance practice (TPM) has a positive effect on cost performance.

3.1.1.2. Impact of TPM on Flexibility

Customers in the contemporary business world have constantly changing demands for products that should meet the highest quality standards at a favourable price (Sahoo, 2019). This, coupled with the volatile supplier requirements, directly impacts the value chain. In addition to the above, globalisation as a complex concept requires businesses to strategise on the most efficient ways to conduct their daily activities. TPM is a science of machinery health that integrates production and maintenance through promoting team working, good working practices, adopting novel technologies and continuous improvement of machinery through preventive as well as autonomous maintenance practices (Kaur et al., 2013).

TPM promotes good working conditions by involving everyone in the maintenance process, which eventually impacts the performance flexibility. The adoption of new technologies and promoting team working can positively influence decision-making in a firm, which, thus, reduces the gap of competition between firms. TPM, being a production-driven methodology, optimises reliability in equipment and ensures efficiency in plant utilisation by involving and empowering employees (Kumar et al., 2014). Involving employees in the firm practices and decision-making, helps designing jobs in a manner that enhances operational flexibility to match consumer specifications. This allows achieving the organisational objectives through ensuring optimal conditions of assets (Hooi and Leong, 2017).

From a theoretical perspective, since TPM is a production-driven approach for improving the operational performance of an operation, it can respond directly to the needs of customers. The approach improves the flexibility and reduces the costs of operations since all stakeholders are involved in preventive and scheduled maintenance (Marian and Soare, 2020). Additionally, as long as the equipment is in a good working order, changing the production schedules and volumes can be effectively accomplished (Kaur et al., 2013). Accordingly, empirical research on the practice of TPM has investigated the different performance measures and considerations that improve the operational performance of organisations. Wickramasinghe and Perera (2016) explored the application of TPM in the manufacturing industry. Their findings indicated a positive association with aspects of manufacturing performance such as product quality, delivery, and volume flexibility. Planned maintenance enhances the flexibility of operations through strategies that raise the capacity of firms to change (Kumar et al., 2014; Mishra et al., 2019).

The effectiveness of TPM as a tool for enhancing operations is associated with the capacity to influence volume flexibility. Motivation management and small group activities are considered the key aspects in ensuring that all employees actively contribute to the required improvements. Wickramasinghe and Perera (2016) focused on the implementation of lean manufacturing strategies in manufacturing firms. One of the key benefits of implementing lean manufacturing was that it improved the level of volume flexibility. This refers to the capacity of the firms to respond to changes in demand and inventory levels and, thus, to maintain low inventory and reduce stock outages. With TPM interventions, firms can alter their production schedules, especially when having different aspects of program implementation. Differing views have been presented by empirical evidence such as the work from McKone et al. (2001) who found

that the implementation of TPM was not significantly associated with flexibility. McKone et al. indicated that the potential reason for the non-significant relationship was the way in which flexibility was measured. Additionally, they argued that the small proportion of maintenance costs in relation to the total operating costs was a key factor in influencing the association.

TPM gives the ownership of the machines or equipment's to the employees and develops a feeling of pride in them that enables them to handle the equipment in more efficient manner as well as modifying its working according to the demand fluctuations. Organisations can gain competitive advantage in the market if they are able to respond according to the changing demands of the markets and sometimes these demands are so high that the ones who successfully grab this opportunity can beat the markets operational flexibility is quite necessary to enjoy such kind of success, while TPM helps organisation attaining such kind of operational flexibility (Mendez and Ramon, 2017). Amoako-Gyampah (2019) argue that TPM contribute positively to enhancing the operational flexibility of the organisation because when equipment's and machines are maintained appropriately, they are always in well working conditions that they can provide the desired operational performance levels even during uncertain times.

Moreover, TPM enhances the operational efficiency of the equipment's and machines by eliminating the operating issues, defects, and delays so organisations also become able to attain the operational flexibility. When equipment's and machines are always ready to produce without any delays and defects then customers' demands, and needs could be better fulfilled. Hence operational flexibility could be achieved through effective implementation of the TPM (Thakur and Panghal, 2021).

Another important aspect of flexibility in production operations is the need to change the product mix and other similar aspects of product volume and type. This plays a major role in influencing the capacity of the firm to respond to changes in the preferences and expectations of customers. Belekoukias et al. (2014) found that TPM had a significant impact on product mix and volume flexibility. However, this was associated with the view that the investigated firms did not implement the interventions effectively as expected. In the same vein, Hallgren and Olhager (2009) presented earlier findings that support the hypothesis that TPM improves organisational flexibility and capacity to respond in terms of product mix and volumes (Mishra et al., 2019). The relevance of flexibility as an aspect of production operations and firms in general is that it enables a firm to void static aspects of their operations that may have a negative

effect on their overall outcomes by preventing responsiveness to customer expectations and the operating environment in general. Thus, based on above discussion the proposed hypothesis is as follows:

H1b: *Total productive maintenance practice (TPM) has a positive effect on flexibility performance.*

3.1.2. The impact of TQM on Operational Performance

3.1.2.1. Impact of TQM on Cost

Pham (2020) defines TQM as a management concept that manages people and processes in an organisation to ensure the satisfaction of internal and external customers in all stages. TQM cumulatively amplify business gains through promoting its growth and good relationships, increasing operational performance, mainstreaming communication, and increasing market share. Pham (2020) further observed that TQM is a tool that promotes quest for excellence in an organisation through reducing defects. Reducing defects in products and processes saves the extra costs incurred during the process remedy, which enhances customer trust.

TQM has been cited as one of the most important strategies for enhancing the overall operational excellence and quality for a firm. This is because it promotes the adoption of good working relationships and teamwork. The overall tenet of TQM adoption in organisations is to reduce defects and provide a production environment where the right quality is achieved at all levels of an organisation. It enables customer-oriented strategies to be adopted in responding to the needs and requirements of customers, which, hence, reduces the need for reworks and waste associated with scrap. Defect reduction in products and processes are achieved through focusing on employee training and customer-oriented production. Products are designed to meet customer requirements equally reduce the adjustment costs. Kober et al. (2012) asserted that implementing TQM in SMEs has positively impacted their performance while reducing rework and scrap. The amount of rework and scrap in firms adversely impacts firm costs. An increased investment in the prevention costs (TQM practices implementation) leads to the reduction of appraisal costs, internal and external failure costs, and overall costs through persistent quality control measures (Gupta, et al. 2021).

Persistent quality control measures and consistent quality levels are adopted by organisations to enable controlling and improving the quality of production to meet customer demands. By

eliminating defects in products and processes, the implementation of TQM encourages a low-cost production environment since there are no losses associated with waste. Therefore, TQM can improve the cost performance of an organisation through eliminating reworks and waste (Wakchaureet al., 2014).

The integrated manufacturing programs consisting of TQM as a key component of how they approach continuous improvement can play a major role in addressing industrial cost issues. The optimisation of costs for an organisation involves reducing the unnecessary, and enhancing the quality of, production in the first instance. In empirical studies such as Kaynak and Hartley (2009), the implementation of TQM alongside just in time (JIT) inventory management has had a positive and significant effect on lowering the cost, and ensuring the speed, of delivery (Kaynak & Hartley, 2009). The benefits of interventions are increasing the effectiveness of operations and lowering the costs by eliminating redundant processes. In an empirical study that was conducted by Wicaksono and Bambang (2019), the effect of TQM on quality costs was evaluated for SMEs in Java. The findings indicated that the different aspects of TQM, such as the focus on customers, obsession with quality, continuous improvement, training, and benchmarking, had positive and significant effects on cost efficiency for manufacturers. This allows focusing on customer needs and elimination of production waste, which leads to a cost-effective organisation (Ahmad et al., 2019).

Developing the understanding of the customers' wants, demands and expectations has become vital for organisations. When the organisations are successfully directing all of their efforts towards attaining such kinds of goals the resulting operational performance is always according to the expectations of customers (Gupta, et al. 2021). Satisfied customers are always ready to pay more for fulfilling their expectations again and again that result in increased sales and revenues of the organisations. It also restrains inventory pileup due to rejected and returned products and unsold items as well as saves the warranty claims, recall costs and product liability costs (Amoako-Gyampah, 2019). Moreover, when organisations take careful consideration of customer requirements, the outcome is always according to the customers' specifications thereby eliminating defective products, associated cost and eventually saves the time as well. Hence, customer focus significantly contributes in enhanced operational performance by eliminating the internal and external failure costs along with assurance of highly demanding products that would be sold out immediately (Chaurey et al., 2021).

TQM is based on the belief that it's better for organisations to bear the prevention costs because prevention of defects is better than dealing with failures, hence the cost of defect prevention is cheaper for organisation than the cost of quality failures (Gupta, et al. 2021). In other words, if organisations are able to eliminate defects level, then the total quality costs incurred by the organisation are also reduced. Some important quality costs incurred by organisations are prevention cost, internal failure cost, appraisal cost and external failure cost that are some kinds of extra burden on the organisations while implementation of the TQM practices helps eliminating all these kinds of costs (Abdallah and Nabas, 2018). A study by Amoako-Gyampah (2019) debated that organisations have to incur cost in order to make efforts for reducing poor quality termed as the prevention cost. It includes improvements in the designs of the products to make them durable, providing training to the employees for enabling them to perform a good job, certifying suppliers for ensuring the supply of high-quality products and services by the suppliers.

Under systematic management approach, organisational operational performance is enhanced through better products, improved responsiveness, shortening of cycle times for novel products and services, effective marketing, production, and engineering strategies. Moreover, by elimination of errors, wastage and defects, the organisations are also capable of enhancing operational performance by having sufficient control on costs (Mishra et al., 2018). The above findings indicate that, with effective implementation of TQM in the management of organisational operation, the efficiency of cost saving can be understood to be the main expected benefit. Keeping the focus on continuous improvement and teamwork for addressing customer concerns and lowering potential faults, reduces the costs incurred by an organisation in serving its customers (Sila, 2012). The expectation is that with the implementation of TQM strategies in organisations, cost effectiveness is a guaranteed outcome. Managers who focus on different aspects of TQM enhance all aspects of cost control (Gupta, et al. 2021). It is hypothesised from the available literature that TQM implementation improves organisational performance in terms of cost management and optimisation. Thus, based on above discussion the proposed hypothesis is as follows:

H2a: Total quality management (TQM) has positive significant influence on cost performance.

3.1.2.2. *Impact of TQM on Flexibility*

In today's business environment, customers' requirements about products vary greatly with the increased number of customer's numbers and their changing preferences. High competition due to industrial expansions is immensely pressuring on the firm creativity. Innovation is the new normal in the field of business as firms aim to increase their market share and expand their operations. Zehir et al. (2012) concluded that TQM provides opportunity for improving a firm's innovative capability and providing goods that meet the quality standards of consumers. Enhancing innovation capacity of firms enables them to provide goods and services that fairly compete in the global marketplace. Firms' ability to create new products in a timely manner and stay relevant depends on three key success factors: management leadership, customer focus and continuous improvement (Gupta, et al. 2021). TQM is a management concept that bridges the gap between the management and its employees. Bridging the gap between employees and management enables a firm to make informed decisions that lead to integrate the employee thinking in product design and customer feedback to launch customer-oriented products. This eventually improves flexibility performance in the firm. Studies on this matter have focused on the effectiveness of TQM strategies and tools in developing organisations that are able to perform in a flexible manner (Phan et al., 2019).

The issue of flexibility, in relation to the impact of TQM, is associated with the fact that consumers are increasingly seeking diversity and variety in their product choices. The application of TQM in these settings provides an opportunity for organisations to focus on the issues of interest to the customers. A key aspect of TQM is responsiveness to customer needs and expectations, which should be considered in making decisions on what and how-to produce (Gupta, et al. 2021). The focus of organisations on customer complaints provides more opportunities to produce high quality goods. It also allows organisations to improve their operational performance by better responding to customer needs.

TQM involves training and education as well as benchmarking other best practices and persistently improving production quality. Such practices and persistence play a beneficial role in enhancing the ability of an industrial firm to better compete in its market by allowing more flexibility in changing its product mix and well as volumes, which, in turn, helps to achieve its strategic objectives. Flexible product mix manufacturing can be considered as a key component of organisational success because of the impact it has on enabling modular production

strategies and processes for responding to varying demands. In a study from Iqbal et al. (2018), the application of TQM was evaluated in conjunction to the use of a just-in-time (JIT) production strategy. It was found that TQM application involved a pull production system consisting of small lot sizes and low set up time. TQM principles were found to be highly beneficial in enabling the agile production strategies that were reliant on common external infrastructure and direct relationships with the targeted customers.

Operational flexibility is found to be closely associated with the TQM practices implementation since the two concepts have some common orientations. Organisations adopt TQM practices for ensuring improved products' quality, reduced costs, and enhancement of customer satisfaction. Likewise, organisations try to attain flexibility performance that may eliminate unnecessary expenses and excessive production costs (Abdallah and Nabas, 2018). Amoako-Gyampah (2019) argues that TQM practices provide organisations' higher adjustment ability in their strategies and structural administration, according to the changing market requirements. Moreover, it is found that TQM emphasises on moving beyond the firm-centric approach towards expansion of quality management practices to up stream and down stream supply chain partners so that organisations could be able to cope with high velocity business environments.

Consistently, Gurahoo et al. (2018) evaluated the determinants of service quality in the service sector in line with the application of TQM. They identified that the most important aspects of TQM practices for the investigated firms were the demand-led production and joint new product development. This provides a highly beneficial tool for addressing firm flexibility through virtual supply chains and deployment of flexible workforce (Alolayyan et al., 2013). One of their key findings indicated that flexible or agile manufacturing required lean implementation strategies in order to achieve the expected benefits for organisations. Organisations seek to achieve flexibility in production by avoiding unnecessary expenses and having high levels of stock. Continuous improvement in daily routines is encouraged through TQM implementation. A key aspect of this relationship is that TQM provides higher adjustment capability in response to market requirements (Dubey, et al., 2021).

The effectiveness of these strategies is also associated with the administration of structures that enable quality management to be extended to other stakeholders in the supply chain outside the firm. Both upstream and downstream supply chain actors must be involved in the implementation of TQM (Flynn and Flynn, 2005). These findings present the view that

effective implementation of TQM is likely to have a positive influence on the flexibility of organisations. Thus, based on above discussion the proposed hypothesis is as follows:

***H2b:** Total quality management (TQM) has positive significant influence on flexibility performance.*

3.1.3. The Impact of TQM on TPM

TPM is a firm management tool that informs decision-making and helps in job designing, problem solving, employee recruitment and selection, performance appraisal and promotion. By performing these functions for both small and large firms, a lot of positive impacts have been realized: improved firms' performance, increased market share and employee satisfaction. While some firms implement TPM as their main management tool, others integrate both TQM and TPM as their management tools, which have brought about significant positive results to the respective firms. Konecny (2011) hypothesised that the conjoint implementation of TPM and TQM presents more positive results when compared to the implementation of either one. Implementing both practices in combination leads to increased benefits due to the fact that they complement each other in ensuring quality results for firms. Kaur et al. (2013) held that TPM adds to TQM practices and techniques to ensure zero defects in equipment. TQM and TPM share commonalities regarding to employee involvement in the management practice, cross functional approach, and continuous improvement. TQM is, thus, an important tool that aids TPM in many managerial aspects within the industrial firms.

From a theoretical perspective, the implementation of TPM evolved with the concepts of lean management and they both share similar elements. This is because TPM is meant to reduce or eliminate equipment failure. Many scholars consider TPM as an innovative practice to machine maintenance that complements TQM and other manufacturing strategies, including continuous improvement and JIT production (McKone and Schroeder, 2001). This means that applying the two strategies is simultaneously meant to enhance the capacity of an organisation to respond to its challenges and improve its output quality. Thus, it is clear from the conceptual aspects of the two methodologies that they are similar in many ways, especially their focus on involving all employees and stakeholders in addressing quality and operational integrity (Dubey, et al., 2021). While TQM is customer centric in which it seeks to optimise the overall quality offered to customers, TPM maximises equipment effectiveness by engaging all employees. Planned maintenance is applied as a key element of the strategy to ensure that all equipment work

effectively, while employees are highly motivated to undertake autonomous maintenance activities. TPM eliminates breakdowns in machinery and equipment, while TQM focuses on reducing defects and waste by ensuring that the right products and other aspects are achieved without the need for reworking in order to reduce cost and time.

The interest in TQM as an approach that enhances TPM implementation is based on the thought that both approaches are effective in enhancing firms' operational performance through holistic initiatives that address all critical production and quality aspects. According to Konecny and Thun (2011), TQM and TPM are relatively similar approaches for addressing organisational issues related to operational performance. A key element of the association is that TPM and TQM should be implemented before the adoption of lean production strategies (Dubey, et al., 2021). There are two sets of critical factors of implementing TPM and TQM that influence their effectiveness and applicability. These include universally significant implementation factors such as leadership, strategic planning, and process management. These approach specific factors are also critical to the implementation of the two strategies such as effective equipment management and customer satisfaction. Ahmad et al. (2015) studied the effect of TPM as a mediator of the link between TQM and business performance in the automotive industry. The results indicated that TQM was a key factor in mediating the association. TQM was found to have an indirect effect of 0.25, meaning it has a high mediation effect. The findings of Ahmad et al.'s (2015) study, as well as other studies such as the one from Konecny and Thun (2011), indicate that TQM is positively correlated with the application of TPM in organisations.

Marian et al. (2020) indicate that TQM practices not only focus on improving the quality of products through enhancing quality inputs while reducing costs as well, but it also emphasises on achieving the operational efficiency for ensuring the effective utilisation of organisational resources. Hence, in the production processes, the equipment's and machines are among the most significantly contributing resources of the organisations, so enhancing the efficiency of these resources is also emphasised by TQM (Dubey, et al., 2021). Therefore, TQM emphasise on incorporation of TPM in the production process so that instead of wasting time of detecting issues when they actual occur in the production process and cause delays, these should be controlled at initial stages. So TPM enables employees to keep their equipment's in good working conditions so that delays and defects could be eliminated that is the prime focus of TQM practices. Hence, effective implementation of TQM practices contributes significantly in

TPM (Chaurey et al., 2021). Moreover, the available literature from different organisational settings indicates that the implementation of TPM is correlated with that of TQM. Although they can be implemented in a conjoined manner, TPM is a mediator of TQM and, hence, it can be hypothesized as such in influencing the outcomes of organisations. Thus, based on above discussion the proposed hypothesis is as follows:

H3: TPM is significantly supported by TQM.

3.1.4. The Impact of Supplier/Internal Integration on TPM

3.1.4.1. Supplier Integration and TPM

TPM aims at maximising customer satisfaction by reducing equipment defects, continuously improving the production process, and involving employees throughout the process. Research on the moderating role of supply chain integration in the impacts of TPM is scant (Dubey, et al., 2021). The supply chain is a source of important information that enables firms to manage consumer needs and preferences. Suppliers provide the information on the changing tastes and preferences of customers to firms that, in turn, interpret and design their products to meet these tastes (Habidinet al., 2018). Suppliers also enhance the innovation capacity of firms by providing the necessary resources that are needed to exceed the usual production norms. Firms involve suppliers in all stages to better manage their competition. The changing business environment with the entry of many businesses has resulted in an immense pressure on firms. Suppliers complement TPM in designing operations that enables producing unique products that increase the market share of firms (Jain et al., 2018). Firms that do not implement novel management techniques have poor access to market information and, thus, have insufficient supplying quality. As a result, they design their work practices without considering other important stakeholders, which, eventually, leads to high production defects and costs. Thus, involving supplier's enables firms maintain quality products and increase customer satisfaction.

TPM holistically addresses all organisational aspects of autonomous maintenance through empowering and engaging employees at all levels. According to TPM practices, employees should be motivated and empowered to take matters in their own hands when addressing the key issues in their organisations. In order to achieve the full potential of TPM and TQM, an organisation must ensure that the understanding of related strategies is not limited to a specific

managerial level. The integration of other stakeholders in the supply chain provides a beneficial framework for ensuring that the proposed interventions are applicable at all levels of the organisation (Modgil and Sharma, 2016).

Integrating all stakeholders in the management of an organisation is critical to the achievement of organisational goals. In the case of TPM, supplier integration can be considered as a critical factor in addressing the overall goals of organisations. This is because it presents a situation where the suppliers can be actively involved in the formulation of strategies and development of solutions that work within the whole organisation. Operational performance is associated with TPM and supplier's integration as the approach provides a suitable opportunity to address all issues affecting the entire value chain of the product or process (Dubey, et al., 2021). The four main constructs of TPM include information tracking, disciplined maintenance, and housekeeping and operator involvement. In Modgil and Sharma's (2016) study TPM implementation was shown to conduct a critical role in improving plant level operational excellence. TPM was also found to have a significant impact on technology management, product innovation and research and development research and development. These issues can be significantly addressed through the active involvement of all stakeholders such as suppliers and customers.

Supplier integration allows the flow of information from the suppliers to the different units of the organisations including the production facility as well. The flow of information in this relationship is two ways, like it's not limited to gain the knowledge and expertise of the suppliers, but it also incorporates the involvement of suppliers at decision making stage of product development (Thakur and Panghal, 2021). In this way, suppliers are well aware about the production capacity and production efficiency of the organisation, so supplier integration enhances the effectiveness of TPM in a way that suppliers provide their valuable opinions about handling of inventory for improving the operational efficiency of equipment's and machines. Moreover, the availability of inventory also has significant impact on the TPM practices like the delays in inventory may lead to machine shut down for short- or long-term basis while the flow of information between suppliers and the production employees helps eliminating such deficiencies and contributes significantly to enhancing the effectiveness of TPM (Mendez and Ramon, 2017).

It is known that active involvement of suppliers in firm operations increases their responsivity and effectiveness in addressing market needs. In this case, the effectiveness of the TPM

approach, as well as the outcomes it provides for organisations, is evaluated in relation of the involvement of suppliers. Considering that supplier involvement is a key predictor of TQM effectiveness, it is expected that the same level of impact and excellence can be achieved with TPM. This suggests that integrating suppliers into the operations and strategies of a firm can have a significant and positive effect on TPM. Since TPM involves equipment maintenance for consistent performance, suppliers can play a key role in providing the necessary support for enhancing organisational outcomes. Thus, based on above discussion the proposed hypothesis is as follows:

H4a: Supplier integration has positive effect on TPM.

3.1.4.2. Internal Integration and TPM

TPM has been known for the great success it has brought to firms. Consistent positive results have been reported for many firms across the globe on the importance of TPM. Such impacts include increased profitability due to reduced operation costs and improved product quality. While these results continue to emerge, there has been constant change being experienced in the field of business that require more rigorous and strategic steps to remain relevant over time. Swink and Schoenher (2015) posited that internal integration enables firms to gather important business information, interpret it and translate it to improve the internal functions to influence strategic decisions. TPM practices rely on involving employees in managerial roles. Internal integration aids the role of TPM in enhancing the participation of employees in the decisions and processes that aim to optimise resource allocation and ensure maximum profits for firms. Failure to involve employees through internal integration causes resources loss, which leads to increased production costs. Internal integration is argued to increase interfaces in the internal functions to simultaneously improve communications and encourage collaborative efforts to achieve higher business profitability.

The issue of internal integration is associated with the performance of organisations because it promotes information management and knowledge sharing. Having effective integration across the organisation reduces the potential or relevance of organisational silos, where departments and functions are isolated from each other. Internal integration has been cited as a beneficial tool for promoting integration and collaboration within an organisation. In relation to TPM, internal integration is expected to be a key contributor as it can promote the capacity of an organisation to adopt a holistic approach (Swink and Schoenher, 2015).

The conceptual link between TPM and internal integration is that they both require the collection of internal information across the organisation and that they used by stakeholders to address critical issues in the organisation. It is expected that effective integration of all internal functions and strategies can enhance the performance of an organisation in responding to its core issues such as maintenance practices, planning operations and responsiveness to market needs. Further, it is logically expected that internal integration supports the practice of TPM. By having all internal functions and stakeholders working together, the elements of TPM can be implemented in the operations such as knowledge sharing, effective maintenance planning and involvement of employees in autonomous maintenance. The focus of TPM is to empower employees within an organisation to contribute to maintenance and development of a fault-free operating environment. This requires involving all stakeholders in determining the occurrence of faults and maintenance challenges to mitigate them, which results in effective production planning and low maintenance and capital expenses.

Internal integration allows the flow of information among different departments and functions of the organisation and contributes significantly to eliminating production deficiencies. When different functions work in coordination then the magnitude of delays and defects reduces (Mendez and Ramon, 2017). Moreover, as in the case of TPM the employees are made responsible for the maintenance and cleanliness of their equipment's and machines, so internal integration allows them keeping in touch with the maintenance staff and getting their guidance when required. So, such flow of information and collaboration among employees and other departments contributes significantly to enhancing the effectiveness of TPM (Chaurey et al., 2021).

Wakchaure et al. (2014) indicate the critical need to integrate TPM across organisations. This integration of stakeholders raises the probability to success through effective interventions in the suitable choice of products (Dubey, et al., 2021). Effective internal integration is, therefore, necessary as a tool for promoting the implementation of TPM. Since the modalities of TPM require an active involvement of all internal stakeholders, having a high level of internal integration can improve the capability of TPM to promote organisational success. The ability of a firm to effectively respond to internal challenges in maintenance and ensuring high efficiency of the production lines is strongly impacted by the level of internal integration. The proper collaboration among employees and departments in addressing challenges also

determines the degree to which TPM can be implemented to derive organisational benefits. Thus, based on above discussion the proposed hypothesis is as follows:

H4b: Internal integration has positive impact on TPM.

3.1.5. The Impact of Supplier/Internal Integration on TQM

3.1.5.1. Supplier Integration and TQM

Garvin (1987) observed that a firm's capability to produce quality products lies in its ability to collaborate with the supply chain. Garvin argued that those firms that cannot establish strong relationships with their major suppliers cannot be consistent in producing quality products. Managing the supply chain during the implementation of TQM practices results in optimal standards of products and lead to customer satisfaction. Rosenzweig et al. (2003) observed that when specific suppliers emphasise their own goals at the expense of other suppliers, suboptimal quality of processes and products is usually the result. On the contrary, when suppliers work in harmony and collaborate together, the technical and task specific know-how accumulates in a firm, which improves its operational performance. Supplier integration makes it easy for TQM to manage the quality of products by ensuring the technical know-how is shared between the various suppliers and the firm for optimal performance. Another important aspect of supplier integration is the timely interpretation of information that is received and transformed into creative designs that guarantee customer satisfaction through quality products.

The need for consistency in producing high quality goods and following a demand-driven approach, in which inputs from customers are considered to be a major element, is highly important. TQM focuses on following quality across the entire value chain by ensuring that faults are eliminated, and costs are reduced through minimising the associated reworks of operations. The market is a key component of the inputs considered in the implementation of TQM because it requires stakeholders to work together in determining what is expected by the customers and providing the necessary internal resources to accomplish it (Sahoo and Yadav, 2020). In addition, Rosenzweig et al. (2003) observed that high intensity of integration can positively influence the product quality and, consequently, customer satisfaction. High intensity of integration among the main stakeholders can have a significant impact on implementation of effective TQM strategies within organisations.

The critical issue in this respect is the significance of integrating suppliers into the internal processes and the strategies adopted for TQM implementation (Dubey, et al., 2021). Since TQM focuses on ensuring that the right commodities are produced at the right time and with no defects, the suppliers play a key role. They must provide material and supplies that are effective in responding to the needs of the firm and its customers. Lack of responsiveness by the suppliers can be costly for organisations because it would increase the costs of remedying defects and returning material to reach the right specifications (Flynn and Flynn, 2005). This means that suppliers need to be directly involved and integrated in internal operations and strategies of the organisation to ensure that it operates properly and contributes to the effectiveness of TQM.

The core purpose of implementing manufacturing practices like TQM is to enhance the quality of products and attaining efficient operational performance according to the demands and expectations of customers. On the other hand, SCM is practiced by the organisations for controlling manufacturing processes for enhancing product quality and eliminating the risks of recalls as well as building a strong customer brand (Amoako-Gyampah, 2019). Hence, both approaches have a similar purpose of quality enhancement and customer satisfaction. The only difference between both elements is the way to approach the quality i.e., within the organisation in case of manufacturing practices (TQM) and across the supply chain in SCM (Marian et al., 2020).

SCM and the TQM have been considered as two most significant approaches for firms like services, manufacturing, and small and medium enterprises (SME). These approaches have become the vital ingredient for organisational success and attaining competitive advantage in the industry. Initially these two approaches were not considered to be adopted at a same time in an organisation due to similarity of purpose but with the passage of time, the integration of these two approaches in an organisation proved to strengthen the competitive advantage of the organisation as well as enhanced customer satisfaction (Talib et al., 2011). Hence, in recent era, applications of TQM principles in supply chains have become common practices especially in the field of SCM (Marian et al., 2020).

Considering the critical role played by suppliers in enabling the smooth running of operations and providing material that enhances the quality of production, the integration of suppliers in an organisation is critical to the success of TQM. Empirical research on this link is relatively limited and most of the related studies focused on how the interventions promote organisational

success and manufacturing excellence. The role played by integrating suppliers requires further evaluated to better understand it and provide a clear view of how it acts as a moderating factor in promoting organisational success (So and Sun, 2010). The integration of suppliers in the development of TQM strategies and frameworks can be highly beneficial in developing a powerful organisational environment where the material supplied is in line with the needs of the entire value chain. Suppliers contribute to the reduction of the associated costs as well as the quality challenges that may arise for the firm. In this respect, the hypothesis is that effective supplier integration for a firm is a significant promoter of TQM. Thus, based on above discussion the proposed hypothesis is as follows:

H5a: Supplier integration has a positive effect on TQM.

3.1.5.2. Internal Integration and TQM

TQM as a managerial concept includes all works conducted to reduce defects in equipment through autonomous maintenance. In so doing, TQM relies on the active involvement of employees to fix problems that emerge on a day-to-day basis to minimise machinery and equipment failure. Internal integration enhances the communication of internal functions by creating several interfaces for collaboration. These communication interfaces enhance JIT production management, which, thus, increases the firm's operational performance. Swink and Schoenherr (2015) argued that firms that achieve high levels of internal integration experience increased profits than their counterparts. Internal integration in a firm may foster strong relationships between different functions and promote preferable organisation outcomes. TQM builds on these established relationships to better drive quality management initiatives through harnessing and utilising feedback to design work practices for maximum profits. Swink et al. (2007) mentioned that internal integration lowers the need for asset buffers such as inventory through improved planning and increased coordination among functions that are involved in the TQM practice. Internal integration, therefore, can be argued to positively influence the role of TQM in maximising firm profits.

Chaudhary et al. (2018) indicate that internal integration enhances the effectiveness of TQM practices implemented in the organisation. The main focus and emphasis of TQM practices is to improve the quality of products while reducing waste, costs and delays. Therefore, when management systems or different functions within the organisation are integrated, then each function lines up towards a single goal enhancing the overall operational performance of the

organisation. Internal integration incorporates coordinated efforts which are much higher than the sum of individual parts which not only improves efficiency but also effectiveness of the organisational operations (Mishra et al., 2018).

Abdallah and Nabas (2018) indicate that internal integration have significant positive impact on the implementation of TQM practices in the form of enhanced quality, safety and resulting operational performance. It also helps eliminating risks associated with the production efficiency as well as helps eliminating defects at different stages of product development. Another significant positive impact of internal integration is the uniform understanding of organisational objectives, policies, resources, and processes that enhances efficient utilisation of organisational resources (Thakur and Panghal, 2021). Moreover, the implementation of quality management practices requires training and assessment of employees for effective implementation, so internal integration enhances the effectiveness of training through appropriate implementation of practices by the flow of information among employees from different functions of the organisation. Integration of multiple functions establishes cohesive processes, resources, objectives that align with the system and improve accountability that ultimately helps improving productive efficiency (Amoako-Gyampah, 2019).

Internal integration involves the consideration of all key stakeholders in determining the value that is expected in the final output and working together to achieve it (Dubey, et al., 2021). Internal integration refers to the adoption of strategies that work across organisational boundaries or departments. The focus of internal integration is the elimination of departmental silos that may adversely affect the achievement of the overall goal of the organisation. Organisational success relies on the presence of strong working relationships among stakeholders to enable the accomplishment of internal organisational goals. Success and excellence require effective accomplishment of internal collaboration among stakeholders to ensure that they work together in responding to the changing needs of customers.

Alongside the goal of internal integration for organisational success, firms use TQM as a tool for ensuring that they address their quality needs and issues across the value chain. The focus on ensuring total quality is associated with the goal of achieving the highest probable outcome using lesser resources. The focus of the research is on the significance of responding to organisational needs by integrating all employees and getting them to work together. How strongly the employees are integrated in an organisation can have a serious impact on organisational goals and excellence in aspects such as quality, cost, and responsiveness.

The link between the internal integration of the different stakeholders and the application of TQM can be significant in determining the accomplishment of organisational excellence. As indicated in prior studies, such as Ahmad et al (2019), TQM is strongly linked to business performance. This, hence, contributes to the success of TQM as a tool that promotes high responsiveness to organisational needs by eliminating unnecessary costs and wastage across all levels. With effective internal integration, all internal stakeholders work together without departmental boundaries to achieve shared excellence goals. In this case, functional boundaries are no more perceived as barriers to knowledge sharing and the implementation of strategies for improving quality along the value chain. In this way, it can support the development of effective TQM strategies and applications to generate the expected excellence levels such as improved quality, low costs, and high profitability.

Thakur and Panghal (2021) argued that internal integration improves the consistency of management systems in the organisation which eliminates the complexity of management system and helps employees developing a better understanding of organisational objectives. Employees having good understanding of organisational objectives work together to achieve these objectives; like optimising resources and processes of the organisation that result in reduced costs. It also helps reducing the time taken for performing certain activities by employees as well as eliminating disruptions and delays that also contribute to cost reduction.

Considering that TQM requires the integration of all stakeholders in the development of effective strategies for addressing quality issues across the organisation through customer focus and continuous improvement, it should be strongly associated with, and supported by, the level of internal integration in organisations. This supports the hypothesis that TQM is significantly supported by the level of internal integration in organisations through better communication, cooperation, and knowledge sharing. Thus, based on above discussion the proposed hypothesis is as follows:

H5b: Internal integration has a positive effect on TQM.

3.1.6. The Impact of Supplier Integration on Operational Performance

3.1.6.1. *Supplier Integration and Cost Performance*

In a study conducted by Rosenzweig et al. (2003), it was established that the higher percentage of revenues received from new products is resulted from the increased supply chain integration. Rosenzweig et al. (2003) added that integrating suppliers in the development of new products reduces the costs and speeds up the lead time. Integrating suppliers enhances the firm's competitive capacity, which results in better business performance. Supplier integration allows firms to achieve cost leadership due to several reasons. First, an integrated business environment provides firms with market intelligence. Suppliers can support firms to acquire market intelligence through their higher access to information during the interaction with other suppliers and customers. Market intelligence helps firms to redesign and redevelop their products to fit with customer requirements. Besides, market intelligence helps firms to strategise on how to overcome competition. On another perspective, the integration of suppliers reduces the firm inventory, which reduces the cost of operations. In so doing, the firm's overall performance is enhanced through redirecting the money invested in inventories to the research and development of other productive products and practices.

Supplier integration is critical for the current business environment where the involvement of stakeholders enables organisations to respond to the needs of their customers. Businesses have become increasingly competitive and, therefore, require clear frameworks for integrating and enhancing value. One such area of cooperation and integration is supplier integration where firms develop inter-connected relationships with their suppliers through knowledge exchange and SCM information systems. Instead of simply integrating within the organisation, knowledge exchange and the expansion of the supply chain to involve upstream and downstream players can noticeably enhance organisational performance and operational performance. Firms nowadays seek excellence across all fronts and forms with issues such as quality management and timely delivery being central to the accomplishment of goals (Konecny and Thun, 2011). To achieve these benefits, organisations must cooperate with their suppliers to ensure that the right quality is delivered, and changes are effectively communicated from the targeted customers. Communication and coordination across the value chain are significant in ensuring that customer demands, and preferences are comprehended by all internal stakeholders including suppliers to ensure the delivery of the right material. Having an ineffective approach to supplier integration can result in additional costs to firms because of

stock outages and inventory costs. Receiving the wrong supplies can result in a situation where the production does not meet the market standards, which dissatisfies customers.

TQM practices are not only focused on enhancing the quality of products but also have significant emphasis on reducing the costs to improve profitability. Hence, supplier integration contributes significantly towards quality management practices because being in mutual beneficial relationship suppliers also strive for saving costs at different stages of product development (Mishra et al., 2018). Supplier integration not only helps reducing the delivery costs but also helps controlling the quality issues, availability problems and delays leading to a better service for customers. Moreover, development of a good communication flow between suppliers and manufacturers results in developing a thorough understanding of the end products which enhances suppliers' ability to serve the business accordingly (Marian and Soare, 2020). However, considering the adverse effects that ineffective supplier integration can have on organisational goals and excellence, it is apparent that improving it enhances the overall organisation. The critical issue for organisations that apply TQM is the need to ensure that the right quality is provided at the right time. Managers also focus on the need to have their facilities in perfect working conditions and their deliveries arriving in time to avoid long lead times. Ineffective integration with suppliers and other stakeholders presents additional costs for firms in terms of defective goods and reworking arrangements (Kaynak and Hartley, 2009). Effective supplier integration, on the other hand, is likely to promote positive outcomes for organisations in terms of operational costs reduction and unnecessary expenses elimination. Supplier integration is expected to be an effective approach for responding to organisational needs and customer requests. It can have a positive impact on costs by saving on enormous capital expenditures of firms. Appropriate supplier integration lowers the costs incurred by firms and, hence, raises their profit numbers. It is then hypothesised that supplier integration has a significant positive impact on the cost performance of firms. Thus, based on above discussion the proposed hypothesis is as follows:

H6a: Supplier integration has positive impact on cost performance.

3.1.6.2. Supplier Integration and Flexibility Performance

The current volatile business environment requires flexibility in operations to gain competitive advantages and improve overall performance. Literature on flexibility performance in firms has been increasing in the recent past with the research work of Tiwari et al. (2015), extending

this research to include the flexibility of supply chain components. The flexibility of processes is a current and urgent centre of focus based on the nature and pace of change in the business environment. The ability of firms to respond to these changes determines their success level. In order for firms have flexible internal operations, key external capabilities must be developed. Supplier integration provides firms with the external capabilities that enable responding to desired changes appropriately (Huo et al., 2018). Supplier integration involves having a high level of flexibility in the management strategies about how suppliers are engaged or selected. A flexible approach to supplier selection represents significant considerations and benefits for organisations to attain the best quality and the highest value for money. Supplier flexibility links the focal firm and its suppliers, which enables it to efficiently react to variances within the procurement process through aligning its resources and routines with suppliers. Inflexibility of the supply chain leads to increased lead time and hinders the firm's ability to respond to the changes in customer tastes and preferences.

Supplier integration has a positive association with aspects of manufacturing performance such as delivery time, product quality and volume flexibility. Having clear integration with suppliers provides firms with an environment that allows higher capability of maintaining the needed supplies at all times. Clear integration with suppliers also contributes to effective performance management because it means that all aspects of the supply chain work together in responding to customer needs. This is one of the most important aspects of SCM that influences firms' adoption of strategies such as TPM and TQM that rely on managing the entire organisation. Planned maintenance promotes the flexibility of operations through strategies that enhance the capacity of the firm to change (Kumar et al., 2014).

Marian and Soare (2020) argue that companies having good supplier relationships along with integrated internal systems have enhanced flexibility to respond according to the customers' changing requirements as well as fluctuations in the market and industry. Organisations' ability to be adaptive and flexible to varying market situations is significant in attaining the competitive advantage in industry. Integration at supplier and internal levels facilitate organisations to respond so quickly and effectively that could never be possible without such integration. Supplier integration helps organisations in attaining the inventory efficiency like only carrying the needed inventory while ensuring the further availability as soon as it is required by the production facility. It helps eliminating carrying costs as well as saving the space at manufacturing facility. Organisations could not enjoy such benefits in the absence of

supplier integration because in such case they may have to face certain delays as well (Chaurey et al., 2021).

Flexibility performance has a critical role in responding to changing market conditions such as increase in demand or variations in customer preferences. With supplier integration, a firm is able to determine the relevant strategies to ensure that it responds to changes in the perceptions of customers. Integrating suppliers in operations and decision-making increases the capacity of the firm to adopt various product volumes and types. Supplier integration represents a highly beneficial tool for addressing flexibility for firms through virtual supply chains and dynamic workforce (Alolayyan et al., 2013). Organisations seek to achieve flexibility in their production strategies by avoiding unnecessary expenses or excessive stock. Continuous improvement in daily routines is encouraged through TQM implementation and high adjustment capability in response to market requirements. From the available literature, it is expected that implementing supplier integration strategies noticeably improves flexibility performance. It is also expected that effective integration of suppliers in the management of firms, especially those involved in manufacturing, allows more efficient performance. There are many dimensions of flexibility performance, but it is apparent from the literature that integrating suppliers in the organisation and involving them in decision-making can improve both the volume flexibility and product mix flexibility. Including flexibility in the selection of suppliers can influence the performance of an organisation in responding to market changes, which adds more agility to operations. Thus, based on above discussion the proposed hypothesis is as follows:

H6b: Supplier integration has positive impact on flexibility performance.

3.1.7. The Impact of Internal Integration on Operational Performance

3.1.7.1. Internal Integration and Cost Performance

Internal integration increases the internal capabilities of firms, which directly impacts their profit through reducing production costs. Ragatz et al. (1997) observed that coupling suppliers in internal capabilities provides avenues for knowledge and skills sharing among stakeholders, which reduces the time and cost of converting a business idea into a product. Rosenzweig et al. (2003) indicated that highly integrated firms with the capacity to provide competitive prices in the initial stages of a new product development process are able to overcome competition and derive high revenues from their new products. Roth (1996) observed a high cost-effective process and higher revenues in the firms with tightly integrated firms when compared to their

competitors. The ability of firms to develop better systems of communication, corporation and coordination between the functional departments help to reduce the costs of new productions. Reducing these costs of internal operations allows the firms to have a competitive position and results in increased sales. For example, Walmart was able to reduce around 20% of their production costs by simply mainstreaming the internal systems (Rosenzweig et al., 2003).

Internal systems are critical to the achievement of organisational goals because of the need to work together in order to avoid inefficiencies. Since internal systems promote the performance of firms by enabling the value chain to operate effectively, they allow responding to the various weaknesses of inefficient coordination that may adversely affect organisational outcomes. Inadequate coordination and communication among internal stakeholders such as departments can influence the operational performance of the operation or organisation. The functional areas of an organisation need to coordinate their activities to ensure that they effectively communicate and respond to the challenges of operations. Open communication systems and strategies can influence the costs incurred by firms through minimising the likelihood of errors (Flynn and Flynn, 2005).

Some of the most important factors to be considered in determining the profitability of an operation are its capacity to maintain efficiency while reducing costs. Cost effectiveness of an operation arises from having highly integrated and coordinated internal functions. The internal functions of a firm need to work together in coordinating responsiveness and communicating how to better compete according to market changes. The cost performance of a firm presents a significant point of competitiveness because it allows using the available resources in the most effective way to create more value for money (Phan et al., 2019). The element of internal integration involves ensuring that all employees are on board and can work together to achieve quality and excellence. With close coordination and cooperation among internal stakeholders, firms benefit from better responsiveness to customer needs and lower costs associated with errors and stock outages. Their employees can coordinate activities such as maintenance and order processing to avoid delays that may cause loss of customers. Thus, based on above discussion the proposed hypothesis is as follows:

H7a: Internal integration has positive impact on cost performance.

3.1.7.2. *Internal Integration and Flexibility Performance*

A firm is stated to have two capabilities for internal integration: coordination and planning. While coordination refers to the ability of firms to integrate across organisational processes, planning is where firms are able to combine their activities with the information provided by the members of the supply chain. Although internal integration is prioritised to manage the supply chain, there is an urgent need for specific firms to align intra- and inter-organisational resources with suppliers. For example, the structure of the supply chain could be aligned with organisational resources. Roberts and McKay (1998) held a contrary opinion that internal integration is an important, yet not a beneficial, approach for the overall firm performance. Subramani (2004) observed that internal integration improves the efficiency of firms' operations such as controlling inventory, order processing and information exchange. Buyers and suppliers may exchange critical information regarding unforeseen problems that interrupt the procurement process and lead to a less productive business environment (McKone-Sweet and Lee, 2009; Danese, 2013). Thus, performance operations are influenced by higher level of internal integration in a firm.

Internal integration is essential to achieve organisational excellence where employees and departments must work together in order to avoid lack of responsiveness to customer needs and inefficiencies in the operation. Internal coordination contributes to the performance of firms by enabling the value chain to operate effectively. Employees have a significant role in responding to organisational challenges and inefficient coordination that may adversely affect organisational outcomes. Inadequate coordination and communication among internal stakeholders such as departments can influence the flexibility of operations in responding to emergent issues, which affects the operational performance of organisations. The functional areas of an organisation need to coordinate their activities to ensure that they effectively respond to emergent changes and new needs. Having better information and knowledge flow can influence the costs incurred by firms through reducing the errors and increasing the flexibility of their approaches and strategies to respond to market needs (Flynn and Flynn, 2005).

The internal functioning of a firm needs to work together in coordinating responsiveness to changes and communicating how to compete in the market. The element of internal integration involves ensuring that all onboard employees cooperate to achieve quality, responsiveness, and

excellence. Close coordination and cooperation among internal stakeholder's means that firms benefit from better responsiveness to customer needs and lower costs associated with errors and stock outages. With highly integrated internal stakeholders, firms can effectively respond to changes in demands and other elements of production strategies such as small lot sizes and better coordination of maintenance activities. With internal integration, firms can make their operations more agile and redeploy employees and other resources to other products to ensure higher profitability. Internal integration provides a beneficial framework for organisations to take advantage of the resources at their disposal to develop products that are essential to customers. The relevant strategies for firms to ensure that they can effectively respond to the needs of their customers involve extending the supply chain and promoting information exchange to enable determining the demand levels and how to address them. These approaches of agile manufacturing were found to have a significant and beneficial effect on the performance of firms as they emphasise flexibility as a competitive advantage. Thus, based on above discussion the proposed hypothesis is as follows:

H7b: Internal integration is positively impacting flexibility performance.

3.2. Model Development

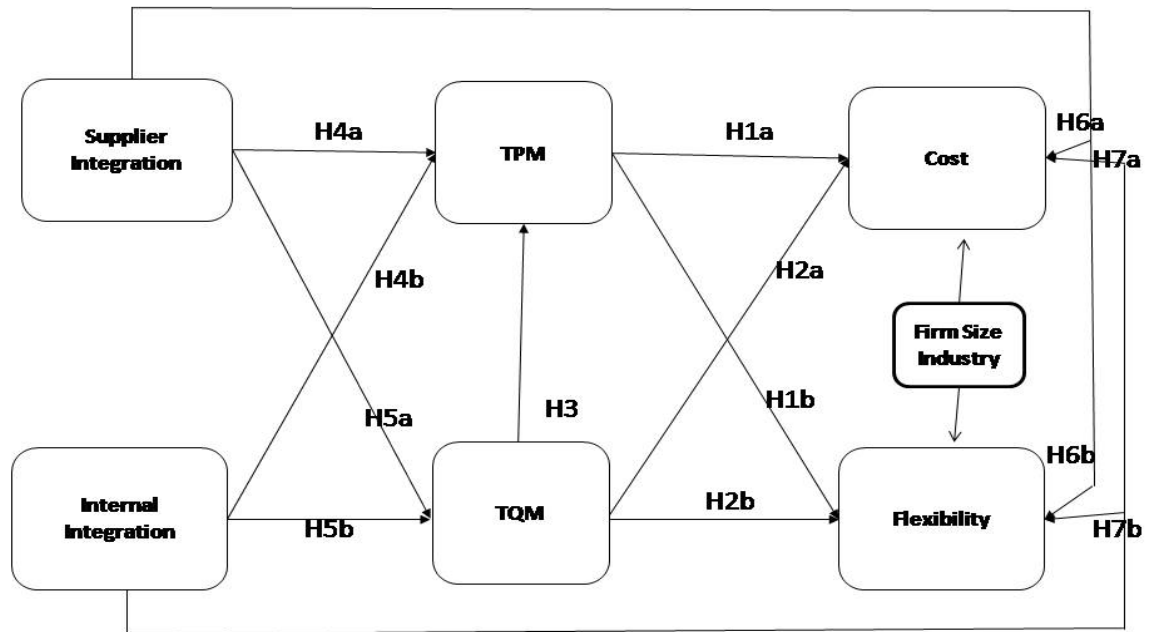


Figure 3-1: Research Model

Figure 3.1 indicates the proposed research model of the current research, where it exhibits the proposed relationships among the study's variables. Thus, based on the above proposed relationships; the study's hypotheses are discussed in the coming section.

<i>Impact of TPM on cost</i>	<i>H1a: Total productive maintenance practice (TPM) has a positive effect on cost performance.</i>
<i>Impact of TPM on flexibility</i>	<i>H1b: Total productive maintenance practice (TPM) has a positive effect on flexibility performance.</i>
<i>Impact of TQM on cost</i>	<i>H2a: Total quality management (TQM) has positive influence on cost performance.</i>
<i>Impact of TQM on flexibility</i>	<i>H2b: Total quality management (TQM) has positive influence on flexibility performance.</i>
<i>Impact of TQM on TPM</i>	<i>H3: TPM is positively supported by TQM.</i>
<i>Supplier integration and TPM</i>	<i>H4a: Supplier integration has positive effect on TPM.</i>
<i>Internal integration and TPM</i>	<i>H4b: Internal integration has positive impact on TPM.</i>
<i>Supplier integration and TQM</i>	<i>H5a: Supplier integration has positive effect on TQM.</i>
<i>Internal integration and TQM</i>	<i>H5b: Internal integration has positive effect on TQM.</i>
<i>Supplier integration and cost performance</i>	<i>H6a: Supplier integration has positive impact on cost performance.</i>
<i>Supplier integration and flexibility performance</i>	<i>H6b: Supplier integration has positive impact on flexibility performance.</i>
<i>Internal integration and cost performance</i>	<i>H7a: Internal integration has positive impact on cost performance.</i>
<i>Internal integration and flexibility performance</i>	<i>H7b: Internal integration has positive impact on flexibility performance.</i>

Table 3-1: Interrelationships of variables under study

3.3. Chapter Summary

Through relying upon the previous discussion, it is found that the above chapter concerns with displaying the various models and hypotheses, those are developed in this research in an attempt to accomplish the core aim and objectives. Additionally, the chapter also concentrates on elaborating the theoretical backgrounds behind each of these models in order to provide a comprehensive view of the logic behind developing the different research hypotheses. The next chapter discusses the methodology and design that were used to guide the effort of collecting and analysing the empirical data as a base to obtain the research findings, contributions, and conclusion.

Chapter 4 Research Design and Methodology

This chapter provides a detailed description of the research methods, those are used to identify, gather, process, and analyse the needed data of this research for the aim of testing hypotheses, answering the study's enquiries as well as fulfilling its core aim and objectives. The chapter is technically describing the entire procedure that is used to get credible information that can stand as new contribution for the advancement of related knowledge. It is divided into several sections: research philosophy, approach and method, sampling and data collection, pilot study, questionnaire design, non-response bias, measures, and conclusions.

4.1. Research Philosophy

A research philosophy represents the conceptual guidelines on how research data should be collected, analysed, and used to produce findings and conclusions for the development of theory (Galli, 2019). The basic assumption of a research philosophy is that it can transform what is considered to be true into what is known to be true. In other words, the purpose of a research philosophy is to transform things that scholars believe to be true into things that are evidently true. The popular philosophies in management research are the interpretivism school of thought and the positivism school of thought. This study leans towards the positivist philosophy, which relies on an epistemological stance that posits that reality can be only apprehended directly with no subjective bias from the researcher. It embraces an objective ontology that assumes the reality to be independent of human minds. Thus, in order to conduct research on a phenomenon related to social sciences, the researcher should decide on his underlying philosophical stance towards the research and chooses the research method based on it.

In general, in order to understand the problems and reality of the basic research elements, many perspectives of the underlying philosophy can be considered. These views differ in terms of defining the reality of phenomena and how to gain knowledge about them. For that, different philosophies can be considered such as positivism, interpretivism, critical realism and post structuralism (Poster, M., 2019). These philosophies differ in the way they answer the three fundamental questions that arise about ontology, epistemology, and methodology. These questions are: what is reality? What is the nature of knowing about the reality of this phenomenon? How is this knowledge obtained? By selecting a philosophical stance, the researcher implicitly follows specific research design to gain knowledge within the theoretical

boundaries of that philosophy. In addition, the required research approach and other relevant research methods are revealed.

As this research follows the fundamental rules and basis of the positivist paradigm, it favours using the quantitative method to investigate the research problem (O'Gorman and MacIntosh, 2014). In these methods, the researcher assumes that the necessary knowledge about the studied phenomenon can be obtained objectively and without any subjective bias. It should be noted that social phenomena, especially those that deal with human behaviour, cannot be observed in a completely objective way and without being exposed to the researcher's bias. Quantitative and qualitative research methods represent the two main types of methods in social sciences. They use correct and real knowledge rather than knowledge based on belief, which, in turn, allows for avoiding personal practice (Collis and Hussey, 2013).

The debate over the nature of quantitative research is a living debate with the philosophical movement of ontology and epistemology. Nevertheless, the question arises as to why the ontology and knowledge of science should be found in the form of a plan or program that is sufficient in research is not accepted. Often ontological and epistemological issues tend to integrate and connect with each other. This research follows the positivism approach, as this is based on the cause-and-effect relationship. Moreover, the deductive reasoning is adopted for this study which means, the theoretical framework was proposed on the basis on theories and then hypothesis was formulated and after that data was gathered through research questionnaire or survey. The data technique was quantitative analysis. As the study needs to highlight the role of manufacturing practices and their combined effect with the SCM practices to enhance the operational performance of Jordanian industrial firms. So, the study chose to collect quantitative data from the concerned respondents i.e., working in Jordanian industries. The quantitative data collected from these respondents help identifying the effect of variables under study. Therefore, this study employs sampling technique and data was analysed through SPSS and AMOS.

4.1.1. Ontology

The ontological stance is a study of tangible aspects, in that the researcher's knowledge of the phenomena under investigation is certain based on experience (Scotland, 2012). As this research studies how the Jordan industrial firms perform in diverse aspects and conditions, objective ontology is adopted to study this phenomenon. The findings of a survey on these

entities may help reach the research objectives and objectively address the reality about the phenomenon concerned.

Ontological questions concern the nature of reality of a phenomenon (Bahari, 2010). These questions are about both physical reality and social reality. In social sciences and research, ontological assumptions include low-depth realist ontology, conceptual realist ontology (Stanley and Mackenzie-Davey, 2012), cautious realist ontology (Magnus, 2018), deep realist ontology (Patomäki and Wight, 2000) and idealistic ontology (Hall and Wicaksono, 2020). Ontology refers to the general beliefs about what things or essences exist in the real world that refers to a range of beings and relationships accepted within a specific scientific field. Research ontology also refers to the assumptions of a particular approach of social research about the nature of social reality (Ansari et al., 2016) and the theory of social entities and everything that exists for research. The ontology of research reflects the researcher's attitude about the nature of the world (Maarouf, 2019) and her/his approach about what the social world constitutes and how it should be studied. In short, research ontology justifies a particular set of beliefs about the nature of social reality.

4.1.2. Epistemology

Epistemology is the theory of knowledge and perception in science (Lemos, 2020). It clarifies what is the acceptable knowledge in any system of thought and how research findings can be judged. In other words, it clarifies what makes the findings of one research more appropriate than the findings of another in the same field. Epistemology raises the question of what tests must be passed in order for data and information to be considered legitimate knowledge, and what classification of subjects can be recognised. Epistemology concerns the relation between the knower or the researcher and the real-life truth (Roots, 2007). In social science research, epistemological assumptions include empiricism, rationalism, constructivism, contractism and neorealism (Krause and Williams, 2002).

Epistemology studies the nature of knowledge and how and what can be known (Krause and Williams, 2002). It can be considered as a justification for knowledge. In short, epistemology of research refers to assumptions about how to recognise whatever is accepted. Moreover, the epistemology of this study or the knowledge claim is concerned with how the researcher gathers data for analysis and consumption (Scotland, 2012). This study deliberately focuses on objectivism, which is the assumption that reality is independent and distinct from human minds and their subjective assumptions. Objectivism provides reliability as it suggests that the results

of the research are consistent. It also provides external validity as it allows gaining replicable and applicable results in other contexts. In conclusion, the research philosophy here aims at deducing evidence from the response of many stakeholders and top managers, which is a gained based on quantitative methods.

4.2. Research Approach

Based on the methods of knowledge acquisition, the two main research approaches are the inductive and deductive research approaches (Kothari, 2004). The inductive research approach moves from specific to general (i.e., collecting specific observations and drawing general conclusions from them). The deductive research approach moves from general to specific (i.e. starting from general statements and data collection while drawing specific conclusions from them). While inductive research involves collecting and interpreting qualitative data, deductive research incorporates collecting and interpreting quantitative data (Walliman, 2011).

The Deductive approach is applicable in this study considering the broad scope that the research questions are covering, which could require receiving different treatments for the different questions (Woiceshyn and Daellenbach, 2018). The deductive approach expects the researcher to commence her/his investigation from a position where there is a theory that requires testing. This theoretical grounding is the reason why the deductive approach commences with a hypothesis that includes a null and alternative query. The idea behind this is that the new study tests whether the existing theory remains valid or not, then proposes a new set of knowledge with respect to the theory.

The inductive approach, on the other hand, concentrates on the data and the findings that are produced from it, which then validates existing theories or creates the base for new theories. This focus on data and results is the reason why the inductive approach relies on questions that confine the scope of the research into areas that are addressable to the researcher. While the deductive approach works better within the context of quantitative research, the inductive approach is mostly associated with qualitative studies. Owing to the depth of the study and the nature of the research variables, as expressed in the research questions, the researcher prefers applying the elements of the deductive approach for data collection.

4.3. Research Method

Methodology concerns how research is conducted (Katz, 2015); it is the collection and analysis of assumptions, principles, and methods of implementation in a particular approach to research.

The methodological part of the research consists of the methods, techniques, instruments, and tools for gathering and analysing data. Research methods are mediators and consequences of research practice. In simpler terms, a methodology justifies the way data is collected and analysed. Therefore, this research adopts the quantitative method in order to emphasise the gathering of larger sets of measurable data (Park and Park, 2016). This entails the collection of objective data that can easily undergo statistical analysis using data collection tools such as questionnaires. These tools deliberately design questions to stimulate answers from respondents, which fit into categories and allow the researcher to assign numerical values. This leads to produce measures that can be interpreted by the researcher in descriptive and inferential analysis.

The quantitative method is appropriate for this study for various reasons. First, it is convenient for the researcher to investigate and determine the relation between various independent and dependent variables. Second, the quantitative method allows the researcher to conduct a descriptive investigation in which subjects are considered only once to establish associations and inform the industry and policy. There is no need to go to the extent of identifying causality as happens with action research. All that is necessary at this stage for the identification of an existent to non-existent relation between the independent and dependent variables. Third, the quantitative method fits with the objective ontology that is adopted in this research with a convergent focus aiming at fulfilling the study objectives (i.e., convergent reasoning and ideas). Fourth, quantitative results drawn from a large sample are generalisable to a population. This means that the discovered relationships between the independent and dependent variables is applicable to all Jordanian industries, the middle east region and possibly to countries beyond that region. For instance, if the sample chosen in this research responds affirmatively to the question on whether the manufacturing practices has influenced the operational performance the Jordanian industrial firms, then it is possible to generalise this to other sectors in the country and possibly draw similar conclusions in other countries with similar economies and market dynamics. Hence, the quantitative method provides solutions to various challenges that the study is seeking to address, which adds to existing knowledge about the phenomenon under investigation.

The aim of this study is to measure the relation between the manufacturing practices (i.e. TPM, TQM) with the focus on the moderation effect of SCM supplier integration and internal integration on the operational performance (i.e. cost, flexibility) of the Jordanian industrial

sector. To achieve this aim, the study relies on the survey as a tool for collecting the data of this research. According to Polland (2005), the survey is a suitable way to collect information under the following conditions: (1) when the objectives are quantitative as is the case in this research; (2) when the information is specific to the respondent and (3) when the researcher has prior knowledge of possible responses.

The responses of the sample members can be obtained through both the traditional written responses and the electronic responses. This saves time and cost and allows higher response rates during the Covid-19 pandemic, which facilitated better access to the required data and more representing results of this study.

Survey research requires the researcher to be interested in developing the research tool. It requires the questionnaire to be carefully chosen and that the questions to be clear and comprehensive for respondents to clearly answer the questions to understand the relationships between variables. The population of this research consists of the Jordanian industrial firms with the sample is selected as the middle and top managers in the targeted firms.

4.4. Questionnaire Development

Survey development involves several key steps that enable the researcher to move from the point of having plain concepts and constructs to administering the survey. It happens as two broad simultaneous processes with planning happening at the same time as administrative preparation (Green and Frantom, 2002).

In planning, several steps should take place. The researcher must clarify the study purpose, decide on results usage, formulate the research questions, and create a timeline. Then, she/he should prepare the tool content, create an outline for topics, identify the constructs and consult the literature. Then, the researcher must compare her/his survey with other existing surveys or consider an already tested data tool to guide the process. Later, the researcher needs to prepare and gather the tool elements either by borrowing from existing tools or by creating her/his own. After that, the researcher should pre-test the survey to understand the robustness of the tool. Finally, the researcher must revise the survey draft and conduct a pre-test study to verify its reliability and validity before data collection.

In administration, a number of processes will also apply. First, the researcher needs to plan survey administration, which involves defining a target population and deciding on the best

way to reach it. Second, the researcher needs to determine the sample size. This involves creating a sample frame, choosing sampling strategies and selecting a sample. Third, the researcher needs to arrange connecting with the respondents by preparing survey invitations, tracking non-responsive participants and keep on communicating for the follow-up stage. Once this is complete, the researcher is in a suitable position to administer the survey.

In this research, questionnaire is used as a data collection instrument to gather the required data to reflect the participants' attitudes towards the relationship between research variables. However, there are various methods are used to determine the validity of the instrument. These methods are apparent validity, content validity, concurrent validity, predictive validity, and construct validity. Each of the above methods is used for specific purposes (Yang et al., 2014). Content validity is used to confirm the content of the instrument. Concurrent validity is used when a standard questionnaire is available. Predictive validity is used to predict a future phenomenon. Structural validity is used to structurally adapt the tool to previous theories. In order to adapt the appearance and determine the content range of the questionnaire, the method of content validity is used.

As the study adopts a survey method, the questionnaire was used as a tool for collecting data from individuals. It is important when designing the questionnaire that the researcher focuses on clarifying the independent, dependent, and intermediate variables. The survey has been carried out in a series of stages: in the beginning, the questions related to manufacturing practices (i.e., TPM, TQM), operational performance (i.e., cost, flexibility) and SCM (i.e. supplier integration, internal integration) were formulated. Then, the questionnaire was presented to a group of academics with specialisation in the same field of study to express their opinions on the formulation of the questions and suggest the necessary amendments. Later, the questions were tested with a sample of four respondents and, as a result of the test, the content of the questionnaire was revised and amended to be clearer and more comprehensive.

4.5. Questionnaire Design

When designing the questionnaire, the following recommendations were considered according to (Dillman et al., 2014):

- 1- *Simplicity*: the questions asked in the questionnaire were as simple as possible so that participants can easily understand the meaning and answer accordingly. For this, the researcher avoided mentioning any technical terms or specialised expressions that are

difficult to understand. To reduce ambiguity in the closed questions, the answers were provided as a range of standard options such as: strongly agree, agree, neutral, disagree and strongly disagree. In this type of scale, it is assumed that the middle option is a neutral point between the options. In some cases, it is possible that the construction of a response scale is not balanced.

- 2- *Length and appearance of the questionnaire:* it should be noted that the longer the questionnaire, the lower the response rate. There is no specific rule for the number of questions in the questionnaire and, thus, it depends on the intelligence and literacy of the respondents and their interest in the subject. The appearance of the questionnaire is also effective in increasing the response rate. The use of question boxes, the type of questionnaire and the clear and explicit expression of the questions help respondents to pay more attention and encourage them to response. Moreover, it is better to use the same scale throughout the questionnaire if the response scale has a certain shape, such as a -point grading scale.
- 3- *Translation process:* it requires observing its own rules and principles. It should be noted that in addition to translating the questionnaire, cultural adaptation of the meaning of the phrases is very important. This is because in order to adapt the questionnaire to the cultural context of the Jordanian society, semantic and content equality of phrases must be carried out. Appropriate translation with cultural adaptation allows comparing the concepts in both target and reference societies. The values, concepts and meanings reflected by the tool may be different from one culture to another. Therefore, the translation quality and validity of the translated version of the tool have an important role in the validation of the translated version.
- 4- *Questionnaire preparation:* the questions and instructions related to the questionnaire should be clear and understandable for respondent. The questionnaire should be written or expressed in a way that encourages respondents to participate. Instructions, questions, and tips should be designed in such a way to make participants interested in continuing their cooperation until returning the completed questionnaires.
- 5- *Purpose and problem statement:* the first step in implementing the questionnaire of the current research was to prepare a list of goals that the researcher intends to achieve. It is almost impossible to prepare a questionnaire without having a clear understanding of

what is required to be achieved. The purpose of the questionnaire is to describe the behaviour, attitudes, practices, thoughts, and feelings and also to test the relationship between two or more variables.

- 6- *Determining the community and selecting the sample:* determining the community in most cases is done based on several points such as subject importance, generalizability and the researcher interest and information resources. In this study, exploring the full research context is not easily possible due to the large community size. In addition, to describe the characteristics of the community and test the related hypothesis, it is not necessary to study all the members of the community. Instead, sample members were selected from the community to be treated as representatives. Before distributing the questionnaire in its final form to the sample, the researcher ensured that the design of the questionnaires is following the advice obtained from the experimental sample. This was achieved through modifying the questions to design a clear questionnaire so that the sample members can easily answer.

4.6. Pilot Study

A pilot study is a feasibility pre-research study that a researcher conducts with the intention to testing and verifying the strength of the data collection instruments (Arain et al., 2010; Lowe, 2019).). It is also a useful way to test the reliability of the target population (In, 2017). It is a way of trying to tease out any possible problems that may arise when the main survey begins. In this ‘rehearsal’ survey, the researcher gets an opportunity to test the methods, design, tools, personnel, target, and population. Piloting enables the researcher to test some important aspects of the study. First, she/he gets to test the sample and its size and whether the recruitment methods are manageable and practical. Second, she/he gets to test the collected data at analysis stage to determine whether the proposed approach is suitable to the study and whether results are reasonable in the given context. Finally, the results of the pilot test enable the researcher to check the reliability and validity of the data collection’s tool. However, through a Cronbach’s-Alpha test, the researcher will be able determining whether the questions in the survey are appropriate to be in the survey and fit with each other. The test indicates which question is weakening or strengthening the entire questionnaire, enabling the researcher to alter, improves or eliminates the problem questions before the main survey takes place. The pilot questionnaire was distributed among the Jordanian industrial firms and has received responses from all four firms, which allowed bettering developing the main questionnaire.

Cronbach's alpha measures the stability of values, as it is clear from the empirical study the study that the values and that they confirm the internal consistency of the elements/items. To ensure the reliability of the study, Cronbach's Alpha test was conducted in order to ensure that the values give reliable results for the research, where the values should be ≥ 0.6 (Taber, 2018), Table 4-1 show the pilot study figures.

Variables	TPM	TQM	SCM - SI	SCM - II	Cost	Flexibility
Original no. of items	5	9	15	7	4	6
Refined no. of items	4	4	8	4	3	3

Table 4-1: Pilot study

4.7. Sample and Data Collection

The target population for this study consists of managers in the Jordanian industrial firms who have a rich knowledge about people, processes, problems, and solutions that affect the industrial sector. The data about the industrial firms where those managers work was obtained from the Ministry of Industry, Trade and Supply and Amman Chamber of Industry. Amman Chamber of Industry has significant contribution in this research. The chamber supported the research in data collection phase. The names and number of industrial companies working in Jordanian economy are provided. The study incorporated SMEs and MNCs working in Jordan. The reason for incorporation of these two different types of organisations is to highlight the effects of manufacturing and supply chain practices on the operational performance of all organisations operating in Jordan, so that the true scenario can be depicted. The researcher was given the approval for data collection. Then the researcher adopted random sampling technique. The employees of the organisations working in Jordanian industries were randomly targeted as a target population of this study. Then the consent of these employees/managers is taken. After that, the researcher targeted those managers who are willing to take the online survey. The researcher relies on the hosting portal of the online survey to stratify the sample as it has built-in stratification features that enable the sample to reflect the targeted population; thereby, increasing the credibility and enabling the generalisability of the obtained findings (Acharya et al., 2013). As the online questionnaire was not enough to collect the data needed for this research due to a relatively poor response rate, the questionnaire was traditionally distributed through visiting the targeted industrial firm and collecting their responses. After excluding eight incomplete responses, 224 completed responses were received from the

targeted managers as a result of distributing both the traditional and online questionnaires (Table 4-2).

Firm size	Frequency	%
10-49	114	50.9
50-250	73	32.6
251-1000	18	8.0
Over1000	19	8.5
Total	224	100
Annual of firm	Frequency	%
less than 1 million	120	53.6
1-7 million	83	37.1
More than 7 million and less than 40 million	20	8.9
More than 40 million	1	0.4
Total	224	100
Revenue of firm	Frequency	%
less than 1 million	119	53.1
1-7 million	56	25.0
More than 7 million and less than 40 million	49	21.9
Total	224	100
Industries	Frequency	%
pharmaceutical and medical	36	16.1
Chemical	51	22.8
Construction	23	10.3
engineering and metal	49	21.9
electrical and automotive	25	11.2
textiles, leathers, and clothing	19	8.5
Food	21	9.4
Total	224	100

Table 4-2: Sample Characteristic.

4.8. Non-response Bias

The non-response bias occurs when some of the participants in a survey are distinctly different from the rest of the sample, which makes them neglect the study because it is not applicable to them (for example, old people in a survey about night clubs) (Sedgwick, 2014). This bias is usually a result of poor targeting methods that mislead researchers to individuals who may not be suitable for the study. In some instances, the survey may not reach its intended target (for example, an invite email that ends up in a potential respondent's spam folder). To minimise non-response bias, the researcher ensured that the targeting methods are precise (for example,

using direct social media channels such as WhatsApp to reach participants, share links to the hosted survey and maintain a good rapport thereafter to stimulate good answers). Furthermore, all questions were close ended in keeping with the quantitative design, which respondents find easier to handle than open-ended questions.

The **Kolmogorov-Smirnov** test is a test for the dependent variables. Here the test was extracted for all the variables, and they should be non-statistically significant:

	tpm1	tqm1	suppl1	flex1	inter1	cost1
N	224	224	224	224	224	224
Normal Mean	5.1763	5.4464	4.5965	5.3244	4.8906	5.3854
Parameters ^{a,b} Std. Deviation	1.12086	1.05320	1.4222	1.11068	1.20945	1.23508
Most Extreme Absolute	.138	.142	.084	.153	.083	.115
Differences Positive	.067	.070	.046	.070	.055	.101
Negative	-.138-	-.142-	-.084-	-.153-	-.083-	-.115-
Kolmogorov-Smirnov Z	2.059	2.131	1.250	2.284	1.238	1.717
Asymp. Sig. (2-tailed)	.000	.000	.088	.000	.093	.005

Table 4-3: One-Sample Kolmogorov-Smirnov Test

Statistics from the Mann-Whitney test for all variables are described as shown in the following table 4-4:

tpm1		N	Mean Rank	Sum of Ranks	Test Statistics	cost1	flex1
cost1	3.00	5	5.30	26.50	Mann-Whitney U	11.500	1.500
	7.00	9	8.72	78.50	Wilcoxon W	26.500	16.500
	Total	14			Z	-1.510-	-2.851-
flex1	3.00	5	3.30	16.50	Asymp. Sig. (2-tailed)	.131	.004
	7.00	9	9.83	88.50	Exact Sig. [2*(1-tailed Sig.)]	.147 ^a	.002 ^a
	Total	14					
tqm1							
cost1	To some extent	3	7.00	21.00	Mann-Whitney U	15.000	5.500

	To a very great extent	15	10.00	150.00	Wilcoxon W	21.000	11.500
	Total	18			Z	-.936-	-2.061-
flex1	To some extent	3	3.83	11.50	Asymp. Sig. (2-tailed)	.349	.039
	To a very great extent	15	10.63	159.50	Exact Sig. [2*(1-tailed Sig.)]	.426 ^a	.039 ^a
	Total	18					
inter1							
cost1	3.00	9	9.33	84.00	Mann-Whitney U	39.000	20.000
	7.00	14	13.71	192.00	Wilcoxon W	84.000	65.000
	Total	23			Z	-1.564-	-2.836-
flex1	3.00	9	7.22	65.00	Asymp. Sig. (2-tailed)	.118	.005
	7.00	14	15.07	211.00	Exact Sig. [2*(1-tailed Sig.)]	.141 ^a	.005 ^a
	Total	23					
suppl1							
cost1	3.00	3	4.83	14.50	Mann-Whitney U	8.500	7.500
	7.00	11	8.23	90.50	Wilcoxon W	14.500	13.500
	Total	14			Z	-1.560-	-1.502-
flex1	3.00	3	4.50	13.50	Asymp. Sig. (2-tailed)	.119	.133
	7.00	11	8.32	91.50	Exact Sig. [2*(1-tailed Sig.)]	.225 ^a	.170 ^a
	Total	14					

Table 4-4: One-Sample Mann-Whitney U Test

It is indicated from the table that when the independent variable is TPM, and the dependent variables are COST and FLEXIBILITY; the value of the Asymp. Sig. for the first column was 0.131, which is not statistically significant, i.e., we accept the null hypothesis. As for the second column, it was 0.004, which is a statistical function, that is, we reject the null hypothesis and accept the alternative, meaning there are statistically differences. When the independent variable TQM was on the dependent variables it is indicated that the cost is not statistically significant, while the flexibility is statistically significant. When the independent variable internal is on the dependent variables we note that the cost is not statistically significant, but for the flexibility, the value was 0.005. When the independent variable is supplier integration

on the functions, it is indicated that both; cost and flexibility are not statistically significant. The Mann Whitney test is for samples that are less than 50 small samples.

4.9. Common method Bias

Common method bias describes the measurement error that is compounded by the sociability of respondents who want to provide positive answers (Chang, v. Witteloostuijn and Eden, 2010). This research collected data on both the independent and dependent variables from the same respondents at one point in time, thus raising potential common method variance as false internal consistency might be present in the data. therefore, it is essential to check the percentage of variance through the common method bias test using Harman's single-factor.

There is no problem with the common method bias test using Harman's single-factor in this data since the total variance extracted by one factor is 30.418% and it is less than the recommended threshold of 50%.

Total Variance Explained						
Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	7.956	30.601	30.601	7.956	30.601	30.601
2	2.592	9.968	40.570			
3	1.630	6.270	46.840			
4	1.473	5.667	52.507			
5	1.231	4.734	57.241			
6	1.105	4.249	61.490			
7	.895	3.441	64.931			
8	.866	3.331	68.262			
9	.757	2.912	71.174			
10	.735	2.827	74.001			
11	.652	2.507	76.508			
12	.618	2.377	78.884			
13	.610	2.344	81.229			
14	.549	2.113	83.341			
15	.508	1.953	85.294			
16	.492	1.892	87.187			
17	.469	1.805	88.992			
18	.440	1.692	90.684			
19	.410	1.576	92.260			
20	.377	1.450	93.710			
21	.330	1.271	94.981			
22	.318	1.222	96.203			
23	.276	1.061	97.264			
24	.258	.993	98.257			
25	.229	.880	99.138			
26	.224	.862	100.000			

Table 4-5: Common Method Bias

4.10. Measures

To safeguard the credibility of this research process, the following measures were pursued by the researchers (Dooly et al., 2017). First, an invitation message was sent to respondents that contain a detailed description of their rights and duties regarding the research. Second, consent was embedded within the questionnaire that participants must approve before they begin answering the survey questions. Third, the questionnaire was maintained confidential, and the identities of participants were kept as anonymised. Forth, the respondents were asked to indicate the extent of their agreement with statements according to the actual situation using a

Likert scale where 1 indicates ‘strongly disagree’ and 7 indicates ‘strongly agree’. Theoretical literature has been used to develop resolution measures (Table 4-6).

Constructs	Measuring items	Main Supporting References
TPM	TPM was studied four items on (1-7) Likert scale “Completely dissatisfied” to “completely satisfied” (Madanhire, and Mbohwa, 2015)	(Madanhire, and Mbohwa, 2015); (Nakajima, 1988); (Ahuja, 2008)
TQM	TQM was studied four items (1-7) Likert scale “completely dissatisfied” – “completely satisfied” (Albuhisi, Abdallah 2018)	(Albuhisi and Abdallah, 2018); (Yeh, 2011); (Al Fayez, 2014); (Mafini and Poove, 2013); (Flynn et al., 2010)
Supplier Integration	Supplier integration was studied four items (1-7) Likert scale “not at all” – “To a very great extent”.(Flynn, 2011).	(Flynn et al., 2010); (Craighead et al., 2007)
Internal Integration	Internal integration was studied four items (1-7) Likert scale “not at all” – “To a very great extent”. (Flynn, 2011).	(Flynn et al., 2010); (Blackhurst et al., 2005) (Elkins et al., 2005)
Cost	Cost was studied four items (1-7) Likert scale from “No emphasis” – “extreme emphasis”. (Ward and Duray, 2000).	(Hitt, Wu and Zhou, 2014); (Sharples, 2009); (Ostwald and McLaren, 2004); (Bener and Glaister, 2010); (Ward and Duray, 2000)
Flexibility	Flexibility was studied four items (1-7) Likert scale from “strongly disagree” – “strongly agree”. (Flynn, 2011).	(Flynn et al., 2010); (Mendes and Machado, 2015); (Vokurka, Lummus, and Krumwiede 2007)

Table 4-6: Support Literature

4.11. Chapter Summary

This chapter has discussed the leading philosophies and approaches in management studies and selected the most suitable ones for this research. The chapter also clarified the methods and techniques on how the research community and its sample members were identified and approached. The chapter further described the design and process used to prepare the survey as a data collection tool that fits with the quantitative method that was adopted in this research. In the next chapter, the focus is on explaining the steps pursued to conduct the descriptive data analysis of the research.

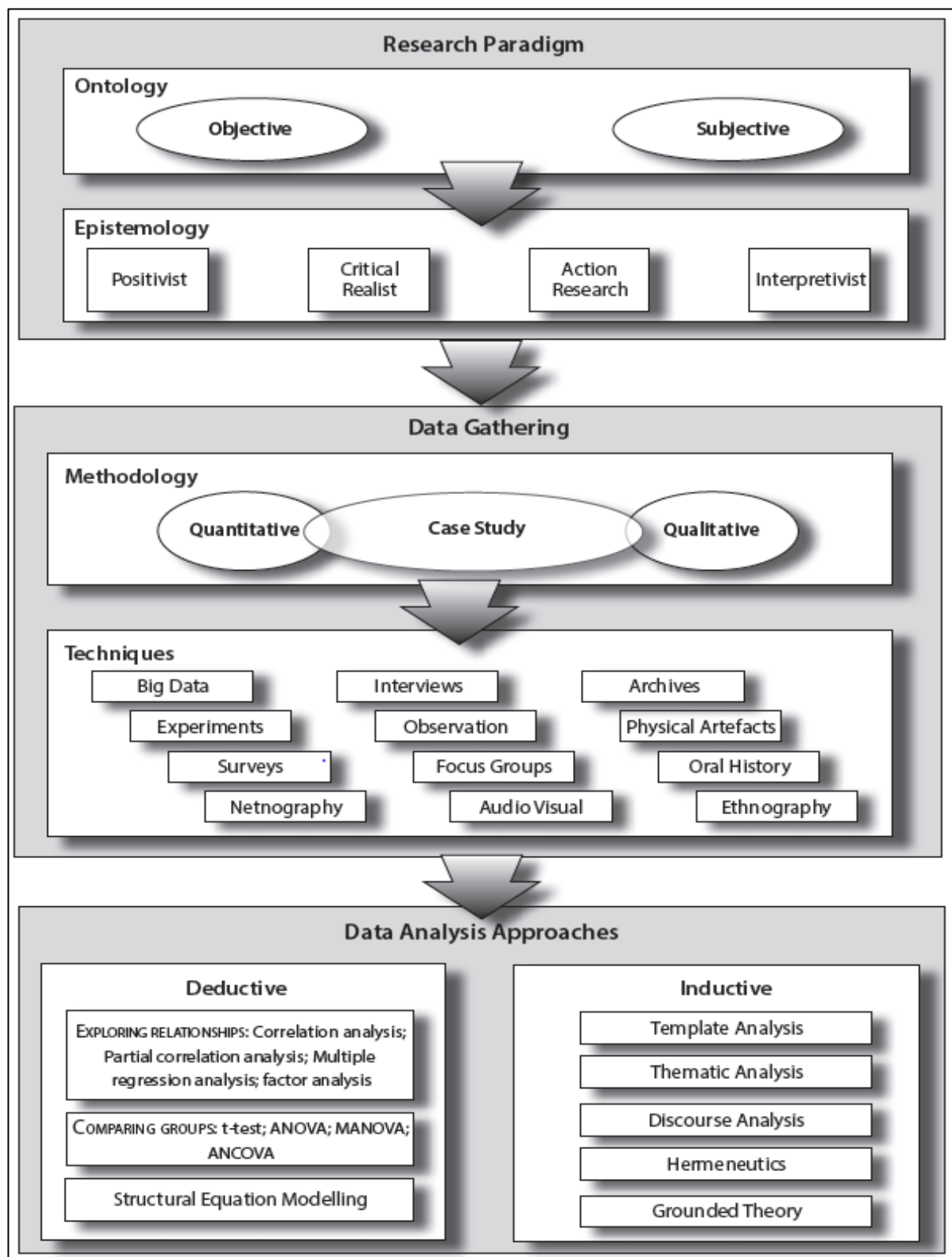


Figure 4-1: Research Methods for Business and Management (O'Gorman and MacIntosh, 2014).

Chapter 5 Data Analysis

The main aim of this research is scrutinizing the effect of manufacturing practices (TPM, TQM) on operational performance (cost, flexibility) when moderated by SCM (supplier integration, internal integration). To attain this aim, the study concentrates on compelling data through developing a questionnaire of twenty-two items. This questionnaire was distributed to a sample of 441 industrial firms and achieved a response rate of 51%; in that, 243 responses were obtained, and after excluding nineteen non-analysable questionnaires, 224 were fit for statistical analysis. These questionnaires included the research data that was analysed through the 'statistical package for social sciences (SPSS)' and AMOS quantitative analysis software. This package was used to conduct the various tests of validity, stability, normality, exploratory and confirmatory factor analysis, and path analysis that were employed to examine the collected data related to the research variables. By doing so, the chapter discusses the descriptive analysis of the research data to extract the results and test the hypotheses that are later presented.

5.1. Missing Data and Outliers

The collected data from the Jordanian industrial firms was analysed by different tests to obtain results about the study hypotheses. It was subjected to different types of tests and focused on the completed questionnaires to obtain accurate results. When the design of the questionnaire was made, the researcher tried to develop questions that may allow representing answers to obtain the needed data to reach conclusions. For that, some questions were excluded and some of which were modified to be easier and clearer. Every invalid questionnaire was excluded due to its incompleteness, the refusal of some respondents to provide sufficient information about their firms (for example, information about cost, revenue, size) and the difficulty in reaching some firms during the covid-19 pandemic. The completed questionnaires were used to allow accurate results by using a 7-point Likert scale. This design, through conducting the exploratory factor analysis, helped to check for outliers. As a result, no outliers were found which increased the validity of the data and the obtained results.

5.2. Descriptive Statistics

Before starting the data analysis, the researcher tried to extract the arithmetic averages, standard deviations, range and kurtosis to know the cause-effect relationships between the variables (i.e.,

independent, dependent, and moderator) and to identify the impact of each of them. However, table 5.1 provides a summary of the different types and values of variables.

Variable Type	Variable Name	Scale Values
Independent variables	TPM	Four components of the variable were measured
	TQM	Four components of the variable were measured
	Supplier integration	Eight components of the variable were measured
	Internal integration	Four components of the variable were measured
Dependent variables	Cost	Three components of the variable were measured
	Flexibility	Three components of the variable were measured

Table 5-1: Summary of Variable Types and Values.

5.2.1. Independent Variables

As previously explained, two independent variables for manufacturing practices were employed (TPM, TQM). This section describes how the industrial firms perceive their ability to apply manufacturing practices to its dimensions (TPM, TQM).

- ***Total production maintenance (TPM)***

The first independent variable in this research is TPM, which contains four items. While table 5.2 shows the descriptive analysis related to all TPM items.

Items	Mean	Std. Deviation
“Decision making, problem solving and job design”	5.35	1.30

“Recruitment, selection and promotion”	5.06	1.56
“Performance appraisals”	5.22	1.44
“Management support system in the form of motivation and incentive compensation system that recognises and rewards employee on merit”	5.12	1.59

Table 5-2: Descriptive Analysis – TPM.

- ***Total Quality Management (TQM)***

The second independent variable in this research is TQM, which contains four items. While table 5.3 shows the descriptive analysis related to all TQM items.

Items	Mean	Std. Deviation
“Employees are actively involved in improving products, services, and processes”	5.38	1.38
“Employees are encouraged to fix problems they find”	5.42	1.34
“Training in specific work skills (technical and vocational) given to employees throughout the company”	5.44	1.49
“Managers and supervisors ensure that all employees receive training that helps them understand how and why the company does what it does”	5.52	1.31

Table 5-3: Descriptive Analysis -TQM.

- ***Supplier Integration***

The first moderation variable in this research is supplier integration, which contains four items. While table 5.4 shows the descriptive analysis related to all supplier integration items.

Items	Mean	Std. Deviation
The participation level of our major supplier in the process of procurement and production	4.73	1.69
The participation level of our major supplier in the design stage	4.79	1.73

Our major supplier shares their production schedule with us	4.40	1.87
Our major supplier shares their production capacity with us	4.60	1.85
Our major supplier shares available inventory with us	4.21	1.98
We share our production plans with our major supplier	4.62	1.90
We share our demand forecasts with our major supplier	4.76	1.73
We share our inventory levels with our major supplier	4.63	1.80

Table 5-4: Descriptive Analysis – Supplier Integration.

- ***Internal Integration***

The second moderation variable in this research is internal integration, which contains four items. While table 5.5 shows the descriptive analysis related to all internal integration items.

Items	Mean	Std. Deviation
“Data integration among internal functions”	5.04	1.46
“Enterprise application integration among internal functions”	4.46	1.81
“Integrative inventory management”	4.98	1.53
“Real time searching of the level of inventory”	5.06	1.41

Table 5-5: Descriptive Analysis – Internal Integration.

5.2.2. Dependent Variables

As discussed in the previous chapters, two dependent variables were selected for this study: cost and flexibility. In this section, an explanation of the descriptive analysis results of the dependent variables is provided in order to detect the role of these variables in influencing the performance of industrial firms.

- ***Cost***

The first dependent variable in this research is cost, which contains three items. While table 5.6 shows the descriptive analysis related to all cost items.

Items	Mean	Std. Deviation
“Reduce inventory”	5.29	1.58
“Increase capacity utilisation”	5.45	1.57
“Increase equipment utilisation”	5.40	1.61

Table 5-6: Descriptive Analysis – Cost.

- **Flexibility**

The second dependent variable in this research is flexibility, which contains three items. While table 5.7 shows the descriptive analysis related to all flexibility items.

Items	Mean	Std. Deviation
“Our company can quickly introduce new products into market”	5.33	1.44
“Our company can quickly respond to changes in market demand”	5.28	1.31
“The lead time for fulfilling customers’ orders (the time which elapses between the receipt of customers’ order and the delivery of the goods in short”	5.35	1.51

Table 5-7: Descriptive Analysis – Flexibility.

5.2.3. Control variables

This section discusses the control variables, which are utilized in this research. These variables are including the firm size and the industry type.

- **Firm Size**

As described in chapter five, firm size was investigated by using the four-point Likert scale, ranging from ‘10-49’ to ‘over 1000’ (Table 5.2). Respondents were asked about the number of employees in their firms. Table 5.8 shows descriptive analysis of this control variable.

Items	Mean	Std. Deviation
Firm Size	1.74	0.93

Table 5-8: Descriptive Analysis – Firm Size.

- **Industry Type**

Seven options were used to investigate the industry type (i.e., pharmaceutical and medical, engineering and metal, chemical, construction, electrical and automotive, textiles, leathers and clothing, food. Table 5.9 summarises the descriptive analysis of this control variable.

Industry Type	Frequency	%
pharmaceutical and medical	36	16.1
Chemical	51	22.8
Construction	23	10.3
engineering and metal	49	21.9
electrical and automotive	25	11.2
textiles, leathers and clothing	19	8.5
Food	21	9.4
Total	224	100

Table 5-9: Descriptive Analysis – Industry Type.

5.3. Normality, Validity and Reliability

In order to obtain accurate results, and before conducting hypotheses testing, a set of tests were conducted to verify and ensure that the data entered into the analysis software is valid for analysis and enough to fulfil the objective of the study.

5.4.1. Normality

The normal distribution is a symmetrical distribution around the mean; it shows that the data close to the mean is more frequent than the data far from the mean. It can be confirmed that the data was subjected to a normal distribution by skewness and kurtosis. The value of deviation was acceptable for the set of test scores normally distributed as it is close to zero, and when it deviates from zero, the positive value indicates the possibility of a positive skewed distribution, and when it is negative, it indicates the probability of a negative skewed distribution, and the kurtosis values are considered to represent a normal distribution when the values are close to one (Brown, 1997).

The normal distribution of all variables tested to use the statistical analysis, firstly, the independent variables that describe two components were selected (TPM, TQM, Supplier integration, Internal integration). In table 5.10, the values of the mean, standard deviation, skewness, kurtosis. This means that it follows a normal distribution.

Independent Variable	Mean	Std. Deviation	Skewness	Kurtosis
TPM	5.17	1.12	-.47	-.60
TQM	5.44	1.05	-.74	.05
Supplier integration	5.59	1.42	-.25	-.52
Internal integration	4.89	1.20	-.32	-.24

Table 5-10: Normality – Independent Variables.

Secondly, the dependent variables that describe two the components that were selected (cost, flexibility). In table 5-11, the values of the mean, standard deviation, skewness.

Independent Variable	Mean	Std. Deviation	Skewness	Kurtosis
Cost	5.38	1.23	-.47	.12
Flexibility	5.32	1.11	-.72	.16

Table 5-11: Normality–Dependent Variables.

Finally, for the control variables in table 5.12, the kurtosis value of the firm size indicates a form close to the normal distribution. It can be mentioned that the firm size is normally and reasonably distributed because it is fixed.

Independent Variable	Mean	Std. Deviation	Skewness	Kurtosis
Firm Size	1.74	0.93	1.17	0.494

Table 5-12: Normality – Firm Size.

5.4.2. Reliability and Validity

For the purpose of verifying the stability of the study measures, the internal consistency method was used by applying Cronbach's Alpha coefficient to test the internal stability of the study tool. It is known that the stability values of the variables must be greater than 0.6 (Pallant, 2005), table 5.17 shows the acceptable results of the reliability test for the study variables.

5.4.3. Exploratory Factors Analysis (EFA)

EFA is employed for the reduction of number of variables to a small number named as factors. This mathematical process is employed aiming at simplifying the correlations among different

variables under study (Plucker, 2003). Initially in EFA factors are extracted through different extraction techniques while PCA (principal component analysis) is among the most widely employed extraction techniques that generate principal factors, so current research has also employed EFA on 224 observations and 22 items.

The main focus of EFA is the validity of tools incorporated in the study, to ensure sufficient loading of factors of each variable of interest. For this the acceptable factor value is greater than 0.5 and for Eigen value it should be greater than 1 (Laher, 2010). Results of EFA for current research are reported in table 5.15.

Exploratory factor analysis could be performed in two ways:

1. First method incorporates exploring and obtaining the number of factors that affect the variables of interest as well as identifying the number of factors due for that variable. For example, incorporating all the information in analysis for obtaining new factors by restructuring the study questionnaire and then analysing it again to obtain new results. Then making comparison of the results of initial questionnaire and the second questionnaire.
2. In the second method, all paragraphs associated with variables are entered separately, while ensuring that paragraphs associated with one variable refer to that particular variable. It confirms the credibility and validity of the construction of associated paragraphs ensuring that the values for all parameters of the clauses should be greater than 0.4. The current research has incorporated first method which is as follows:

	Component					
	Supplier	Internal	TPM	TQM	Flexibility	Cost
TPM 01 decision making, problem solving and job design	.144	.066	.680	.232	.024	.034
TPM 02 recruitment, selection, and promotion	.172	.353	.601	.117	.104	.029
TPM 03 performance appraisals	.172	.041	.738	.191	.200	.018
TPM 04 management support system in the form of motivation and incentive compensation system that recognizes and rewards employee on merit	.074	.035	.733	.109	.196	.202
TQM 01 EI employees are actively involved in improving products, services, and processes	.160	-.034-	.147	.787	.099	.042
TQM 02 EI employees are encouraged to fix problems they find.	.120	.294	.078	.648	.173	.028
TQM 03 T&D training in specific work skills (technical and vocational) given to employees throughout the company.	.052	.071	.316	.676	.209	.026
TQM 04 T&D managers and supervisors ensure that all employees received training that helps them understand how and why the company does what it does	.131	.281	.204	.630	.120	.136
SI 01 The participation level of our major supplier in the process of procurement and production	.724	.178	.173	.142	-.038-	.064
SI 02 The participation level of our major supplier in the design stage	.690	.188	.233	-.058-	.000	.037

SI 03 Our major supplier shares their production schedule with us	.795	.232	.092	.051	-.030-	.017
SI 04 Our major supplier shares their production capacity with us	.758	.022	.075	.112	.084	.112
SI 05 Our major supplier shares available inventory with us	.715	.208	.027	.102	.292	.055
SI 06 We share our production plans with our major supplier	.779	.125	.014	.091	.079	.127
SI 07 We share our demand forecasts with our major supplier	.728	.135	.078	.089	.269	.133
SI 08 We share our inventory levels with our major supplier	.759	-.001-	.113	.152	.125	.078
II 01 Data integration among internal functions	.200	.631	-.033-	.276	.092	.050
II 02 Enterprise application integration among internal functions	.401	.665	.032	.165	.158	.029
II 03 Integrative inventory management	.125	.787	.172	.023	.070	.110
II 04 Real-time searching of the level of inventory	.181	.648	.212	.093	.278	.076
Flex 1 Our company can quickly introduce new products into the market	.178	.105	.135	.248	.631	.098
Flex 2 Our company can quickly respond to changes in market demand	.160	.114	.129	.189	.761	.088
Flex 3 The lead time for fulfilling customers' orders (the time which elapses between the receipt of customer's order and the delivery of the goods) is short	.067	.259	.218	.097	.658	-.053-

COST 01 Reduce inventory	.179	-.012-	.009	.030	.198	.747
COST 02 Increase capacity utilization	.134	.041	.068	.022	.046	.837
COST 03 Increase equipment utilization	.056	.186	.147	.118	-.114-	.664

Table 5-13: Factor Extraction.

It is noted from the previous table that all values were acceptable, as they were higher than 0.50, so it can be said that the test results for EFA they were acceptable.

- **KMO for all Variables**

The tests of adequacy and relevance of data are considered, so the value of Kaiser-Meyer-Olkin “KMO” must be greater than 0.50, and the value of Bartlett's sig. must be less than a 0.05 value (Babae, 2010).

KMO and Bartlett's Test		
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.875
Bartlett's Test of Sphericity	Approx. Chi-Square	2419.426
	df	325
	Sig.	.000

Table 5-14:KMO and Bartlett's Test.

It is clear from the previous tables that all values were acceptable because the Kaiser-Meyer value 0.87 the value high, and significant.

Through the values shown in the previous tables, it is clear that the KMO value was 0.87 and the Bartlett' value was 0.0001, which are acceptable. However, table 6.13 shows that after exploring the factors, which are affecting the variables and identifying the number of factors to which the variables are due (i.e. entering all items in the analysis at once to obtain new factors based on the variables) (Decoster, 1998), all variables were confirmed to pass the factor loading test since the values must be greater than 0.5. In detail, the six variables in the study were included in the factor loading test and all of them have passed: Four independent variables (TPM, TQM, Supplier Integration, and Internal Integration), two dependent variables (cost. flexibility).

It is clear from the previous tables that all values were acceptable, and this confirms the validity of the construction of the study tool and that the study paragraphs are suitable for factor analysis and, thus, confirms the sufficiency of the sample.

5.4.4. Confirmatory Factor Analysis (CFA)

Based on the results of the exploratory factor analysis (EFA), the confirmatory factor analysis (CFA) was used to verify the structural validity of the scale. This procedure is common in the confirmatory factor analysis to determine the selected model. Thus, if the confirmatory factor analysis achieves the assumed model of measurement, the conformity quality indicators are achieved. It is possible to judge the sincerity of its expressions or its dimensions, and in this analysis, the saturations with which it is reduced are rejected regression weights; validity or saturation coefficients on the latent variable should be about 0.5 (Hair et al., 2010).

Confirmatory Factor Analysis:

			Estimate
TPM 01	<---	TP	.597
TPM 02	<---	TP	.628
TPM 03	<---	TP	.746
TPM 04	<---	TP	.668
TQM 01	<---	TQ	.634
TQM 02	<---	TQ	.631
TQM 03	<---	TQ	.681
TQM 04	<---	TQ	.700
SI 01	<---	SUPP	.725
SI 02	<---	SUPP	.672
SI 03	<---	SUPP	.780
SI 04	<---	SUPP	.727
SI 05	<---	SUPP	.758
SI 06	<---	SUPP	.769
SI 07	<---	SUPP	.767

			Estimate
II 01	<---	Inter	.585
II 02	<---	Inter	.787
II 03	<---	Inter	.684
II 04	<---	Inter	.669
SI 08	<---	SUPP	.734
FLEX 01	<---	FLEX	.638
FLEX 02	<---	FLEX	.702
FLEX 03	<---	FLEX	.591
COST 01	<---	COS	.636
COST 02	<---	COS	.782
COST 03	<---	COS	.522

Table 5-15: Estimate for Variables

Table 5.16 shows the results of the confirmatory factor analysis using AMOS software, were according to Arbuckle (2012), the cut-off indices for good fit should be:

$$Chi^2/DF < 5, GFI > 90, CFI > 90, RMSEA < 0.08, AVE \geq 0.5$$

Chi ²	DF	Chi ² /DF	CFI	IFI	RMSEA
454.150	284	1.59	.923	.924	0.05

Table 5-16: The Results of the Confirmatory Factor Analysis.

As it can be clearly seen in the above table, all the tested values were appropriate, which means that the confirmatory validity of the study tool items was acceptable. Therefore, the results of the confirmatory factor analysis confirm that the research model is fit. As both the exploratory and confirmatory factor analysis tests have been done and the results were acceptable, it is now the time to test the reliability of the variables and factors. For that, the Cronbach's Alpha test was conducted to check which variables and factors are reliable to include in the research and which should be excluded (Table 5.17).

Construct	Indicator	Item loadings (from CFA)	Cronbach's alpha	Composite Reliability
-----------	-----------	-----------------------------	------------------	-----------------------

TPM	TPM1	0.597	.850	.65
	TPM 2	0.628		
	TPM 3	0.746		
	TPM 4	0.668		
TQM	TQM 1	0.634	.854	.76
	TQM 2	0.631		
	TQM 3	0.681		
	TQM 4	0.7		
SI	SI 1	0.725	.928	.89
	SI 2	0.672		
	SI 3	0.78		
	SI 4	0.727		
	SI 5	0.758		
	SI 6	0.769		
	SI 7	0.767		
	SI 8	0.734		
II	II 1	0.585	.865	.78
	II 2	0.787		
	II 3	0.684		
	II 4	0.669		
Cost	Cost 1	0.636	.835	.69
	Cost 2	0.782		
	Cost 3	0.522		
Flexibility	Flex 1	0.638	.835	.68
	Flex 2	0.702		
	Flex 3	0.591		

Table 5-17: Reliability Results of the Research Variables.

It is clear from the above table that the values of loadings for items of TPM range between 0.59-0.74 which are in acceptable range, while the values for Cronbach's alpha (0.85) and the composite reliability (0.65) are also in acceptable ranges. The values of loadings for items of TQM range between 0.63-0.70 which are also in acceptable range, while the values for Cronbach's alpha (0.854) and the composite reliability (0.76) are also in acceptable ranges. The values of loadings for items of supplier integration range between 0.67-0.78 which are in acceptable range, while the values for Cronbach's alpha (0.928) and the composite reliability (0.89) are also in acceptable ranges. The values of loadings for items of internal integration range between 0.58-0.78 which are in acceptable range, while the values for Cronbach's alpha (0.865) and the composite reliability (0.78) are also in acceptable ranges. The values of loadings for items of cost range between 0.52-0.78 which are in acceptable range, while the values for Cronbach's alpha (0.835) and the composite reliability (0.69) are also in acceptable ranges. The values of loadings for items of Flexibility range between 0.59-0.70 which are in

acceptable range, while the values for Cronbach's alpha (0.835) and the composite reliability (0.68) are also in acceptable ranges.

It is clear from the above table that the validity and reliability values for all study items were acceptable, and this means that the values are suitable for analysis and extraction of results. With this reliability test is completed, it is now clear that all the needed tests to check the normality, validity and reliability of the research data and variables were conducted and all of them were acceptable. This allows moving on to the next testing step that includes path analysis for test hypothesis.

5.4.5. Path Analysis

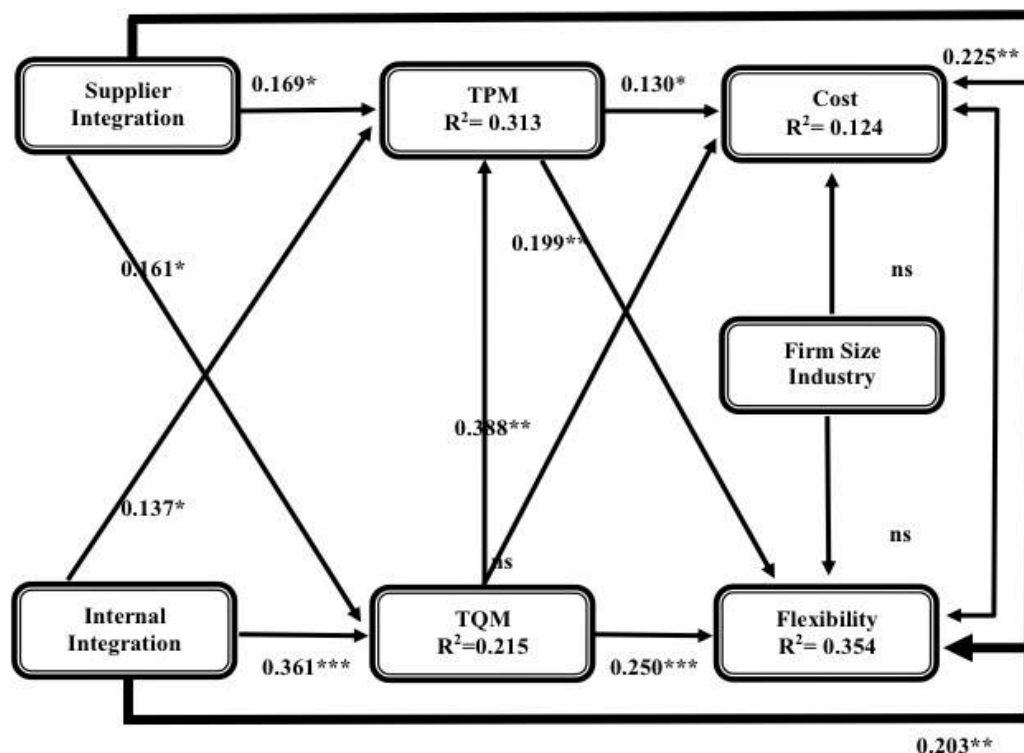


Table 5-18: Results of Structural Model

			P
TQM	<---	SI	.019
TQM	<---	II	***
TPM	<---	SI	.010
TPM	<---	II	.045
TPM	<---	TQM	***
Cost	<---	firm size	.106

Flexibility	<---	firm size	.300
Cost	<---	TQM	.567
Flexibility	<---	TQM	***
Flexibility	<---	TPM	.002
Cost	<---	TPM	.085
Flexibility	<---	SI	.067
Cost	<---	SI	.004
Flexibility	<---	II	.003
Cost	<---	II	.602
Cost	<---	industry	.984
Flexibility	<---	industry	.132

Table 5-19: Result of Regression

†p < 0.1; *p < 0.05; ** p < 0.01; *** p < 0.001 NS—not significant

Notes: Level of Confidence for Confidence Intervals: 95; number of Bootstrap Resample: 5,000 There is a good model fit, with acceptable values - $X^2/df = 1.462$ (p=ns); CFI = 0.993; RMSEA = 0.045; GFI = 0.992; IFI = 0.993; NNFI = 0.961.

5.5. Hypotheses Test Results

The following table indicates the results of proposed hypotheses of current research indicating that which proposed relationships are supported or not supported?

Hypotheses	Result
H1a: TPM has a positive effect on cost performance.	Support
H1b: TPM has a positive effect on flexibility performance.	Support
H2a: TQM has a positive effect on cost performance.	Not support
H2b: TQM has a positive effect on flexibility performance.	Support
H3: TQM has a positive effect on TPM.	Support
H4a: Supplier integration has a positive effect on TPM.	Support
H4b: Internal integration has a positive effect on TPM.	Support
H5a: Supplier integration has a positive effect on TQM.	Support
H5b: Internal integration has a positive effect on TQM.	Support
H6a: Supplier integration has a positive effect on cost performance.	Support
H6b: Supplier integration has a positive effect on Flexibility performance	Support
H7a: Internal integration has a positive effect on cost performance.	Not support
H7b: Internal integration has a positive effect on Flexibility performance	Support

Table 5-20: Hypothesis Test

Results reported in the table indicate that among the thirteen proposed hypotheses only two are not supported i.e., TQM and internal integration are not found to have significant relationship with cost performance of the organisation. It means if organisations adopt TQM and go for internal integration then such decisions would not affect the cost performance of organisational operations. Further two hypotheses are marginally supported i.e., the relationship of TPM with cost performance and the relationship of supplier integration with flexibility performance are marginally supported. It means although these relationships exist but TPM and supplier

integration influence cost performance and flexibility performance of the organisation respectively but to some extent only. Moreover, the other nine hypotheses are supported indicating the relationships of variables in a way that cost performance is found to be positively affected by supplier integration while flexibility performance is found to be positively affected by TPM, TQM and internal integration. On the other hand, TPM is found to be positively affected by TQM while supplier integration and internal integration are found to have positive effect on TPM and TQM. All these findings indicate the interrelationships of variables in a way that the operational performance of organisations is influenced by manufacturing practices and supplier and internal integration play significant role in enhancing operational performance through the implementation of manufacturing practices. It means that if organisations adopt and implement manufacturing practices to enhance their operational performance then they should also consider establishing supplier and internal integration for attaining desired outcomes of operational performance of organisations.

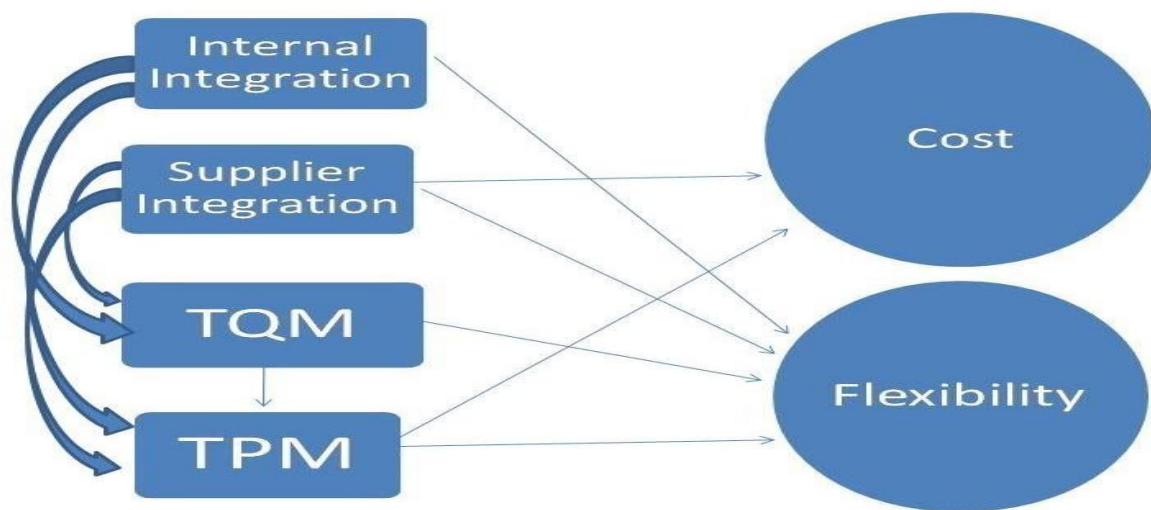


Figure 5-1: Final Framework

5.6. Discussion

It is important that manufacturing practices receive academic attention, as this study focused on knowing the impact of manufacturing practices on cost and flexibility, in industrial companies in the Hashemite Kingdom of Jordan, where it was clarified how to design the model in the previous chapters, the current study clarified the role of four independent variables (TPM , TQM , supplier , internal) on Two dependent variables (cost , flexibility), where a set of hypotheses was designed, data were collected and a description of the statistical analysis was extracted, and then a hypothesis test was extracted to know the effect of each variable on the other.

This chapter contains a presentation of the results of the study and a discussion of the meaning of the results, as it presents the limits and limitations, and future studies.

5.6.1. Overview of the Study

The current research focuses on identifying the key determinants of operational performance in the industrial sector, by exploring the impact of the manufacturing practices (TQM, TPM) on operational performance (Cost, Flexibility) while studying the role of supplier integration and internal integration, in the Jordanian industrial sector. The focus of this research is on the significance of internal and external integration in companies as a strategy for improving operational performance and other outcomes. These include the assessment of the identified benefits derived from the adoption of TPM and TQM in organisations to enhance operational performance and excellence. It focuses on these innovations in the industrial sector, which are designed to improve efficiency and operational performance of manufacturing processes. It is essential to conduct empirical research to identify how these manufacturing practices influence suppliers and internal integration as well as operational performance in the industrial sector.

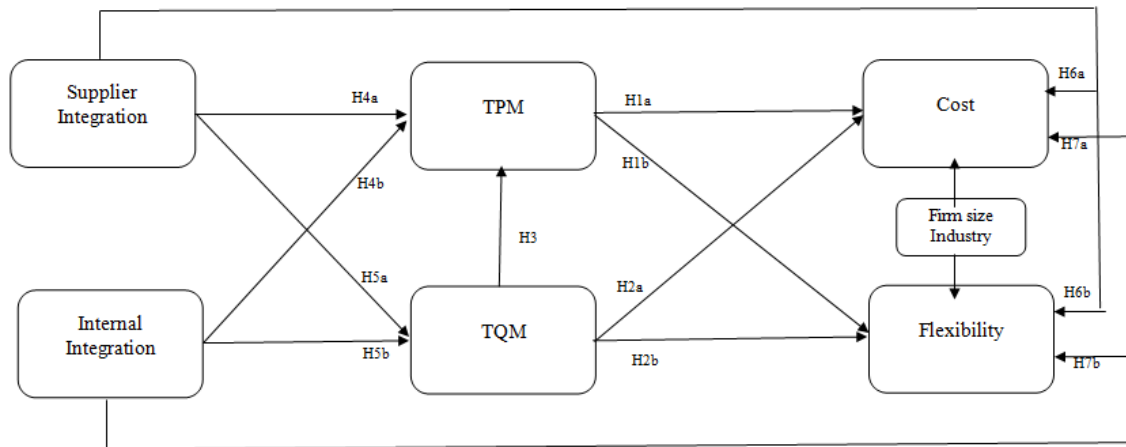


Figure 5-2: Model of Study

In an attempt to understand the relationship between manufacturing practices on operational performance, the research model was developed and tested through the use of survey data collected by Jordanian industrial companies, where the results path analysis showed the relationship between the variables, in relation to the results explained in the previous chapter.

It has been found that TPM has an impact on the cost, and this means that the more attention is paid to TPM it will help reduce the costs that industrial companies are affected by, As for TQM quality is important for any company because it aims to strengthen the competitive position of companies, in light of recent changes in technology and means of communication. In this study, TQM did not prove an effect on costs. As for the effect of TQM on TPM it clarified the existence of Effect, as for the elasticity variable, the TPM and TQM variables proved their effect on flexibility in this study, this study also prove the impact of the supplier on flexibility and cost, where the effect on flexibility was marginally effect, but on cost the effect was positive, on the other hand, the effect of Internal on flexibility was positive, but the effect is effect on cost. Figure 5-3 shows the model summary.

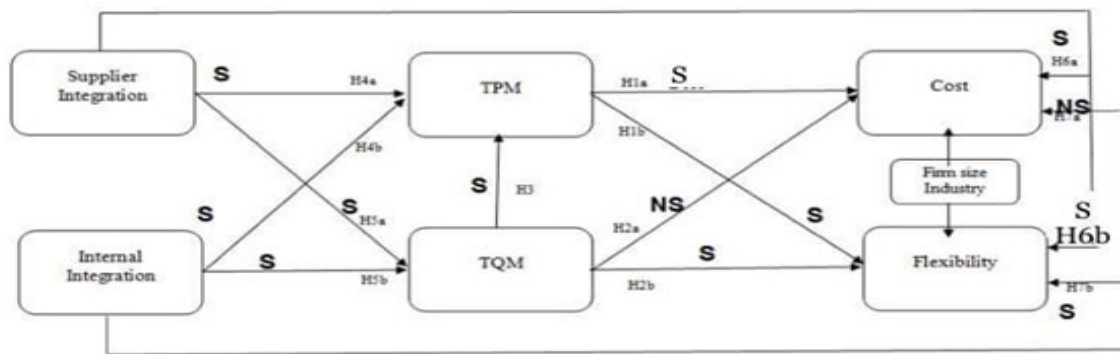


Table 5-21: Model Summary

S: Support NS: Not Support

Such results not only indicate the direction and the nature of relationships between the variables of interest; rather these results have significant implications for literature (theoretical implications) as well as for Jordanian industrialists (practical implications).

5.6.2. Theoretical Implications

Current research has significant theoretical implications as it has contributed significantly to the literature of Resource based view. Resource based view is no more limited to have organisations' focus on its internal resources only, rather it also emphasises on considering external sources of information as a significant resource for organisations that can contribute in enhancing organisational performance (Huo et al., 2016). Hence, in light of this view current research contributes further in the resource based view in a way that the findings of current research indicate that although manufacturing practices like TQM and TPM are implemented by organisations to manage and maintain the internal resources of the organisation to have positive outcomes from the operational performance of the organisation. But suppliers could also contribute as a significant resource so along with internal resource management; organisations should also focus on their external resources. Among all other external resources, suppliers are one of the most significant resources of organisation and they could enhance their operational performance through integration of this resource i.e. supplier integration.

Moreover, as the resource based view emphasises that the resources of the organisation are the key determinants of organisational performance, so the findings of current research contribute in this view that suppliers as a significant resource of organisation also contribute towards operational performance of organisations and could also create a competitive advantage for the

organisation if firms go for integrated supply chains especially for supplier integration and the internal integration. More specifically the findings of current research contribute theoretically to the literature of resource-based view in following ways.

5.6.2.1. *The Impact of TPM/TQM on Operational Performance*

The resource-based view is based on the concept that organisational performance is determined by the resources of the organisation especially the way organisation utilises its resources for having a competitive edge and enhancing organisational performance. More specifically the resource-based view emphasises on the internal strengths and weaknesses of the organisation that could contribute in the development of the core competencies for organisation. But these competencies could be achieved only when these resources are not only identified theoretically but organisation should be able to utilise and gain benefit of these resources through practical implementation. The motivation behind this study is the realisation that Jordanian manufacturing organisations must raise the quality of their output if they are to increase productivity and become more competitive on the world market. Although several research have been conducted to examine the relationship between TQM and operational performance globally, very little, if anything.

As the RBV focuses on the core competencies of the organisations by effective and efficient utilisation of its resources, hence the adoption and implementation of manufacturing practices like TQM and TPM also help organisations attaining such competencies. The findings of current research indicate that TQM is found to have significant positive effect on the flexibility performance while TPM is found to have significant positive effect on cost and flexibility performance of the organisation. It indicates that the manufacturing practices can help organisations in gaining the core competencies through effective and efficient utilisation of organisational resources. RBV has the major focus on the efficiency of organisations that how efficiently they utilise their resources, so the findings contribute that manufacturing practices help organisations in effective and efficient utilisation of these resources.

TQM mainly focuses on elimination of waste as well as elimination of errors in the manufacturing and business operations of the organisations. So, the elimination of waste and errors lead towards effective and efficient utilisation of resources and enables the organisation to get better outputs from same inputs. TQM not only focuses on the manufacturing operations of the organisations, but it also emphasises on the stakeholders of the organisations like

efficient management of the supply chain at both ends as well as training and development of the employees of the organisation, so the findings of the study that TQM significantly positively contributes in operational performance of the organisation contribute that under the resource based view, organisations could better utilise their resources through implementation of the TQM practices in the organisation. TQM offers a procedure for managing dissatisfied customers, making necessary repairs, and preventing similar reoccurrences in addition to the quality assurance that consumers will receive what they want. the value of happy customers and the high cost of acquiring new consumers as compared to retaining existing ones. Customer satisfaction and revenue are directly correlated, according to business study (Nayal et al., 2021).

These findings of current research are consistent with the findings of previous studies; Yunis et al. (2013) indicated that TQM plays significant role in enhancing the operational performance of organisations. The author argued that in recent dynamic business environment, the TQM practices have become the vital need of organisations. Organisations now not only need to compete in their local markets, but globalisation has expanded the competition in international markets as well. Therefore, operational performance plays significant role in attaining organisational effectiveness and efficiency and enables the firm in international markets as well, while TQM practices help organisations in enhancing their operational performance. Zeng et al., (2015) indicated that TQM plays significant role in enhancing the operational performance of organisations in different perspectives but the most important one is its focus in enhancing the value of outcomes and gaining the competitive advantage. Nilda et al., (2017) indicated that organisations make significant efforts to have their operational performance according to the markets' demands and wants, and TQM helps organisations in directing all their efforts for successful achievement of such goals.

Current research also highlights the role of TPM which emphasises on the involvement of employees in the effective and efficient maintenance of the tools and machineries being utilised by the employees, so the tools and machines being the significant resources of organisation could be better utilised by the implementation of TPM in the organisation. As the findings of current research indicate that TPM significantly positively affects the operational performance of the organisation, so these findings contribute in RBV literature in a way that the tangible resources of the organisations i.e. the tools and machines could be best utilised and could

contribute in the enhancement of operational performance of organisation through implementation of the TPM practices in the organisations.

These findings of current research are also consistent with the findings of previous studies; Chandran (2015) indicated that TPM improves the effectiveness and efficiency of production and manufacturing processes of the organisations. Madanhire and Mbohwa (2015) indicated that TPM optimises the operational efficiency of organisations through incorporation of employees in maintenance practices. Singh et al. (2013) indicated that TPM helps organisations enhancing their operational efficiency through ensuring the appropriate maintenance of tools and machines involved in the production operations and secondly through the implementation of correct procedures and corrective measures for enhancing the operational performance of organisation. Similarly, Al Maazmi and Al Hamadi (2021) reported that TPM helps organisations in enhancing their operational performance not only in terms of increased number of units but also in the form of improved quality of the products. In fact, TPM ensures the effective maintenance of equipment by involvement of concerned employees hence ensuring the elimination of breakdowns, accidents and defects. When such disruptions in the production process are eliminated the operational efficiency of the production units is ensured hence enhancing the operational performance of the organisation (Gupta et al., 2021).

In this way the findings of current research contribute theoretically in a way that initially the resource-based view only emphasizes on the fact that the resources of organisation should be utilised in most efficient manner, but the findings of current research also indicate the ways that how these resources could be best utilised and how the outcomes of these resources could be enhanced i.e., through adoption and implementation of the manufacturing practice (TPM, TQM).

5.6.2.2. *The Impact of Supplier/Internal Integration on TPM/TQM*

As the findings of current research indicate the significant relationships between the independent variables of study i.e. TPM, TQM, supplier and internal integration, so these findings also contribute theoretically in the literature of RBV. The findings indicate that supplier integration has significant positive effect on TPM, and TQM and similar positive effect of internal integration is found on TPM and TQM as well. As mentioned earlier that the major focus of RBV is the effective utilisation of the resources of organisation and initially the RBV was based on focuses on the internal resources of the organisations, but with the passage

of time the view has also extended that resources are not limited to the internal factors of the organisations, but the flow of information should also be among the internal and external stakeholders of the organisations (Huo et al., 2016).

Therefore, in light of this view suppliers are also considered as a significant resource of organisations that not only contribute to providing effective information about the market and materials but also facilitate the organisations in efficient management of inventory that leads to have financial and non-financial benefits for the organisations. (Gupta et al., 2021). So, the findings of current research contribute that the effective and efficient utilisation of resources could be achieved through the implementation of manufacturing practices that help organisation in enhancing the outputs through elimination of errors and improving the efficient utilisation of tools and machines by the employees, while this effectiveness and efficiency is further enhanced through internal and supplier integration.

Although TPM and TQM enhance the efficiency of business operations, but this efficiency could never be achieved if these practices are implemented at individual departmental levels, rather internal integration i.e., the collaborative efforts of the internal departments of the organisations would lead to get better results in the form of elimination of errors and better performance of tools and machines through efficient maintenance. The integration of departments facilitates the flow of information among the concerned individuals, and they could better resolve issues through effective discussion and feedback from the concerned authorities and individuals (Gupta et al., 2021).

Hence, as the resource-based view emphasises on the effective and efficient utilisation of organisational resources so internal integration helps achieving this efficiency by enhancing the outcomes of TPM and TQM practices. Moreover, effective and efficient utilisation also implies to get better outcomes from available inputs, so supplier integration helps organisations having significant control on their inputs i.e., inventory management etc. As the TQM focuses on eliminating errors and wastes of the organisation so that enhanced outputs could be gained through similar inputs and TPM focuses on the utilisation of tools and machines and their maintenance to improve their productive efficiency, so supplier integration contributes positively in a way that it facilitates organisations to have significant control on the inventory levels hence helps eliminating unnecessary levels of inputs (Gupta et al., 2021). Similarly, when inventory levels are just according to the requirements of the organisation and the

demand of the markets, then excessive production is also controlled, and tools and machines are not over run hence TPM is efficiently implemented as well.

Therefore, in this way supplier and internal integration contribute positively towards TPM and TQM and also contribute in RBV literature because these integrations help organisations in effective and efficient utilisation of organisational resources. Again, as discussed earlier that RBV emphasises on effective utilisation of resources, so the findings of current research contribute by providing ways of such effective utilisation. Moreover, the findings of current research also contribute theoretically to the literature of manufacturing practices and SCM as these two approaches are always compared by the researchers in different perspectives, but their joint effects are always ignored by the previous studies. Marian et al., (2020) while making comparison among the manufacturing practices (TPM, TQM) and the SCM reported that manufacturing practices (TPM, TQM) aim at enhancing the quality of outputs through enhancement of operational performance while SCM helps controlling the manufacturing processes with a focus of quality enhancement. Thus, both approaches have shared purpose of quality enhancement approached in different ways. Similarly, Talib, et al. (2011) indicated that SCM and TQM have become vital for organisations' success hence organisations adopting any of these approaches could achieve their goals in better way than the ones who are not implementing such approaches. These studies lack reporting the combined effect of these approaches as well as have not focused on the facts that if organisations are adopting both approaches in a single setting then how they are able to enhance their operational efficiency. So, the findings of current research contribute to filling this gap.

On the other hand, some previous studies (Nilda et al. 2017; Yunis, 2013; Rachmat, 2019) have reported the interrelationship of supplier and internal integration with the manufacturing practices, so the findings of current research validate the findings of these previous studies. Nilda et al. (2017) reported that the effectiveness of TQM practices is enhanced by internal integration in a way that when different functions within the organisation work collaboratively then they all line up for achievement of a single goal of quality improvement. Yunis (2013) reported that manufacturing practices are adopted by the organisations to cope with the challenges of market dynamics; hence supplier integration improves the effectiveness of these manufacturing practices by providing appropriate information, control of costs and consolidating the processes in efficient manners. Rachmat (2019) indicated that implementation of manufacturing practices requires training of employees for ensuring the

effectiveness of these practices, so internal integration improves the effectiveness of such training by facilitating the flow of information among different functional units of the organisation. Hence, all such studies have reported the effects of supplier and internal integration on manufacturing practices (TPM, TQM), which are also validated by the findings of current research, but current research further contributes in literature that how these two approaches jointly contribute towards improvement of the operational performance of the organisation.

5.6.2.3. *The Impact of Supplier/Internal Integration on Operational Performance*

Findings from this research indicate that supplier integration is found to have significant positive effect on cost and flexibility performance of the organisation, while internal integration is found to have significant positive effect on flexibility performance of the organisation. Operational performance is considered as an indicator of the success of organisation as it helps achieving the desired and targeted goals, so any factors contributing positively to enhancing this operational performance are also considered as significant resources of the organisation. The major focus of RBV for effective utilisation of organisational resources is further enhanced through the findings of current research in a way that organisational departments and suppliers are the significant resources of organisations, but these could be best utilised through integration. As supplier integration is found to have positive effect on cost performance so it is indicated that through supplier integration organisations could have better control on the costs of their operations.

Cost efficiency could be a competitive advantage for the organisations so through supplier integration organisations could gain the core competencies as emphasised by the RBV. Moreover, supplier integration is also found to have significant positive effect on the flexibility performance of the organisation, and in recent dynamic environment operational flexibility has become essential for the survival of organisations due to frequent changes in the marketplaces. Organisations having operational flexibility could gain the core competencies in making modifications and adjustments in their operations according to the changes in market demands and supplier integration plays significant role in attaining this core competency. Having healthy relationships with suppliers and incorporating them in the decision-making stages could help organisations in achieving the operational flexibility.

In this way, the findings of current research also validate the findings of previous research (Wieland et al., 2016; Gurahoo and Salisbury, 2018; Ahmad et al., 2019). Wieland et al. (2016) reported that internal functional units of the organisations are critical resources for attaining the targeted outcomes of the organisations, but their collaborative efforts are necessary for eliminating the inefficiencies of operations, so internal integration helps organisations in ensuring the collaborative efforts of these functional units which ultimately helps improving the operational performance of organisations. Ahmad et al. (2019) indicated that when internal departments of the organisations are integrated then they all work together for the achievement of desired goals. So, employees' integration creates competitive advantage for the organisation and helps achieving the goals related to cost, quality, and responsiveness. Gurahoo and Salisbury (2018) indicated that supplier integration incorporates different stakeholders working together for achievement of desired value of the outputs. Such integration helps organisations in accurate assessment of the risks and opportunities prevailing in the external environment and helps mitigating the uncertainty of the business environment and contributes positively in enhancement of the operational performance.

Hence, the findings of current research not only contribute in expanding the approaches of resource based view but also contribute significantly to the literature of SCM, TQM and TPM. It also provides insights to the researchers for investigating the relationships among factors in different perspectives.

5.6.3. Practical Implications

Current research has not only significant theoretical implications but also have practical implications for industrial organisations especially for Jordanian industrial organisations. As highlighted by the review of previous literature that initially the manufacturing practices (TPM, TQM) and the supply chain integration are found to have significant contribution towards operational performance of the organisations, but these were focused individually like some organisations were focusing on implementation of manufacturing practices for enhancement of their operational performance or the overall performance of the organisation. While some others were focusing on the supply chain integration for enhancing the operational efficiency of the organisations, but neither researchers nor industrialists have focused on adoption of both strategies in single settings due to the lack of investigation in this area.

In this regard, current research has provided significant insights for the industrialists by highlighting the interrelationships of these practices and strategies. Industrial organisations'

efficiency and success depends on their operational efficiency and in the context of Jordan cost and flexibility are found to be the main determinants of the operational performance of industrial organisations. Hence, current research provides insight to Jordanian industrialists that which factors contribute positively to enhancing cost and flexibility performance of these organisations.

The positive impact of TPM and TQM on operational performance of the organisation as found by the results of current research indicate that Jordanian industrialist could enhance their operational performance by adoption and implementation of manufacturing practices (TPM and TQM). Operational performance efficiency could be a competitive edge for these industrialists when they would have better controls on their internal resources; they are able to eliminate waste and errors in the operations so that they could be able to get higher levels of outputs from similar levels of inputs. Moreover, through the implementation of TPM practices they could enable their employees to effective and efficiently utilising their tools and machines in a way that these could provide enhanced outputs. So, the implementation of manufacturing practices could help them gaining the efficiency in operational performance.

On the other side, the supplier and internal integration is found to have positive impact on the manufacturing practices, so Jordanian industrialists should also focus on developing supply chain integration especially supplier integration and internal integration. Currently in Jordan, supplier and internal integration is not widely implemented in the industrial firms, so the findings of current research provide insight to Jordanian industrialists that they should not only focus on implementation of the manufacturing practices, but they should also focus on supplier and internal integration. Internal integration would help their employees having a better understanding of the operations and to resolve issues in an efficient manner that would ultimately contribute positively to enhancing operational performance of the organisation. Moreover, if supplier integration is developed then they would be able to have better performance in terms of cost efficiency and operational flexibility. Just like all other economies across globe, Jordanian industries are also moving in the dynamic environment so they also need to have operational flexibility so that they can cope with the dynamic business environment. Hence, supplier integration helps attaining this operational efficiency so Jordanian industrialists should focus on developing internal and supplier integration so that they could enhance their operational performance as well.

The current research is expected to help Jordanian industrialists to have a practical implication of the resource-based view in a way that the RBV has the main focus on effective utilisation of organisational resources, so current research provides insights to the industrialists that how they could utilise their internal and external resources in best possible way. In recent dynamic environment, the flow of information at internal levels of organisations as well as from external to internal levels of organisations has become vital for the survival of organisations, and this flow of information could be developed through integration at different levels of the supply chain. As Jordanian economy and Jordanian firms have somehow rigid environments which lack flexibility and integration, so current research emphasises that Jordanian industrialists should focus on integration of supply chain especially internal and supplier integration otherwise they would not be able to enhance their operational performance.

The success of industrial organisations is dependent on the operational efficiency of the organisations, so organisations could not survive while ignoring the important factors that contribute significantly in enhancement of this operational performance. Similarly, in the context of Jordan, as current research has found that TPM, TQM, supplier integration and internal integration are significantly contributing towards operational performance of the organisations, so Jordanian industries could not survive while focusing on manufacturing practices or SC integration separately. Rather they should now focus on adoption of both these strategies at the same time so that they could be able to gain the competitive advantage.

5.7. Chapter Summary

The current research utilized a path analysis in order to test the hypotheses of the model, where the study model was designed, which consists of four independent variables (TPM, TQM, Supplier, and Internal) and two dependent variables (Cost, Flexibility) in the Jordanian industrial companies, where all the results were presented in this chapter, which were acceptable. The results of the analysis will be discussed in detail in the next chapter. Additionally, the chapter explains the results of the descriptive analysis of the data collected from the study sample, which is represented by the Jordanian industrial firms. The chapter explained the logic behind conducting the different tests to check the normality, validity, and reliability of the study data. After running the different quantitative tests using SPSS and AMOS software, all obtained values from the data analysis process were within the acceptable ratios. In that, they obtained values have passed the stability coefficient test of Cronbach's

alpha as well as the exploratory factor analysis and the confirmatory factor analysis tests. Based on this, it can be confirmed that the research model is fit, and the obtained values are valid.

Chapter 6 – Conclusion

After exploring the literature and discussing with academic experts, the quantitative method was favoured to investigate the research phenomenon and to provide generalisable findings and conclusions. For that, both traditional and online forms of questionnaire were distributed to collect data from firms operating in the Jordanian industrial sector. Many firms have responded to the questionnaire, and after the filtration process, 224 responses were adopted. The data from these questionnaires were descriptively analysed and tested (normality, EFA, CFA, regression) using SPSS and AMOS software. Based on these analysis steps, the different research hypotheses were tested, which allowed producing twenty-one findings that were presented in six models. These results were then discussed and thus, collectively answered the main research question:

RQ: What impact do manufacturing practices (TPM, TQM) have on the operational performance of the industrial firms when adopted in line with supplier and internal integration?

This thesis showed that manufacturing practices are very important in the business world, thus, they deserve to be focused on by researchers. The thesis discussed the impact of manufacturing practices (TPM, TQM) on operational performance (cost, flexibility). It also clarified the importance of SCM elements (supplier integration, internal integration) in enhancing this impact in industrial firms. This research has significant contribution in literature as well as in the economic development of Jordanian economy. The literature lacked evidence about reporting the impact of the combined effect of manufacturing practices and the SCM practices to enhance the operational performance of the organizations. This study has filled this gap. It highlighted the effect of TPM, TQM along with supplier integration and internal integration on operational performance of industries. By studying these effects in Jordanian economy, the study has contributed to Jordanian literature and knowledge of the policy makers that how these factors can enhance the operational performance of Jordanian industrial organisations. Although industrial organisations in Jordan are already practices such approaches to enhance their operational performance but the combined effects of these approaches are never implemented and focused by these organisations. Therefore, this study has significant practical implications as well. Following the findings of this research, Jordanian industrialists can focus

on integration and accumulated implementations of the approaches so that enhanced outcomes can be generated through operational efficiencies.

The remainder of this chapter contains a quick overview of the various research findings, its strengths, limitations, and recommendations for future research efforts.

6.1. Overview of Research

In order to find out the impact of manufacturing practices (TPM, TQM) on operational performance (Cost, Flexibility) of the firms while investigating the role of supplier and internal integration in this relationship, current research has incorporated Jordanian industrial firms through collection of primary data from the concerned participants. Although the relationship of manufacturing practices and SC integration has already been studied by previous research in the same field, but these studies lacked investigating the interrelationships of manufacturing practices and SC integration towards operational performance of the organisations. Especially in the context of Jordanian firms, two manufacturing practices i.e., TPM and TQM were identified as significant contributors towards operational performance, and in SC integration, supplier integration and internal integration were found to have more contribution as compared to the customer integration, so current research has focused on these specific factors only. Moreover, for operational performance, the cost and flexibility performance are found to be the key determinants of operational performance of Jordanian industries.

The investigation of the relationships among all these variables indicated that manufacturing practices (TPM, TQM) are positively contributing towards operational performance. Although TQM is not found to have significant relationship with the cost performance, but other variables are found to have significant positive contribution. On the other hand, supplier and internal integration are also found to have positive contribution towards operational performance of the organisations. Among these variables, internal integration is not found to have significant relationship with the cost performance of the organisation, but it is found to have significant relationship with the flexibility performance of the organisations.

While studying the interrelationships of supplier and internal integration with the manufacturing practice, it is found that supplier and internal integration have positive role towards TPM and TQM. So, the findings of current research indicate that Jordanian industries should not only focus on implementation of the manufacturing practices for enhancing their operational performance, but they should also focus on developing internal and supplier

integration. As the current research has investigated these relationships in light of the RBV, so it contributes significantly to literature by highlighting that organisations should not only focus on management and maintenance of the internal resources of the organisation, but these should also consider external factors i.e., suppliers as significant resource and effective utilisation of these resources could enhance the operational performance of these organisations.

6.2. Research Strengths

Every research work contains several strengths and limitations. While limitations will be the focus of later discussion, this section highlights the research strengths and how they provide extra value for its outcomes. This research has covered the Jordanian industrial sector and obtained responses from a large group of firms that are usually difficult to access. The research has gained this access after working on developing several contacts with different Jordanian authorities and institutions in order to guarantee their support. These include the Ministry of Industry, Trade and Supply and Amman Chamber of Industry. This resulted in the ability of this research to access a poorly explored and nearly unlocked sector to dig deeper and understand the various factors that affect its operational performance.

This research has covered an area that is usually less interesting for researchers globally. To explain, while research efforts in this field usually focus on exploring the manufacturing practises and their role in improving the operational performance of organisations in the well-developed, Western economies, this thesis has focused on investigating the Jordanian industrial sector that is considered to be a developing, middle eastern economy. The research, therefore, has produced findings that can provide a steppingstone for other research efforts in the same area or other areas.

Limited studies have focused on manufacturing practises and the operational performance with supply chain integration as a moderating factor. However, the Jordanian industrial sector has been suffering in the last years from the difficulty of securing the industrial logistics, especially with several regional circumstances affecting the ability to import the raw material and export final goods. As a developing country, it was necessary to highlight the importance of properly implementing the manufacturing practices such as TQM and TPM in order to enhance the quality, and reduce the time, of production. With this in mind, this research provides a solution that can help stakeholders in the industrial sector to make thoughtful decisions to produce more unique products with a competitive quality, which enables the Jordanian industrial firms to maximise their profitability.

6.3. Research Limitations

This research has encountered some limitations during the process of data collection. As this research was conducted based on the quantitative survey tool, and as the world today is going through the covid-19 pandemic, more access to data with few more participating firms would increase the reliability and credibility of the findings. This limitation was addressed by distributing online questionnaires to participants in order to secure more responses.

In many cases, some terms and concepts were ambiguous to participants and, thus, required to be further clarified to guarantee the high accuracy of answers. Although some responses were excluded because of the poor answers that occurred from misunderstanding the included terms and concepts, other responses were completed by explaining the intended meanings of these terms and concepts to participants. As in any research discussing such technical topic, this issue is expected to occur, but, fortunately, it was overcoming by adopting the right communications with participants. Another limitation of this research is to not highlight the differences of effect in different kinds of settings i.e., SMEs and MNCs. As the SMEs and MNCs vary in their structures and functioning, so the effects of these manufacturing and supply chain practices will be different in these two different settings. These are not highlighted by this research. These could be highlighted in future research.

6.4. Research Recommendations

The results of this research show that there is an effect of manufacturing practices on flexibility. This, in turn, encourages the first future research avenue about the need to studying the other factors of manufacturing practices that were not addressed in this thesis in order to provide more comprehensive view of this variable. As there was no effect of TPM on cost and flexibility, future works should investigate the reasons for the absence of any relationship between these variables. While this research has focused on the scale of industrial firms, other research efforts should study the same variables in other sectors and areas to find out about the ability of replicating the current findings.

Further, conducting more studies about the other factors of supply chains management such as customer integration rather than supplier integration and internal integration can be useful to check their role, if any; in enhancing the operational performance of industrial firms (the same idea applies to factors related to manufacturing practices). In addition, future research avenues might consider the impact of manufacturing practices on the financial performance, as opposed

to the operational performance, of the industrial firms. Furthermore, the research can be conducted on finding out the impact of these combined practices in different types of organisational settings. The research could also be conducted in finding out the differences in different kinds of industries.

6.5. Chapter Summary

This research has developed a framework about the impact of manufacturing practices (TPM, TQM) on the operational performance (cost, flexibility) of the industrial firms. The research further has focused on the factors of SCM (supplier integration, internal integration) as moderators of this previously explained relationship. The research, thus, has fulfilled its aim of identifying and better understanding the role of having proper manufacturing practices in enhancing the operational performance of industrial organisations, and how this role is promoted with the presence of the right supply chain integration.

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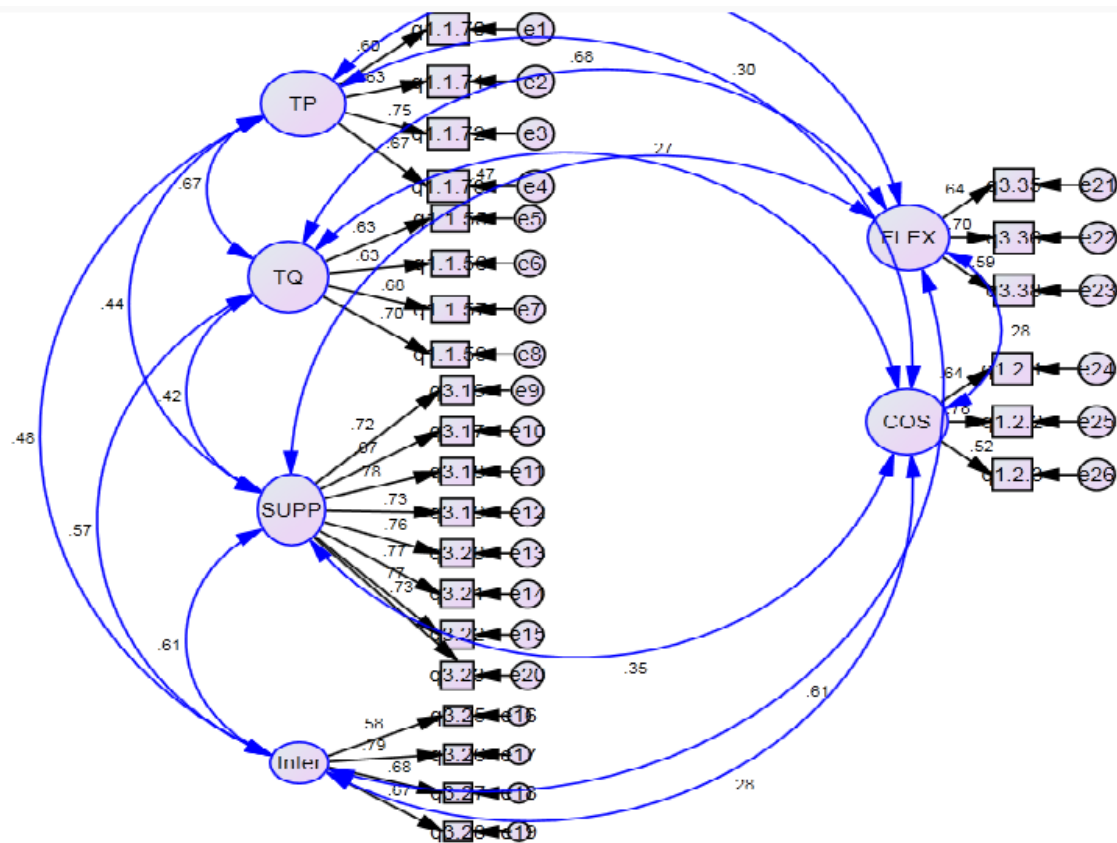
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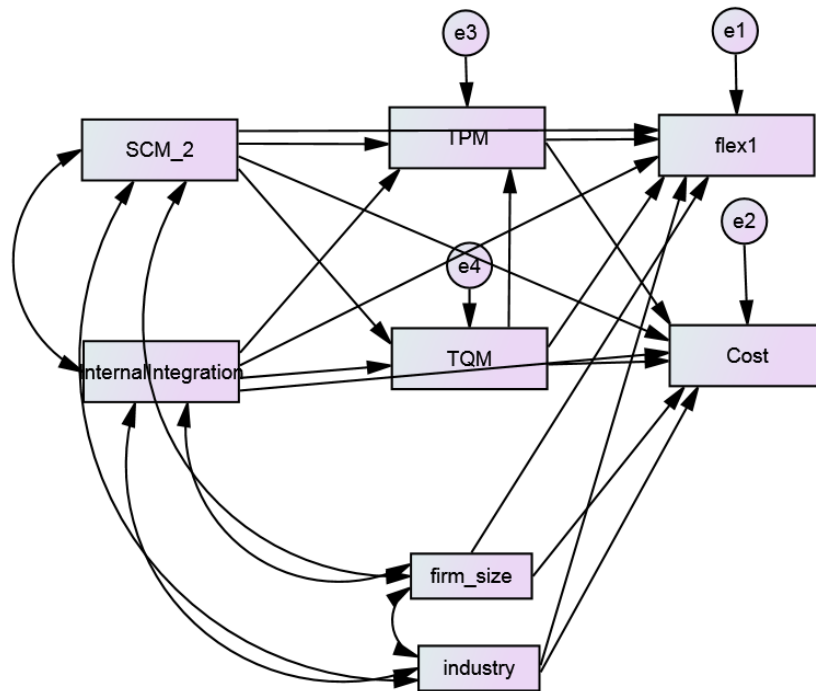
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Appendices

Appendix A 1 – CFA



Appendix A 2: Path Analysis



Appendix B 1: Ethical Approval Letter 1

Ethics - Message (HTML)

File Message Help Tell me what you want to do

Delete Archive Move Reply Reply All Forward Move to: ? Mark Unread Find Zoom

Ethics

SK Smallwood, Karen <karen.smallwood@anglia.ac.uk>
To: Al-Junidi, Mohammad (Postgraduate Researcher)
Cc: Teng, Teng

Follow up. Start by 19 October 2019. Due by 19 October 2019.
You forwarded this message on 09/11/2019 22:01.
This message was sent with High importance.

Thu 17/10/2019 16:06

Dear Student

Principal Investigator: Mohammad Al-Junidi
DREP number: Dear Student
Project Title: Potential Determinants of Productivity in the Jordanian Industrial Shareholding Companies Including the Impact of the Arab Spring Crisis

I am pleased to inform you that your ethics application has been approved with minor amendments by the Faculty Research Ethics Panel (FREP under the terms of Anglia Ruskin University's Research Ethics Policy (Dated 8 September 2016, Version 1.7)).

Please action the points below and submit to your academic supervisor for them to review:

- Obtain a gatekeeper letter(s) for key employees.
- Remove the cover page for the PIS Guidance (this should not be a part of PIS in practice).
- Section 2 of the ethics form will need to be amended...there will be primary data, so you will use data that are not publicly available, gatekeeper letter needed;
- Revise Section 4 accordingly. In Section 4, you states 'The data collected will be carefully gathered and stored through official emails and personal storage devices. In addition to that, the questionnaire procedures will ensure data protection as it will not be shared with other parties and will only be used for the research purpose. Moreover, the data will be collected in a confidential and an anonymous way to keep the research's integrity in accordance with the ethical principles of the research practice.' - How are the data stored through official emails? With personal storage devices, how are these safeguarded? What does 'confidential and an anonymous way' mean? How do you plan to anonymise the data? How do you ensure that the data collected cannot be reversed engineered to identify participants? For how long do you intend to keep the data?
 - You also states that 'several initial arrangements have been made with representatives of the companies addressed in the research in order to ensure that the data collected will remain confidential.' What are these arrangements? Need to be clearly articulated.

Once all the bullet point issues have been addressed, you can start his research whilst awaiting for the gatekeeper letter(s).

Ethical approval is given for a period of 1 year for undergraduates/masters students from the date of this letter.

It is your responsibility to ensure that you comply with Anglia Ruskin University's Research Ethics Policy and the Code of Practice for Applying for Ethical Approval at Anglia Ruskin University available at www.anglia.ac.uk/researchethics including the following.

Appendix B 2: Approval Letter 2



غرفة صناعة عمان
AMMAN CHAMBER OF INDUSTRY

Date: ٢٠١٩/٥/٢٨ التاريخ

Ref. No.: ١٨٨/١/٢٤ الرقم

لمن يهمه الأمر

تحية طيبة وبعد،،،

نود اعلام سعادتك أن الطالب محمد مزيد الجندي، هو طالب دكتوراة في جامعة أنجاليا رسكن / بريطانيا، يقوم باعداد بحث حول "المحددات المحتملة للانتاجية في الشركات المساهمة الصناعية الأردنية بما في ذلك تأثير أزمة الربيع العربي"، حيث سيقوم بتوزيع استبيانات بحثية على عدد من الشركات الصناعية في المملكة.

راجين التعاون معه فيما يتعلق بتعبئة هذه الاستبيانات.

شاكرين ومقدرين لكم اهتمامكم.

وتفضلوا بقبول فائق الاحترام،،

د. نائل الحسامي

المدير العام



م.ص.